BOHR–SOMMERFELD ASYMPTOTICS FOR LIMIT-CIRCLE JACOBI MATRICES

Explicit high-energy eigenvalue laws for every self-adjoint extension of a Jacobi matrix with links $c\,n^{\alpha}(1+(\kappa+\kappa'(-1)^n)/n+\cdots)$, $\alpha>1$: offsets that depend on the extension, rates for balanced links, and the second-order term for shifted powers.

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Abstract. A Jacobi matrix whose off-diagonal entries grow regularly like $c\,n^{\alpha}$ with $\alpha>1$, without a strong alternation between even and odd entries, is in the limit-circle case: it has a one-parameter family of self-adjoint extensions H_t, each with discrete spectrum. The growth of the eigenvalue counting function is known up to multiplicative constants, but no eigenvalue law with an explicit constant was known beyond a few explicitly solvable chains. We prove such a law, with an explicit offset, for every self-adjoint extension of a zero-diagonal Jacobi matrix with links $a_n = c\,n^{\alpha}\big(1+(\kappa+\kappa'(-1)^n)/n+O(n^{-1-\delta})\big)$ in the limit-circle range $2|\kappa'|<\alpha-1$. For $\kappa'=0$ the eigenvalues satisfy the Bohr–Sommerfeld rule

$$N_{\lambda_j} I_\alpha - \frac{\kappa\pi}{2\alpha} = j\pi + t + O\big(j^{-\min(1,\delta)}\big), \qquad N_\lambda = (\lambda/2c)^{1/\alpha}, \quad I_\alpha = \tfrac{1}{2}\,\mathrm{B}\big(\tfrac{\alpha-1}{2\alpha},\tfrac{1}{2}\big),$$

with no quarter-period correction in the normalisation of the extensions fixed by the solutions at $\lambda=0$, and for a shifted power with a curvature term the next coefficient is $(\kappa_2+(2\alpha-1)/24)\,J_\alpha$. For $\kappa'\neq 0$ (dimerised links) the far field is a Bessel problem of order $\nu=1/2-\kappa'/(\alpha-1)$: measured in units of π, the offsets of the two extensions H_0 and $H_{\pi/2}$ are shifted by $+\kappa'/(2(\alpha-1))$ and $-\kappa'/(2(\alpha-1))$ from the balanced values 0 and $1/2$, every other extension follows one of these two laws, and the remainder is $o(1)$. The proofs combine an exact rotation identity for two-site blocks, an adiabatic normal form, a Bessel comparison for the far field, and a discrete Olver-type bound for Airy approximations that we prove from scratch; the last also extends a theorem of Fischer, Burgarth and Lonigro on singular limits from $\alpha>4/3$ to $\alpha>1$. Applications include the spectra of all symmetry-preserving self-adjoint extensions of higher-order squeezing operators and the Krein and Friedrichs spectra of birth–death processes with polynomial rates. For these processes we compute the critical indices of Berg and Szwarc exactly: $(1-\nu)/2$ for the Friedrichs solution and $(1+\nu)/2$ for every other N-extremal solution, all of them determinate at the critical weight. The numerical claims are checked against an independent eigenvalue computation.

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1 INTRODUCTION

A Jacobi matrix whose off-diagonal entries grow fast enough is not determined by its action on finitely supported vectors. If the entries a_n grow regularly like n^α with $\alpha > 1$, without a strong alternation between even and odd entries, and the diagonal is small, every solution of the eigenvalue equation is square-summable (Proposition 2.1 gives the precise range): the matrix is in Weyl's *limit-circle* case, the associated Hamburger moment problem is indeterminate, and the self-adjoint realisations form a circle $\{H_t\}_{t\in[0,\pi)}$ [2, 36]. Each H_t has discrete spectrum, and the spectra of different extensions interlace. Such matrices arise wherever an unbounded operator is written in a basis in which it is tridiagonal: birth–death processes with polynomial rates and their Stieltjes problems [9, 38], the higher-order squeezing operators $a^k + a^{\dagger k}$ of quantum optics for $k \geq 3$ [16], and, as a motivating example from representation theory, the triple product of the generators of three $\mathfrak{su}(1,1)$ representations, which in a recoupling basis is a zero-diagonal Jacobi matrix whose links grow cubically (we do not use this example below). In each case the physically meaningful object is a particular extension, and the question is what its spectrum is.

What was known. The general theory controls the growth of the spectrum but not its position. The eigenvalues of every H_t are the zeros of an entire function of the Nevanlinna matrix, and all these functions have the same order and type [4]. For power-like links the order is $1/\alpha$ [6, 35], and in one concrete case, $a_n = n^p$, Bochkov determined the type [11]. General results bound the counting function of the eigenvalues only up to universal multiplicative constants, among them the recent growth estimates of Pruckner, Reiffenstein and Woracek [31, 33]; see the survey [34], whose Part V discusses exactly this gap. Eigenvalue laws with an explicit constant were known only for explicitly solvable chains: the quartic birth–death rates of Berg and Valent, whose Krein and Friedrichs spectra are known in closed form [9, 39, 5], and the leading term for some cubic rates [23, 24]. Valent's conjecture on the indicator of the Nevanlinna matrix [38, 6] predicts the leading constant for polynomial rates, but says nothing about the next term, which is the one that distinguishes the extensions.

What this paper proves. We prove a Bohr–Sommerfeld law with an explicit offset for every self-adjoint extension of a zero-diagonal Jacobi matrix with links

$$a_n = c\, n^\alpha \Big(1 + \frac{\kappa + \kappa'(-1)^n}{n} + O(n^{-1-\delta})\Big), \qquad \alpha > 1, \tag{1}$$

which is in the limit-circle case exactly when $2|\kappa'| < \alpha - 1$. For balanced links ($\kappa' = 0$) the law comes with a rate, and for shifted powers with a curvature term we also compute the second-order term. For dimerised links ($\kappa' \neq 0$) the law is proved with an $o(1)$ remainder only; its rate and second-order term are open (Section 13). Put $N_\lambda := (\lambda/2c)^{1/\alpha}$, the leading-order turning scale, defined by $2c\,N_\lambda^\alpha = \lambda$ (the site where $2a_n = \lambda$ differs from it by a bounded amount), and

$$\Phi(\lambda) := N_\lambda I_\alpha - \frac{\kappa\pi}{2\alpha}, \qquad I_\alpha = \tfrac{1}{2}\,\mathrm{B}\big(\tfrac{\alpha-1}{2\alpha}, \tfrac{1}{2}\big), \qquad J_\alpha = \tfrac{1}{2\alpha}\,\mathrm{B}\big(\tfrac{1}{2} + \tfrac{1}{2\alpha}, \tfrac{1}{2}\big).$$

The extensions are parametrised by an angle t in a normalisation, fixed by the solutions at $\lambda = 0$, in which t enters the law additively (Theorem 2.2); it is related to Simon's and Yafaev's parametrisations by explicit formulas (Remark 2.2).

Theorem 1.1 — Main theorem

Let J have links (1) with $2|\kappa'| < \alpha - 1$ and $b \equiv 0$, and let $\lambda_j(t)$ be the eigenvalues of H_t, labelled by $\psi(\lambda_j(t)) = j\pi + t$, where ψ is the lifted phase of the regular solution at infinity (Theorem 2.2); for $b \equiv 0$ this means $\lambda_0(0) = 0$ and $\lambda_j(t)$ increases with j and t. As $j \to +\infty$:

(a) if $\kappa' = 0$, then $\Phi(\lambda_j(t)) = j\pi + t + O(j^{-\min(1,\delta)})$ for every t;

(b) if $\kappa' \neq 0$, put $\theta_0 := \kappa'\pi/(2(\alpha-1))$. Then $\Phi(\lambda_j(0)) = j\pi + \theta_0 + o(1)$ and $\Phi(\lambda_j(\pi/2)) = j\pi + \pi/2 - \theta_0 + o(1)$, and every other extension follows the law of $H_{\pi/2}$ if $\kappa' > 0$ and of H_0 (shifted by π for $t > \pi/2$) if $\kappa' < 0$;

(c) if $a_n = c(n+s)^\alpha(1 + \kappa_2(n+s)^{-2} + O(n^{-3}))$, then

$$\Phi(\lambda_j(t)) = j\pi + t + \Big(\kappa_2 + \frac{2\alpha - 1}{24}\Big)\frac{J_\alpha}{N_{\lambda_j}} + O\big(N_{\lambda_j}^{-4/3}\big), \qquad \kappa = \alpha s.$$

Parts (a) and (b) hold with a diagonal $|b_n| \le C(1+n)^\beta$, $\beta < \alpha - 1 - 2|\kappa'|$: in (a) with the labels $\psi(\lambda_j(t)) = j\pi + t$ (the convention $\lambda_0(0) = 0$ is then no longer available) and the rate $O(j^{-\min(1,\delta,\alpha-1-\beta)})$, and in (b) with the offsets holding modulo π.

Part (a) is Corollary 8.3, which needs no condition on the differences of the links; part (b) is Corollary 6.3 with Theorem 6.4; part (c) is Theorem 9.1; the counting functions follow (Corollary 5.3). Three features of the law are worth stating separately.

First, there is no quarter-period correction. Below the turning point each two-site block contributes π to the phase. The Airy turning point contributes its quarter-period $+\pi/4$, and the limit-circle end contributes $-\pi/4$: the turning-point mode takes equal values on the two sites of a block, and the solutions that define the boundary conditions were normalised to equal amplitude at infinity, so in their coordinates the mode sits at $-\pi/4$ (Section 4, Figure 1). The statement that does not depend on any coordinate on the circle of extensions is the law for H_0, the extension defined by the solution P of the equation at $\lambda = 0$ with $P_0 = 1$, $P_1 = 0$: rescaling P does not change the boundary

condition, and for balanced links $\Phi(\lambda_j(0)) = j\pi + o(1)$. The same holds for $H_{\pi/2}$, defined by the second solution R, with $j\pi + \pi/2$. Every other extension needs a coordinate, and the normalisation of P and R to equal amplitude at infinity (by the constant ϖ of Proposition 2.1) is the one in which the band-edge mode sits at exactly $-\pi/4$ for every λ, which is why t then enters additively. In Simon's coordinate τ_S, related by $\tan t = \tan \tau_S/\varpi$, the same law is not additive. So the absence of a quarter-shift is a statement relative to this calibration, which is fixed once by the data at $\lambda = 0$ and not adjusted to any eigenvalue.

Second, the offset $\kappa/(2\alpha)$ is set by the $1/n$ term of the links, and the extension enters only through t/π.

Third, an alternating $1/n$ term changes the far field qualitatively. The two solutions at $\lambda = 0$ acquire block envelopes of different sizes, the equation becomes a Bessel problem of order $\nu = {}^1\!/_2 - \kappa'/(\alpha - 1)$, and the extensions split into the one defined by the solution of smaller block-envelope size (H_0 if $\kappa' > 0$, $H_{\pi/2}$ if $\kappa' < 0$) and all the others. The coordinate-free form of the dimerised law is that the offsets of H_0 and $H_{\pi/2}$, the extensions defined by P and by R, move from their balanced values 0 and $\pi/2$ by $+\theta_0$ and $-\theta_0$, $\theta_0 = \kappa'\pi/(2(\alpha - 1))$; this is visible already in the leading behaviour of the eigenvalues. The second-order constant also has an interpretation: $\kappa_2 J_\alpha$ is the classical correction from the curvature of the links, and $(2\alpha - 1)/24 = (\alpha^2 - (\alpha - 1)^2)/24$ is the Weyl–Moyal term of the symbol $2A(X)\cos p$ (Remark 9.1).

What is new and what is not. The leading constant is the one predicted by Valent's conjecture under regular distribution of the zeros, and the completely regular growth and indicator of the Nevanlinna matrix that follow from the law (Corollary 5.4) can also be obtained from Bochkov's type theorem together with the Berg–Szwarc reduction and a Tauberian argument. The limit-circle criterion $2|\kappa'| < \alpha - 1$ is due to Świderski and Trojan [37]. The factorisation of the transfer matrix into shears is the classical factorisation of the Nevanlinna matrix and the transfer matrix of Kac's canonical system [26]. What we believe is new is the $O(1)$ term and everything beyond it: the offsets and their dependence on the extension, the Bessel far field for dimerised links and its consequences (including the Krein and Friedrichs offsets $(\mathfrak{d} + \mathfrak{e})/(2p^2) \mp \nu/2$ for all polynomial birth–death rates, Corollary 6.5), the rates with their sharpness, the second-order term, and the exact critical indices of Theorem 11.3. These indices bear on two questions of Berg and Szwarc [7]. Their open problem, whether both of their classes A and B of N-extremal solutions are nonempty, is answered for B by explicit measures (Theorem 11.3); for A it follows from the case $r = 2$ of a theorem of Borichev and Sodin, by a construction that is not explicit (Corollary 11.2). We also locate the determinacy threshold of [7, Ex. 4.1] at $c = -2$. We found no eigenvalue law with an explicit offset for any class of limit-circle Jacobi matrices beyond the explicitly solvable chains.

Method. The proof works with one monotone function, the lifted phase ψ of the regular solution at infinity, whose level sets are the spectra of all extensions (Section 2). Three devices compute it. An exact identity turns each two-site block of the shear product into a rotation by $\varphi_m = 2\arcsin(\lambda/2a_{2m+1})$ in a slowly varying frame, so that an adiabatic normal form sums the frame changes (Section 3); a compactness argument handles a window of $O(N_\lambda^{1/3})$ blocks at the turning point (Section 4); and, for dimerised links, a Bessel comparison handles the far field (Section 6). For rates and the second-order term we prove a discrete analogue of Olver's error bounds for Airy approximations (Theorem 7.1), an exact inequality whose only weights are Olver's envelopes. The same bound answers a question of Fischer, Burgarth and Lonigro: their description of essentially singular limits of Jacobi operators holds for growth exponents $\alpha > 1$, not only $\alpha > 4/3$ (Theorem 10.2). Finally, the Bessel boundary values determine the residues of the Nevanlinna functions of a birth–death problem at their poles. Combined with a weighted determinacy criterion, the case $r = 2$ of a theorem of Borichev and Sodin specialised to N-extremal supports (Proposition 11.1), this gives the critical indices $(1 \mp \nu)/2$ of all N-extremal solutions of Valent's indeterminate problems. Berg and Szwarc introduced these indices and proved $a(\mu_F) \le 1/2 \le a(\mu_\sigma)$ for Stieltjes problems, exhibiting members of the classes at the value $1/2$; no index had been computed (Theorem 11.3).

Organisation. Section 2 fixes notation and reduces the spectral problem to the phase ψ. Sections 3 and 4 prove the block identity, the normal form and the turning-point analysis. Section 5 states the first-order law for balanced links and its consequences, and Section 6 treats dimerised links and birth–death processes. Section 7 proves the discrete Olver bound, which Sections 8 and 9 use for rates and the second-order term. Section 10 applies the results to squeezing operators and to singular limits, Section 11 to the critical indices of indeterminate moment problems, Section 12 compares every headline number with an independent computation, and Section 13 lists open problems. Appendix A contains the technical estimates and auxiliary proofs, so that the main text can follow the ideas, and Appendix B the computational cross-checks and the code and data availability. Section 14 says how the work began and how AI tools were used, and Appendix C how the results were reached.

2 THE SETTING AND AN EXACT COUNT

This section fixes the notation, recalls the limit-circle criterion, and turns the eigenvalue problem for every self-adjoint extension into a statement about one monotone function ψ, the phase of the regular solution at infinity. Everything later in the paper is an asymptotic statement about ψ.

2.1 JACOBI MATRICES WITH GROWING LINKS

We index sites by $k \geq 0$ and write the Jacobi equation as

$$a_k u_{k-1} + b_k u_k + a_{k+1} u_{k+1} = \lambda u_k, \qquad k \geq 0, \qquad a_0 := 0, \tag{2}$$

with links $a_k > 0$ ($k \geq 1$) joining sites $k-1$ and k, and real diagonal b_k. The operator J acts on $\ell^2(\mathbb{N}_0)$; $J_{\min}$ is its closure on finitely supported vectors and $J_{\max} = J_{\min}^*$. We write J_0 for the matrix with the same links and $b \equiv 0$. The *regular solution* $p = p(\lambda)$ has $p_{-1} = 0$, $p_0 = 1$; it spans the solutions of the eigenvalue equation at site 0, so λ is an eigenvalue of a self-adjoint extension exactly when $p(\lambda)$ lies in its domain. For two sequences f, g we put

$$W_k(f,g) := a_{k+1}\big(f_k g_{k+1} - f_{k+1} g_k\big), \tag{3}$$

which is independent of k when f, g solve (2) with the same λ.

The solutions at $\lambda = 0$. The zero-diagonal recurrence at $\lambda = 0$ has the solutions P ($P_0 = 1$, $P_1 = 0$) and R ($R_0 = 0$, $R_1 = 1/a_1$). They live on alternate sites:

$$P_{2m} = (-1)^m \prod_{i=1}^{m} \frac{a_{2i-1}}{a_{2i}}, \qquad R_{2m+1} = \frac{(-1)^m}{a_1} \prod_{i=1}^{m} \frac{a_{2i}}{a_{2i+1}}, \qquad P_{2m+1} = R_{2m} = 0, \tag{4}$$

so that $a_{2m+1} P_{2m} R_{2m+1} = 1$ for every m and $W(P,R) = 1$.

Blocks. The analysis is organised in two-site *blocks*: block m consists of the sites $2m, 2m+1$, and its central link is

$$A_m := a_{2m+1}, \qquad N_m := 2m+1.$$

For a spectral parameter λ we write $\mu_m := \lambda / A_m$, $u_m := 2 - \mu_m$ and

$$\varphi_m := 2 \arcsin\big(\min(1, \mu_m/2)\big) \in [0, \pi]. \tag{5}$$

The *discrete Bohr–Sommerfeld sum* is $S(\lambda) := \sum_{m \geq 0} \varphi_m$. It converges when $\sum_m 1/A_m < \infty$, because $\varphi_m \leq \pi \mu_m / 2$.

The hypotheses. The links are compared with $c\,n^{\alpha}$, $\alpha > 1$, $c > 0$. The main class of the paper allows a $1/n$ term with an alternating part:

Hypothesis 2.1 (dimerised power links). $a_n = c\,n^{\alpha}\big(1 + (\kappa + \kappa'(-1)^n)/n + e_n\big)$ for $n \geq n_0$, with $\alpha > 1$, $c > 0$, $\kappa, \kappa' \in \mathbb{R}$ and $|e_n| \leq C_e\, n^{-1-\delta}$ for some $\delta \in (0, 1]$. Finitely many links are arbitrary positive numbers.

The chain is *balanced* if $\kappa' = 0$ and *dimerised* otherwise. Nothing is assumed about differences of e_n. For the first-order law in the balanced case a weaker hypothesis on first differences suffices:

Hypothesis 2.2 (slowly varying perturbation). $a_n = c\,n^{\alpha}(1 + r_n)$ with $\alpha > 1$, $c > 0$, $r_n \to 0$ and $|r_{n+1} - r_n| \leq C\,n^{-1-\delta}$ for some $\delta \in (0, 1]$.

Under Hypothesis 2.2, $r_n = -\sum_{k \geq n}(r_{k+1} - r_k) = O(n^{-\delta})$. Hypothesis 2.1 with $\kappa' = 0$ is contained in Hypothesis 2.2, and so is $a_n = c(n + s)^{\alpha}$ for every shift s. A diagonal enters only through a size condition:

Hypothesis 2.3 (small diagonal). $|b_n| \leq C_b(1 + n)^{\beta}$ with $\beta < \alpha - 1 - 2|\kappa'|$.

We use $\Lambda := \lambda/(2c)$, $N_\lambda := \Lambda^{1/\alpha}$ (the position where $2c\,N^{\alpha} = \lambda$), and the constants

$$I_\alpha := \int_0^\infty \arcsin\big(\min(1, y^{-\alpha})\big)\,dy = \tfrac{1}{2}\,\mathrm{B}\big(\tfrac{\alpha-1}{2\alpha}, \tfrac{1}{2}\big),$$
$$J_\alpha := \int_1^\infty \frac{y^{-\alpha-2}\,dy}{\sqrt{1 - y^{-2\alpha}}} = \frac{1}{2\alpha}\,\mathrm{B}\big(\tfrac{1}{2} + \tfrac{1}{2\alpha}, \tfrac{1}{2}\big). \tag{6}$$

(The first identity follows by writing $I_\alpha = \pi/2 + \int_1^\infty \arcsin(y^{-\alpha})dy$, integrating by parts and substituting $w = y^{-2\alpha}$; the second by the same substitution.) The *explicit phase* is

$$\Phi(\lambda) := N_\lambda I_\alpha - \frac{\kappa\pi}{2\alpha}. \tag{7}$$

2.2 THE DATA AT $\lambda = 0$ AND THE LIMIT-CIRCLE CRITERION

Let $\varepsilon_m := \tfrac{1}{2}\big(\log a_{2m+3} - 2\log a_{2m+2} + \log a_{2m+1}\big)$, a second difference of $\log a$ centred on the even link a_{2m+2}. It measures how far consecutive blocks are from being similar: by (4),

$$\frac{A_{m+1}P_{2m+2}^2}{A_m P_{2m}^2} = \frac{a_{2m+3}a_{2m+1}}{a_{2m+2}^2} = e^{2\varepsilon_m}. \tag{8}$$

Proposition 2.1 — The data at $\lambda = 0$

Assume that $a_n = c\,n^{\alpha}(1 + \kappa'(-1)^n/n + r_n)$, where r satisfies Hypothesis 2.2. This class contains Hypothesis 2.1.

(a) $\varepsilon_m = -2\kappa'/N_m + \tilde{\varepsilon}_m$ with $|\tilde{\varepsilon}_m| \le C\,N_m^{-1-\delta}$.

(b) The limit $\varpi := \lim_{m\to\infty} A_m P_{2m}^2 N_m^{2\kappa'}$ exists in $(0,\infty)$, and $\rho_m := A_m P_{2m}^2/\varpi$ satisfies $|\log(\rho_m N_m^{2\kappa'})| \le C\,N_m^{-\delta}$.

(c) The normalised solutions $\widetilde{P} := P/\sqrt{\varpi}$ and $\widetilde{R} := R\sqrt{\varpi}$ satisfy

$$\widetilde{P}_{2m}^2 = \rho_m/A_m, \qquad \widetilde{R}_{2m+1}^2 = 1/(\rho_m A_m), \qquad W(\widetilde{P},\widetilde{R}) = 1.$$

(d) J_0 is in the limit-circle case if and only if $2|\kappa'| < \alpha - 1$. If in addition the diagonal satisfies Hypothesis 2.3, then $\sum_k |b_k|(\widetilde{P}_k^2 + \widetilde{R}_k^2) < \infty$ and J is in the limit-circle case as well.

Proof. (a) Write $\log a_n = \log c + \alpha \log n + \ell_n$. The second difference of $\alpha \log n$ is $O(n^{-2})$. Since $r_n = O(n^{-\delta})$, $\ell_n = \log(1 + r_n) + \kappa'(-1)^n/n + O(n^{-1-\delta})$, and the remainder has first differences $O(n^{-1-\delta})$ by Hypothesis 2.2. The alternating term contributes $\tfrac{1}{2}\kappa'\big(-\frac{1}{2m+3} - \frac{2}{2m+2} - \frac{1}{2m+1}\big) = -2\kappa'/N_m + O(N_m^{-2})$.

(b) By (8) and (a), the increments of $\log(A_m P_{2m}^2 N_m^{2\kappa'})$ are $2\varepsilon_m + 2\kappa' \log(N_{m+1}/N_m) = 2\tilde{\varepsilon}_m + O(N_m^{-2})$, which are summable with tail $O(N_m^{-\delta})$.

(c) The first identity is the definition of ρ_m; the second follows from $a_{2m+1} P_{2m} R_{2m+1} = 1$; the Wronskian is unchanged by the rescaling.

(d) By (b) and (c), $P_{2m}^2 \asymp N_m^{-\alpha-2\kappa'}$ and $R_{2m+1}^2 \asymp N_m^{-\alpha+2\kappa'}$. Both are summable if and only if $\alpha \pm 2\kappa' > 1$, that is $2|\kappa'| < \alpha - 1$. By Weyl's alternative at the real point $\lambda = 0$ the matrix is in the limit-circle case exactly when every solution at $\lambda = 0$ is square-summable [36, Thm. 3]. Under Hypothesis 2.3, $|b_k|(\widetilde{P}_k^2 + \widetilde{R}_k^2) \le C k^{\beta-\alpha+2|\kappa'|}$, which is summable; the limit-circle property of J then follows from Theorem 2.2(i) below at $\lambda = 0$, which writes every solution of J at $\lambda = 0$ in the bounded coordinates of J_0. □

Remark 2.1. The criterion in (d) is not new. For periodically modulated links of Kostyuchenko–Mirzoev type it is due to Świderski and Trojan [37, Prop. 6.10 with Thm. 6.3]; with period two and no diagonal their condition reads exactly $2|\kappa'| < \alpha - 1$. At the boundary $2|\kappa'| = \alpha - 1$ the matrix can be limit point, as the example with links $(n+1)^{5/2}(n+2)^{1/2}$ and $(n+1)^{1/2}(n+2)^{5/2}$ in [34, §10, remarks to Prop. 10.4] shows ($\alpha = 3$, $|\kappa'| = 1$). Part (b) for balanced chains, and the equal-amplitude normalisation in (c), are the inputs that make the extension phase t below meaningful.

2.3 THE SHEAR FACTORISATION AND THE PHASE AT INFINITY

From now on J is in the limit-circle case, the diagonal satisfies Hypothesis 2.3, and $\widetilde{P}$ and $\widetilde{R}$ are as in Proposition 2.1. Define the shear weights

$$\gamma_k(\lambda) := (\lambda - b_k)\widetilde{P}_k^2 \quad (k \text{ even}), \qquad \gamma_k(\lambda) := (\lambda - b_k)\widetilde{R}_k^2 \quad (k \text{ odd}), \tag{9}$$

and the shears $L(\gamma) = \left(\begin{smallmatrix} 1 & 0 \\ \gamma & 1 \end{smallmatrix}\right)$, $U(\gamma) = \left(\begin{smallmatrix} 1 & -\gamma \\ 0 & 1 \end{smallmatrix}\right)$. Put $v_{-1} := (1, 0)$ and

$$v_k := L(\gamma_k)v_{k-1} \ (k \text{ even}), \qquad v_k := U(\gamma_k)v_{k-1} \ (k \text{ odd}).$$

A shear S with $\det S = 1$ moves the angle of a unit vector $\vartheta \mapsto \vartheta + \delta_S(\vartheta)$ with $\delta_S(\vartheta) := \operatorname{atan2}(\det[v, Sv], \langle v, Sv\rangle)$, $v = (\cos\vartheta, \sin\vartheta)$. For $L(\gamma)$ and $U(\gamma)$, $\det[v, Sv] = \gamma\cos^2\vartheta$ or $\gamma\sin^2\vartheta$, and $\det[v, Sv] = 0$ forces $Sv = v$; so $\delta_S \in (-\pi, \pi)$, δ_S has the sign of γ, and it depends continuously on (γ, ϑ). We call $\vartheta + \delta_S$ the *canonical lift*. The *partial phase* $\psi_k(\lambda)$ is the sum of the first $k+1$ canonical increments along $v_{-1}, v_0, \dots, v_k$, starting from the angle 0.

Theorem 2.2 — Exact count

Let J be in the limit-circle case with $\sum_k |b_k|(\widetilde{P}_k^2 + \widetilde{R}_k^2) < \infty$.

(i) $v_k = \varpi^{-1/2}\big(W_k(p, \widetilde{R}), -W_k(p, \widetilde{P})\big)$, and for every $k \geq 0$

$$p_k = \sqrt{\varpi}\,(v_{k,1}\widetilde{P}_k + v_{k,2}\widetilde{R}_k), \qquad p_{k+1} = \sqrt{\varpi}\,(v_{k,1}\widetilde{P}_{k+1} + v_{k,2}\widetilde{R}_{k+1}).$$

(ii) $\psi_k \to \psi$ locally uniformly on $\mathbb{R}$; ψ is continuous and is a lift of the angle of $v_\infty := \lim v_k \neq 0$.

(iii) ψ is strictly increasing. If $b \equiv 0$ then $\psi(0) = 0$ and ψ is odd. In general $\psi_{J_0 - \operatorname{diag}(b)}(-\lambda) = -\psi_J(\lambda)$.

(iv) For $t \in [0, \pi)$ let w_t be a real tail solution at $\lambda = 0$, that is, a real sequence that satisfies (2) with $\lambda = 0$ for $k \geq 1$ (it need not satisfy the equation at $k = 0$, so Jw_t may be supported at site 0), whose limiting coordinates in (i) are proportional to $(\cos t, \sin t)$. Let H_t be the restriction of $J_{\max}$ to $\{f : \lim_k W_k(f, w_t) = 0\}$. Then H_t is self-adjoint, every self-adjoint extension of $J_{\min}$ is one H_t, and

$$\lambda \in \operatorname{spec} H_t \iff \psi(\lambda) \in t + \pi\mathbb{Z}.$$

(v) If $\psi(\pm\infty) = \pm\infty$ (proved in Theorem 5.1 for balanced links and follows from Theorem 6.2 for dimerised links) then $\operatorname{spec} H_t = \{\lambda_j(t)\}_{j\in\mathbb{Z}}$ with $\psi(\lambda_j(t)) = t + j\pi$, all eigenvalues are simple, and $\#\big(\operatorname{spec} H_t \cap (\lambda', \lambda]\big) = \#\{j : \psi(\lambda') < t + j\pi \leq \psi(\lambda)\}$.

Proof. **Step 1** (*coordinates*)**.** For a solution g of $J_0 g = 0$ and the regular solution p, the recurrences at site $k \geq 1$ give

$$W_k(p, g) - W_{k-1}(p, g) = p_k\big(a_{k+1}g_{k+1} + a_k g_{k-1}\big) - g_k\big(a_{k+1}p_{k+1} + a_k p_{k-1}\big) = -(\lambda - b_k)p_k g_k.$$

For any sequence f and $k \geq 0$, Cramer's rule with $W(\widetilde{P}, \widetilde{R}) = 1$ gives $f_j = W_k(f, \widetilde{R})\widetilde{P}_j - W_k(f, \widetilde{P})\widetilde{R}_j$ for $j = k, k+1$. With $\alpha_k := W_k(p, \widetilde{R})$ and $\beta_k := -W_k(p, \widetilde{P})$ this reads $p_k = \alpha_{k-1}\widetilde{P}_k + \beta_{k-1}\widetilde{R}_k$. On even sites $\widetilde{R}_k = 0$, so $\alpha_k = \alpha_{k-1}$ and $\beta_k = \beta_{k-1} + \gamma_k\alpha_{k-1}$; on odd sites $\widetilde{P}_k = 0$, so $\beta_k = \beta_{k-1}$ and $\alpha_k = \alpha_{k-1} - \gamma_k\beta_{k-1}$. Finally $(\alpha_0, \beta_0) = (a_1\widetilde{R}_1, a_1 p_1 \widetilde{P}_0) = \sqrt{\varpi}\,(1, \gamma_0)$. This is (i).

Step 2 (*convergence*)**.** For a unit vector and $|\gamma| \leq 1/2$ we have $\langle v, Sv\rangle \geq 1 - |\gamma|/2 > 0$ and $|Sv| \geq 1 - |\gamma|$, so $|\delta_S| \leq \pi/2 \cdot \frac{|\gamma|}{1-|\gamma|} \leq \pi|\gamma|$. Since $\sum_k |\gamma_k| \leq \sum_k (|\lambda| + |b_k|)(\widetilde{P}_k^2 + \widetilde{R}_k^2)$ converges locally uniformly, so do ψ_k and v_k. The limit v_∞ is nonzero: the partial products of the shears $L(\gamma_k), U(\gamma_k)$ converge, since each differs from I by $|\gamma_k|$ in norm and $\sum_k |\gamma_k| < \infty$; every partial product has determinant one, so the limit has determinant one, is invertible, and maps $v_{-1} = (1, 0)$ to a nonzero vector. The same argument at a fixed real spectral parameter shows that the coordinates of every solution converge and that the limiting coordinates of two independent solutions are independent.

Step 3 (*monotonicity*)**.** Let $q := \partial_\lambda p$, so $(J - \lambda)q = p$ and $q_0 = 0$. The general Green identity $W_k(f, h) - W_{k-1}(f, h) = f_k(Jh)_k - h_k(Jf)_k$, in which b cancels, gives $W_k(p, q) = \sum_{j \leq k} p_j^2$. Since $\widetilde{P}, \widetilde{R}$ do not depend on λ, the coordinates of q are $\partial_\lambda(\alpha_k, \beta_k)$, and the bilinear form of Cramer's rule gives $W_k(p, q) = \alpha_k\beta_k' - \beta_k\alpha_k'$. The entries are entire and converge locally uniformly, so $\alpha_\infty\beta_\infty' - \beta_\infty\alpha_\infty' = \sum_j p_j^2 > 0$, which is $\psi'(\lambda) > 0$ up to the positive factor $\alpha_\infty^2 + \beta_\infty^2$. For $b \equiv 0$ and $\lambda = 0$ all γ_k vanish. With $\Sigma = \mathrm{diag}(1, -1)$, $\Sigma L(\gamma)\Sigma = L(-\gamma)$ and $\Sigma U(\gamma)\Sigma = U(-\gamma)$; Σ fixes $(1, 0)$ and reverses angles, so the canonical lifts are negated, which is the reflection identity.

Step 4 (*extensions*)**.** In the limit-circle case every self-adjoint extension is the restriction of $J_{\max}$ by a condition $\lim_k W_k(f, w) = 0$ with w a real tail solution at a fixed real point [36, Thm. 2.6 and §4]; we use the point 0. A tail solution is square-summable and Jw is supported at site 0, so $w \in D(J_{\max})$. For two solutions f, w (at any spectral parameters) the bilinear Cramer identity at the sites $k, k+1$ reads $W_k(f, w) = A_k(f)B_k(w) - B_k(f)A_k(w)$, where $(A_k, B_k) = (W_k(\cdot, \widetilde{R}), -W_k(\cdot, \widetilde{P}))$ are the coordinates of (i). The coordinates converge by the convergence step, and $w \mapsto (A_\infty(w), B_\infty(w))$ is a linear isomorphism of the two-dimensional space of real tail solutions at 0 onto $\mathbb{R}^2$ (injective by the determinant-one argument above), so the real directions are labelled by $t \in [0, \pi)$. Now λ is an eigenvalue of H_t if and only if $p(\lambda) \in D(H_t)$, that is if and only if $v_\infty(\lambda)$ is parallel to $(\cos t, \sin t)$, which is $\psi(\lambda) \in t + \pi\mathbb{Z}$. Part (v) follows from (iii) and (iv). □

Remark 2.2 (Provenance and other parametrisations). The product of the shears is the classical factorisation of the Nevanlinna matrix into factors $I + zT_n$ with alternately upper and lower triangular T_n [2, Ch. I], [6, eq. (51)]. In the language of canonical systems it is the transfer matrix of Kac's Hamburger Hamiltonian [26]: a piecewise constant Hamiltonian of rank one whose angles alternate between 0 and $\pi/2$ (zero diagonal) and whose lengths are $\widetilde{P}_k^2$ and $\widetilde{R}_k^2$ in our normalisation; the problem is determinate if and only if the total length is infinite. Our ψ is the lifted angle of the monodromy at

infinity. For $b \equiv 0$, Simon's boundary condition $\lim W_k(f, \cos\tau_S\, P + \sin\tau_S\, R) = 0$ [36] is our H_t with $\tan t = \tan\tau_S/\varpi$, since $P = \sqrt{\varpi}\,\widetilde{P}$ and $R = \widetilde{R}/\sqrt{\varpi}$; and Yafaev's extension J_ω defined by the asymptotics of solutions [40, Thm. 3.9] is H_t with $\omega = e^{-2it}$, because $\widetilde{P}_n$ and $\widetilde{R}_n$ behave like $a_{n+1}^{-1/2}\cos(n\pi/2)$ and $a_{n+1}^{-1/2}\sin(n\pi/2)$ when $\rho_m \to 1$. The point of the normalisation by ϖ is that it makes the phase t appear additively in the eigenvalue law.

Notation. Table 1 collects the symbols that recur across sections. There is one phase, ψ, and every other angle in the paper is a way of computing it: ψ_k along the chain, χ_m in the moving frame of Section 3, and Ω inside the Airy window of Section 4. A few letters are used in more than one place with unrelated meanings that the table separates: J is the Jacobi matrix, J_α a constant and $J_{\pm\nu}$ a Bessel function, while the rotation generator is J; D_m and $D(N)$ are frame matrices, while D alone is an entry of the Nevanlinna matrix. The Airy comparison sequences ψ_A, ψ_B of Section 7 are not phases, and γ denotes the shear weights γ_k here but a local exponent in Sections 6, 8 and 11. For a statement P, $[P]$ is 1 if P holds and 0 otherwise.

3 TWO-SITE BLOCKS AND AN ADIABATIC NORMAL FORM

The shear product of Theorem 2.2 is a product of 2×2 matrices close to the identity, but the turning angle per block is not small: in the region where $\mu_m < 2$ each block turns vectors by an angle of order one. This section shows that each block is exactly a rotation by φ_m in a frame that changes slowly from block to block (Lemma 3.1), computes the frame change to first order (Lemma 3.2), and sums the resulting errors by a normal-form transformation (Proposition 3.3). The proposition is stated for an arbitrary splitting of the frame change into a part that is kept and a part that is estimated; the balanced, dimerised and quantitative applications differ only in that choice.

Throughout, $e_\pm := (1, \pm 1)/\sqrt{2}$, $\mathrm{J} := \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$, $\Sigma_1 := \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$, $\Sigma_3 := \mathrm{diag}(1, -1)$, and $R(\varphi)$ is the rotation by φ. A positive definite symmetric matrix moves every angle by less than $\pi/2$; its lift with this property is unique and is called its principal lift. A matrix g with $\|g - I\| \le \epsilon < 1$ also has a unique lift moving angles by at most $\arcsin\epsilon$, again called principal. Two continuous families of lifts of the same continuous family of maps differ by a constant multiple of 2π; we use this homotopy argument repeatedly to identify lifts.

SYMBOL	MEANING	WHERE
N_λ, Λ	turning scale, $2cN_\lambda^\alpha = \lambda$; $\Lambda = \lambda/(2c) = N_\lambda^\alpha$	§2
A_m, N_m	central link a_{2m+1} and centre $2m+1$ of block m	§2
μ_m, u_m	λ/A_m and $2-\mu_m$; $\mu_m = 2$ at the turning point	§2
φ_m, S	block rotation $2\arcsin(\min(1,\mu_m/2))$ and $S = \sum_m \varphi_m$	(5)
Φ, I_α, J_α	explicit phase $N_\lambda I_\alpha - \kappa\pi/(2\alpha)$ and its constants	(7)
P, R; $\widetilde{P}$, $\widetilde{R}$, ϖ	solutions at $\lambda = 0$; normalised to equal amplitude by ϖ	Proposition 2.1
ρ_m, ε_m	frame normalisation and its logarithmic increment	Proposition 2.1
ψ, ψ_k	phase at infinity and partial phase after site k	Theorem 2.2
t, H_t; α_∞, β_∞	extension angle and extension; coordinates of p at infinity	Theorem 2.2
J, Σ_1, Σ_3, $e_\pm$	rotation generator, reflections, diagonal basis	§3
s_m, σ_m, E_m, D_m	frame parameter, its decrement, block frame, its diagonal part	Lemma 3.1
χ_m	frame angle of block m; $\chi_m \to \psi$	(10)
ℓ, x_m	Airy scale $\asymp N_\lambda^{1/3}$ and Airy variable of block m	(15)
Ω	phase of the Airy function $\mathrm{Ai}(-\zeta)$	Lemma 4.3
$\mathcal{N}_t^\pm$	eigenvalue counting functions on the half-lines	Corollary 5.3
ν, τ, N_{dec}	Bessel order, Bessel variable, site where $\tau = 1$	§6
$D(N)$, Y, Y_0, Q	model frame, Bessel solution matrix, its large-τ limit, amplitude ratio	§6.2
θ_0, θ_t	dimerised offsets, $\theta_0 = \kappa'\pi/(2(\alpha-1))$	Corollary 6.3
$\mathfrak{d}$, $\mathfrak{e}$	sums of the birth–death parameters d_j, e_j	§6.5
E, M, k	Olver's envelopes and the derivative bound	§7
h, x_n, g	lattice step, Langer variable, Langer amplitude	§7
$\mathscr{P}$, $\mathscr{Q}$; $\mathfrak{p}_n$, $\mathfrak{q}_n$	frozen residuals; residual coefficients	Lemma 7.3
Γ	error scale of the rate theorem	(29)
A, B, C, D; G_σ	Nevanlinna matrix; $B + \sigma D$	§11
$\hat{p}_k$, $\mathscr{E}_\sigma$, W_σ	orthonormal polynomials in x, annihilator, residue weight	(33)
$\mathscr{I}_\sigma$, $\mathscr{S}_\sigma$, $a(\mu)$	weighted series, index series, critical index	Proposition 11.1
K_0; C_{res}	$K(1/\sqrt{2})$; residue constant of Theorem 11.3	Example 5.1; (34)

Table 1. *Recurring notation. Throughout, ϑ denotes the angle of a unit vector* $(\cos\vartheta, \sin\vartheta)$ *(in Section 9,* $\arccos Z$*) and $\hat{g}$ a lift of a matrix g to the angles.*

3.1 THE BLOCK ROTATION

Let $b \equiv 0$. The shear weights of block m are $\gamma_{2m} = \lambda \widetilde{P}_{2m}^2 = \mu_m \rho_m$ and $\gamma_{2m+1} = \lambda \widetilde{R}_{2m+1}^2 = \mu_m / \rho_m$, by Proposition 2.1(c).

Lemma 3.1 — Block rotation

Let $T_m := U(\gamma_{2m+1}) L(\gamma_{2m})$ and $D_m := \operatorname{diag}(\rho_m^{-1/2}, \rho_m^{1/2})$. Then $T_m = D_m T_1(\mu_m) D_m^{-1}$ with $T_1(\mu) = \left(\begin{smallmatrix} 1-\mu^2 & -\mu \\ \mu & 1 \end{smallmatrix}\right)$. For $0 < \mu < 2$ put $s = s(\mu) := \tfrac{1}{4} \log \frac{2+\mu}{2-\mu}$ and let $\widetilde{E}(\mu)$ be the symmetric matrix with $\widetilde{E} e_+ = e^{-s} e_+$ and $\widetilde{E} e_- = e^{s} e_-$. Then

$$\widetilde{E}(\mu)^{-1} T_1(\mu) \widetilde{E}(\mu) = R(\varphi), \qquad \varphi = 2 \arcsin(\mu/2),$$

and with $E_m := D_m \widetilde{E}(\mu_m)$ and the lifts described above, $\widehat{E}_m^{-1} \circ \widehat{T}_m \circ \widehat{E}_m$ is exactly $\vartheta \mapsto \vartheta + \varphi_m$. Moreover $\tanh 2s = \mu/2$, so $\sinh 2s = \tan(\varphi/2)$, $\cosh 2s = 1/\cos(\varphi/2)$ and $e^{2s} = \sqrt{(4-u)/u}$ with $u = 2 - \mu$.

Proof. $D_m^{-1} U(x) D_m = U(x \rho_m)$ and $D_m^{-1} L(y) D_m = L(y/\rho_m)$, which gives the first identity. $T_1(\mu)$ has determinant 1 and trace $2 - \mu^2 = 2\cos\varphi$. In the orthonormal basis (e_+, e_-) one computes $T_1 e_+ = (1 - \frac{\mu^2}{2}) e_+ - (\mu + \frac{\mu^2}{2}) e_-$ and $T_1 e_- = (\mu - \frac{\mu^2}{2}) e_+ + (1 - \frac{\mu^2}{2}) e_-$; conjugating by $\operatorname{diag}(e^{-s}, e^{s})$ in this basis turns the off-diagonal entries into $e^{2s}\mu(1 - \mu/2)$ and $-e^{-2s}\mu(1 + \mu/2)$, which are $+\sin\varphi$ and $-\sin\varphi$ exactly when $e^{4s} = (2+\mu)/(2-\mu)$, because $\sin\varphi = \mu/2\sqrt{4-\mu^2}$. In a positively oriented orthonormal basis the matrix $\left(\begin{smallmatrix} \cos\varphi & \sin\varphi \\ -\sin\varphi & \cos\varphi \end{smallmatrix}\right)$ would be the rotation by $-\varphi$; but $\det[e_+, e_-] = -\tfrac{1}{2} - \tfrac{1}{2} = -1$, so (e_+, e_-) is negatively oriented and the same matrix is the rotation by $+\varphi$ in the standard orientation. Both sides of the lift identity are lifts of $R(\varphi_m)$ depending continuously on $(\mu, \rho) \in [0, 2) \times (0, \infty)$; at $\mu = 0$ all matrices are the identity and both lifts are the identity, so they agree everywhere. The hyperbolic identities follow from $e^{4s} = (2+\mu)/(2-\mu)$ and $\mu = 2\sin(\varphi/2)$. □

The *frame angle* of block m is $\chi_m := \widehat{E}_m^{-1}(\psi_{2m-1})$, with $\psi_{-1} := 0$. By the lemma, $\psi_{2m+1} = \widehat{T}_m(\psi_{2m-1})$ becomes

$$\chi_{m+1} = \widehat{F}_m(\chi_m + \varphi_m), \qquad F_m := E_{m+1}^{-1} E_m, \qquad \widehat{F}_m := \widehat{E}_{m+1}^{-1} \circ \widehat{E}_m. \tag{10}$$

If $\mu_M \to 0$ and $\rho_M \to 1$ then $E_M \to I$, so $\chi_M - \psi_{2M-1} \to 0$ and $\chi_M \to \psi$.

Write $s_m := s(\mu_m)$, $\sigma_m := s_m - s_{m+1}$, and recall $e^{2\varepsilon_m} = \rho_{m+1}/\rho_m$ from (8). Then

$$F_m = G_m H_m, \qquad H_m := \widetilde{E}(\mu_m)^{-1} e^{\varepsilon_m \Sigma_3} \widetilde{E}(\mu_m),$$
$$G_m := \widetilde{E}(\mu_{m+1})^{-1} \widetilde{E}(\mu_m) = \cosh\sigma_m\, I - \sinh\sigma_m\, \Sigma_1.$$

The first factor is diagonal in (e_+, e_-) and $e_+ e_+^{\mathsf T} - e_- e_-^{\mathsf T} = \Sigma_1$. For the second, Σ_3 exchanges e_+ and e_-, so in the basis (e_+, e_-) conjugation by $\widetilde{E}$ multiplies its off-diagonal entries by $e^{\pm 2s}$; written in the standard basis,

$$H_m = \exp(\varepsilon_m X_m), \qquad X_m = \cosh(2s_m)\,\Sigma_3 + \sinh(2s_m)\,\mathrm{J}. \tag{11}$$

A one-parameter group $e^{\tau Y}$ moves the angle ϑ of a unit vector $v = (\cos\vartheta, \sin\vartheta)$ with velocity $\det[v, Yv]$, which equals $\cos 2\vartheta$ for $Y = \Sigma_1$, $-\sin 2\vartheta$ for $Y = \Sigma_3$ and 1 for $Y = \mathrm{J}$.

Lemma 3.2 (One block of frame change). If $|\sigma_m| \le 1/2$ and $|\varepsilon_m| e^{2s_m} \le 1/3$, then for every ϑ

$$\widehat{F}_m(\vartheta) = \vartheta + \varepsilon_m \tan\tfrac{\varphi_m}{2} + \mathrm{Re}\big(\Gamma_m e^{2i\vartheta}\big) + \mathscr{R}_m(\vartheta), \qquad \Gamma_m := -\sigma_m + \frac{i\varepsilon_m}{\cos(\varphi_m/2)},$$

with $|\mathscr{R}_m| \le C(\sigma_m^2 + \varepsilon_m^2 e^{4s_m})$ for an absolute constant C.

Proof. By (11) and Lemma 3.1, H_m moves angles with velocity $\varepsilon_m(\tan\frac{\varphi_m}{2} - \frac{\sin 2\vartheta}{\cos(\varphi_m/2)})$ along a flow of length one, and the velocity field and its ϑ-derivative are bounded by $3|\varepsilon_m| e^{2s_m}$; so $\widehat{H}_m(\vartheta)$ equals ϑ plus that velocity plus $O(\varepsilon_m^2 e^{4s_m})$. Likewise $G_m = e^{-\sigma_m \Sigma_1}$ gives $\widehat{G}_m(\vartheta) = \vartheta - \sigma_m \cos 2\vartheta + O(\sigma_m^2)$. The composition adds a cross term bounded by $C|\sigma_m||\varepsilon_m| e^{2s_m}$, which is absorbed. The lift $\widehat{F}_m$ is the time-one map of these flows because both depend continuously on (σ, ε) and agree at $\sigma = \varepsilon = 0$. Finally $-\sigma\cos 2\vartheta - \frac{\varepsilon}{\cos(\varphi/2)} \sin 2\vartheta = \mathrm{Re}(\Gamma e^{2i\vartheta})$. □

The first term, $\varepsilon_m \tan(\varphi_m/2)$, does not oscillate in ϑ: it is a rotation of the frame, a *drift*. Under Hypothesis 2.2 it is summable and harmless; for dimerised links $\varepsilon_m \approx -2\kappa'/N_m$ and the drift adds up to a finite, nonzero phase (Proposition 4.5).

A diagonal. If $b \not\equiv 0$, block m is $T_m^b = U((\lambda - b_{2m+1})\widetilde{R}_{2m+1}^2) L((\lambda - b_{2m})\widetilde{P}_{2m}^2)$, and the computation in the proof of Lemma 3.1 gives $D_m^{-1} T_m^b D_m = U(\mu_m'') L(\mu_m')$ with $\mu_m' = (\lambda - b_{2m})/A_m$ and $\mu_m'' = (\lambda - b_{2m+1})/A_m$; the factors ρ_m cancel exactly. Hence in the frame E_m the block is $R(\varphi_m)(I + \Delta_m)$ with

$$\|\Delta_m\| \le \omega_m := C\, e^{2s_m} \frac{|b_{2m}| + |b_{2m+1}|}{A_m}, \tag{12}$$

and if $\omega_m \le 1/3$, deforming b to 0 shows that its lift is $\vartheta \mapsto \vartheta + \varphi_m + \delta_m(\vartheta)$ with $|\delta_m| \le 2\omega_m$. Thus $\chi_{m+1} = \widehat{F}_m(\chi_m + \varphi_m + \delta_m(\chi_m))$.

Proposition 3.3 — Adiabatic normal form

Let $0 \le m_1 < M \le \infty$ and suppose that for $m_1 \le m < M$ we have $\mu_m < 2$, $|\sigma_m| \le 1/2$, $|\varepsilon_m| e^{2s_m} \le 1/3$ and $\omega_m \le 1/3$. Split $\varepsilon_m = \bar\varepsilon_m + \tilde\varepsilon_m$ and $\sigma_m = \bar\sigma_m + \tilde\sigma_m$ in any way, and put

$$\bar\Gamma_m := -\bar\sigma_m + \frac{i\bar\varepsilon_m}{\cos(\varphi_m/2)}, \qquad \xi_m := -\frac{\bar\Gamma_m\, e^{i\varphi_m}}{2i\sin\varphi_m}, \qquad y_m := \chi_m + \operatorname{Re}\bigl(\xi_m e^{2i\chi_m}\bigr),$$

so that $|y_m - \chi_m| \le |\xi_m|$. Then, for $m_1 \le M' \le M$ and an absolute constant C,

$$\Bigl| y_{M'} - y_{m_1} - \sum_{m_1 \le m < M'} \Bigl(\varphi_m + \bar\varepsilon_m \tan\tfrac{\varphi_m}{2}\Bigr)\Bigr| \le C \sum_{m_1 \le m < M'} \mathscr{E}_m,$$

$$\mathscr{E}_m := |\xi_{m+1} - \xi_m| + |\tilde\sigma_m| + |\tilde\varepsilon_m| e^{2s_m} + \sigma_m^2 + \varepsilon_m^2 e^{4s_m} + \omega_m + |\xi_{m+1}|\bigl(|\sigma_m| + |\varepsilon_m| e^{2s_m} + \omega_m\bigr).$$

Proof. Write $w_m(\vartheta) := \operatorname{Re}(\xi_m e^{2i\vartheta})$ and $\chi_m^+ := \chi_m + \varphi_m + \delta_m(\chi_m)$. By (10) and Lemma 3.2, with the split inserted,

$$\chi_{m+1} - \chi_m = \varphi_m + \bar\varepsilon_m \tan\tfrac{\varphi_m}{2} + \operatorname{Re}\bigl(\bar\Gamma_m e^{2i(\chi_m+\varphi_m)}\bigr) + r_m,$$

where r_m collects δ_m, the change of $\operatorname{Re}(\Gamma_m e^{2i\vartheta})$ between $\vartheta = \chi_m^+$ and $\vartheta = \chi_m + \varphi_m$, the parts of the drift and of $\operatorname{Re}(\Gamma_m e^{2i(\chi_m+\varphi_m)})$ that come from $\tilde\sigma_m$ and $\tilde\varepsilon_m$, and $\mathscr{R}_m(\chi_m^+)$. The coefficient is kept unsplit until the diagonal perturbation has been handled, because only Γ_m, not its retained part, is controlled by the hypotheses: since $\tan(\varphi/2) = \sinh 2s$ and $1/\cos(\varphi/2) = \cosh 2s$ are at most e^{2s}, $|\Gamma_m| \le |\sigma_m| + |\varepsilon_m| e^{2s_m} \le 1$, so the change is at most $2|\delta_m|$; and $|\Gamma_m - \bar\Gamma_m| \le |\tilde\sigma_m| + |\tilde\varepsilon_m| e^{2s_m}$. Hence $|r_m| \le C(|\tilde\sigma_m| + |\tilde\varepsilon_m| e^{2s_m} + \omega_m + \sigma_m^2 + \varepsilon_m^2 e^{4s_m})$, and no bound on $\bar\Gamma_m$ is needed. The choice of ξ_m is the one that solves $\xi_m(e^{2i\varphi_m} - 1) = -\bar\Gamma_m e^{2i\varphi_m}$, so

$$\operatorname{Re}\bigl(\bar\Gamma_m e^{2i(\chi_m+\varphi_m)}\bigr) + w_{m+1}(\chi_m + \varphi_m) - w_m(\chi_m) = \operatorname{Re}\bigl(e^{2i\chi_m}(\xi_{m+1} - \xi_m) e^{2i\varphi_m}\bigr).$$

Adding $w_{m+1}(\chi_{m+1}) - w_{m+1}(\chi_m + \varphi_m)$, which is at most $2|\xi_{m+1}|\,|\chi_{m+1} - \chi_m - \varphi_m| \le C|\xi_{m+1}|(|\sigma_m| + |\varepsilon_m| e^{2s_m} + \omega_m)$, gives $|y_{m+1} - y_m - \varphi_m - \bar\varepsilon_m \tan(\varphi_m/2)| \le C\mathscr{E}_m$. Summing proves the claim. No inversion of $\chi \mapsto y$ is needed. □

The size of ξ_m is governed by $\sin\varphi_m = \mu_m \cos(\varphi_m/2)$ and $\cos^2(\varphi_m/2) = u_m(4 - u_m)/4$:

$$|\xi_m| \le \frac{|\bar\sigma_m|}{2\sin\varphi_m} + \frac{2|\bar\varepsilon_m|}{\mu_m u_m (4 - u_m)}, \qquad 2\sin\varphi_m \ge \mu_m\sqrt{2u_m} \quad (u_m \le 2). \tag{13}$$

The smooth part of σ_m is small in the turning region because s varies slowly in μ there:

$$s'(\mu) = \frac{1}{u(4-u)} \le \frac{1}{2u}, \qquad u = 2 - \mu \le 2. \tag{14}$$

4 THE TURNING POINT AND THE MISSING QUARTER-PERIOD

This section follows the frame angle χ_m from site 0 to a block M well inside the oscillatory region and shows that it equals the discrete Bohr–Sommerfeld sum up to M, plus a drift, with no quarter-period (Proposition 4.5). We first explain the result as a budget of phase, region by region, and then prove the three entries of the budget. The argument is the same for balanced and dimerised links, so we treat them together under the following hypothesis.

Hypothesis 4.1. $a_n = c\,n^{\alpha}\big(1+\kappa'(-1)^n/n+r_n\big)$ with r as in Hypothesis 2.2 and $2|\kappa'| < \alpha-1$; the diagonal satisfies $|b_n| \le C_b(1+n)^{\beta}$ with $\beta < \alpha - 1 - 2|\kappa'|$.

The odd links $A_m = a_{2m+1}$ see the alternating term only through a constant: $A_m = c\,N_m^{\alpha}(1+\hat r_{2m+1})$ with $\hat r_n := r_n - \kappa'/n$, and $\hat r$ again satisfies Hypothesis 2.2. We write $\bar\ell_m := \alpha\log(N_{m+1}/N_m)$ for the increment of $\log(c\,N^{\alpha})$ and $\Delta\hat e_m := \log(1+\hat r_{2m+3}) - \log(1+\hat r_{2m+1})$, so that $\log A_{m+1} - \log A_m = \bar\ell_m + \Delta\hat e_m$ with $|\Delta\hat e_m| \le C N_m^{-1-\delta}$.

4.1 THE PHASE BUDGET

Measure the distance from the turning point on the Airy scale

$$\ell := \big(N_\lambda/(16\alpha)\big)^{1/3}, \qquad x_m := 4\ell^2(\mu_m - 2) = -4\ell^2 u_m. \tag{15}$$

Then $x_m > 0$ on the *hyperbolic* side, where $2A_m < \lambda$, and $x_m < 0$ on the *oscillatory* side. Near the turning point x_m decreases by $\ell^{-1}(1+o(1))$ per block (Lemma A.3), so it is the variable in which the turning point has a fixed shape, and the window $|x_m| < K$ is about $2K\ell$ blocks, that is $O(N_\lambda^{1/3})$ sites, wide. Fix $K \ge 1$, let m_L be the largest block with $x_m \ge K$ and m_R the least with $x_m \le -K$. The blocks then fall into the regions sketched in Figure 1, and each contributes a definite amount to χ.

Below the turning point each block contributes π. For $m \le m_L$ the regular solution has no sign change: it is positive and increasing (Lemma 4.1). The reference solutions $\widetilde P_{2m}$ and $\widetilde R_{2m+1}$, on the other hand, change sign from each block to the next, by (4). In the coordinates of Theorem 2.2, which expand p in $\widetilde P$ and $\widetilde R$, the phase therefore advances by π per block in the following sense: after block m the lifted angle ψ_{2m-1} lies in the open quarter $(m\pi - \pi/2, m\pi)$. The increment of a single block is not exactly π; only the quarter is fixed, and after the hyperbolic region ψ_{2m_L-1} lies in $(m_L\pi - \pi/2, m_L\pi)$. Since $\varphi_m = \pi$ there too, the hyperbolic region contributes exactly its terms $m_L\pi$ of the sum S, up to a position inside one quarter that the window decides.

The Airy window contributes $\Omega(K) - \pi/4$. Across the window the block is a small perturbation of $T_1(2) = -I + \mathcal{N}$, where $\mathcal{N}$ is nilpotent with kernel $e_- = (1,-1)/\sqrt{2}$, and each block acts on the pair (amplitude, slope) as an Euler step of length ℓ^{-1} for the Airy equation $y'' = xy$ (Lemma A.3). The amplitude y is the component along e_-. The regular

solution enters the window as the recessive solution Ai, because it was increasing, and leaves it at $x = -K$ with the Airy phase $\Omega(K) = \arg\big(\mathrm{Ai}(-K) - i\,\mathrm{Ai}'(-K)/\sqrt{K}\big)$, measured from the direction e_-. That direction has angle $-\pi/4$, so (Lemma 4.2)

$$\chi_{m_R} - m_R\pi \;\longrightarrow\; \Omega(K) - \frac{\pi}{4}.$$

In the same window the sum S falls short of π per block by $T_K := \sum_{m_L<m<m_R}(\pi - \varphi_m) \to {}^2\!/_3 K^{3/2}$ (Lemma 4.4), which is the classical action across the window. Hence

$$\chi_{m_R} - \sum_{m<m_R} \varphi_m \;\longrightarrow\; \Omega(K) - \frac{\pi}{4} + \frac{2}{3}K^{3/2} = O(K^{-3/2}), \tag{16}$$

because $\Omega(K) = \pi/4 - {}^2\!/_3 K^{3/2} + O(K^{-3/2})$ (Lemma 4.3). The action ${}^2\!/_3 K^{3/2}$ is already in S, and what is left over is $\pi/4 - \pi/4$.

The two quarter-periods. The $+\pi/4$ in Ω is the Maslov phase of the turning point, the $-\pi/4$ in $\mathrm{Ai}(-\zeta) \sim \pi^{-1/2}\zeta^{-1/4}\cos({}^2\!/_3\zeta^{3/2} - \pi/4)$. The $-\pi/4$ is set by the limit-circle end. At the turning point $\lambda = 2a$ the local solution of $au_{k-1} + au_{k+1} = \lambda u_k$ is the band-edge wave $u_k = \text{const}$, which takes equal values on the two sites of a block. The coordinates of Theorem 2.2 measure a solution against $\widetilde{P}$ and $\widetilde{R}$, which live on alternate sites. The block frame $E_m = D_m\widetilde{E}(\mu_m)$ first rescales them by D_m to equal amplitude on block m, and in that scaling a vector with equal values on the two sites has angle $-\pi/4$ modulo π: it is the direction e_-. For balanced links the frames tend to the identity as $m \to \infty$, and $D_m \to I$ precisely because $\widetilde{P}$ and $\widetilde{R}$ were normalised by ϖ to have equal amplitude at infinity (Proposition 2.1(c)). So the $-\pi/4$ measured at the turning point is the $-\pi/4$ read at the limit-circle end, in the coordinates in which the boundary condition of H_t is imposed. It is the band-edge direction in the staggered equal-amplitude coordinates, not an independent Maslov index of the end: the absence of an extra constant is relative to this normalisation, which is fixed once by the data at $\lambda = 0$. A change of the boundary coordinates changes the angle of a given extension and its offset together, without changing any spectrum. (For dimerised links D_m keeps drifting like $\mathrm{diag}(N_m^{\kappa'}, N_m^{-\kappa'})$, and Section 6 follows that drift to the end.) The cancellation takes place in ψ, before an extension is chosen, which is why every extension obeys a law $j\pi + t$ and none obeys $(j + {}^1\!/_4)\pi + t$.

Beyond the window each block adds φ_m and a drift. On the oscillatory side the adiabatic normal form (Proposition 3.3) applies. Each block adds its rotation φ_m and the frame drift $\bar{\varepsilon}_m \tan(\varphi_m/2)$ with $\bar{\varepsilon}_m = -2\kappa'/N_m$, and the drifts add up to $-\kappa'\pi/(2\alpha)$. This is 0 for balanced links.

Adding the entries gives the near-field law

$$\begin{aligned}\chi_M &= \underbrace{m_L\pi}_{\text{hyperbolic}} + \underbrace{\sum_{m_L\le m<m_R} \varphi_m + \Big(\Omega(K) - \frac{\pi}{4} + T_K\Big)}_{\text{Airy window}} + \underbrace{\sum_{m_R\le m<M} \varphi_m - \frac{\kappa'\pi}{2\alpha}}_{\text{oscillatory}} + o(1) \\ &= \sum_{m<M} \varphi_m - \frac{\kappa'\pi}{2\alpha} + o(1),\end{aligned} \tag{17}$$

where $o(1)$ is taken as $\lambda \to \infty$ and then $K \to \infty$. For balanced links $M = \infty$ and $\chi_\infty = \psi$, so $\psi = S + o(1)$. For dimerised links the frames do not converge, and the far field beyond M is the subject of Section 6.

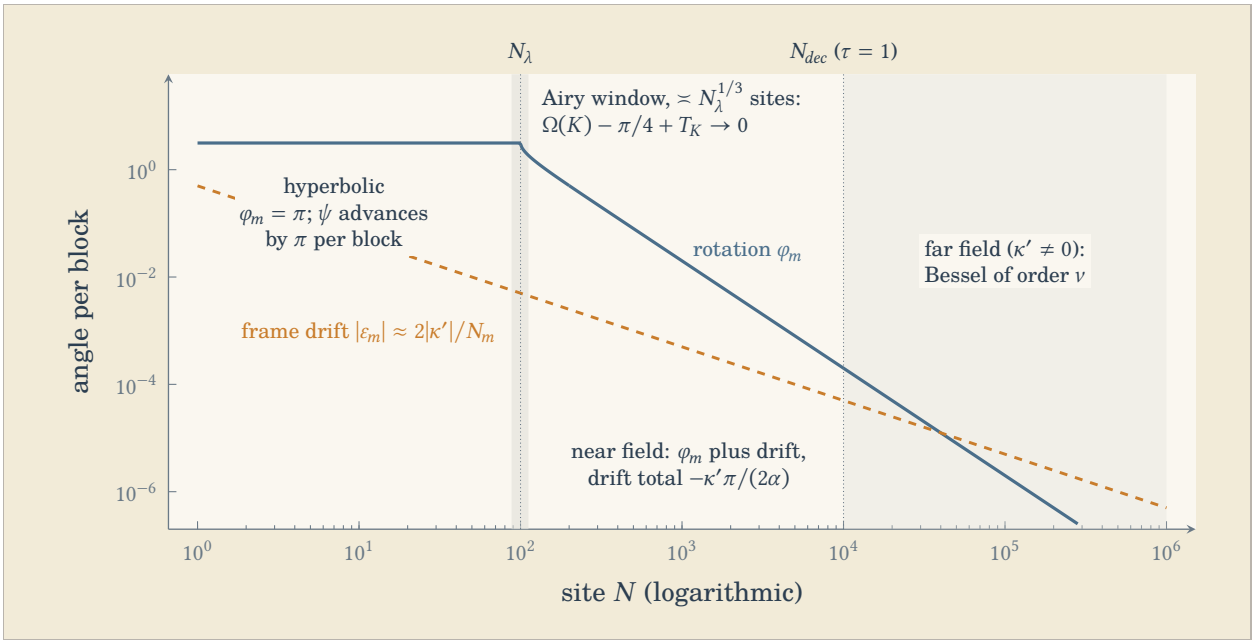


FIG. 1. *The phase budget along the chain, drawn for $\alpha = 2$, $N_\lambda = 100$ and $\kappa' = 1/4$. Below the turning point N_λ each block contributes π to the sum S, and the phase ψ advances by π per block up to a position inside one quarter. In the Airy window the Maslov phase $\pi/4$ of the turning point cancels the $-\pi/4$ fixed by the limit-circle end in the equal-amplitude coordinates set at $\lambda = 0$, so no quarter-period survives in those coordinates. Beyond it each block adds its rotation and a drift. Where the rotation falls below the frame drift, near N_{dec}, a dimerised chain becomes a Bessel problem (Section 6).*

The rest of the section proves the three entries. The block geometry and the error sums that the proofs need are collected in Appendix A.

4.2 THE HYPERBOLIC SIDE

Lemma 4.1 (Monotone region). If $\lambda - b_k \geq a_k + a_{k+1}$ for $0 \leq k \leq k_1$, then $1 = p_0 \leq p_1 \leq \ldots \leq p_{k_1+1}$ and $\psi_k \in (k\pi/2, (k+1)\pi/2)$ for $0 \leq k \leq k_1$.

Proof. Inductively,

$$p_{k+1} = \frac{(\lambda - b_k)p_k - a_k p_{k-1}}{a_{k+1}} \geq \frac{(\lambda - b_k - a_k)\, p_k}{a_{k+1}} \geq p_k.$$

By Theorem 2.2(i), v_k is a positive multiple of

$$\left(\frac{p_{2m}}{\widetilde{P}_{2m}}, \frac{p_{2m+1}}{\widetilde{R}_{2m+1}}\right) \quad (k = 2m), \qquad \left(\frac{p_{2m+2}}{\widetilde{P}_{2m+2}}, \frac{p_{2m+1}}{\widetilde{R}_{2m+1}}\right) \quad (k = 2m+1),$$

whose entries have the signs $(-1)^m, (-1)^m$ and $(-1)^{m+1}, (-1)^m$ respectively, by (4). In both cases the angle of v_k lies in $(k\pi/2, (k+1)\pi/2)$ modulo 2π. All $\gamma_k > 0$, so each

canonical increment lies in $[0,\pi)$; starting from $\psi_0 \in (0,\pi/2)$, the lift ψ_k lies in an interval of length $3\pi/2$ that meets the class $(k\pi/2,(k+1)\pi/2)+2\pi\mathbb{Z}$ only once. □

For $\lambda \geq \lambda_0(K)$ the hypothesis holds with $k_1 = 2m_L$, and therefore $\psi_{2m_L-1} \in (m_L\pi - \pi/2, m_L\pi)$ (Lemma A.2).

4.3 THE AIRY WINDOW

Lemma 4.2 — Airy window

Assume Hypothesis 4.1 and fix $K \geq 1$. Then

$$\limsup_{\lambda\to\infty}\Big|\chi_{m_R} - m_R\pi - \big(\Omega(K) - \tfrac{\pi}{4}\big)\Big| \leq C_0\,\eta_A(K), \qquad \eta_A(K) := (1+\sqrt{K})\,\frac{\mathrm{Ai}(K)}{\mathrm{Bi}(K)},$$

where Ω is the Airy phase of Lemma 4.3 and C_0 is absolute.

The proof works in coordinates adapted to the turning point, in which the blocks become Euler steps of the Airy equation. Write $z_m := D_m^{-1}v_{2m-1}$ and define (X_m,Y_m) by

$$z_m = (-1)^m\big(X_m\,e_- + \tfrac{Y_m}{4\ell}\,e_+\big), \qquad \text{that is} \qquad (X_m,Y_m)^{\mathsf{T}} = (-1)^m C_\ell^{-1} z_m, \quad C_\ell := \big[e_-,\ e_+/(4\ell)\big]. \tag{18}$$

The pair (X_m,Y_m) will track $(y(x_m),-y'(x_m))$ for a solution y of the Airy equation $y''=xy$. A direct computation of $D_m^{-1}v_{2m-1}$ from Theorem 2.2(i) and (4) gives the exact form

$$z_m \;\propto\; (-1)^m A_m^{1/2}\big(p_{2m},\,-c'_m\,p_{2m-1}\big), \qquad c'_m := a_{2m}/a_{2m+1}, \tag{19}$$

with a positive constant of proportionality; no normalisation by ϖ or ρ_m remains. So X_m is, up to a factor, the average of p over the block and Y_m/ℓ its difference. The *Airy angle* of block m is $\varrho_m := \widehat{C}_\ell^{-1}\big(\widehat{D}_m^{-1}(\psi_{2m-1}) - m\pi\big)$, where $\widehat{C}_\ell$ is the lift of C_ℓ with $\widehat{C}_\ell(0) = -\pi/4$, the angle of e_-. Lemma A.3 shows that the blocks act on (X,Y) as Euler steps $I + h_m\mathrm{B}(x_m)$ of the Airy system, with $h_m = x_m - x_{m+1} = \ell^{-1}(1+o(1))$ and $\mathrm{B}(x) = \left(\begin{smallmatrix}0 & 1\\ x & 0\end{smallmatrix}\right)$, that ϱ_{m+1} is the principal lift of one step applied to ϱ_m, and that the window starts with $X_{m_L} > 0$ and $Y_{m_L}/X_{m_L} \geq -C\ell/N_\lambda$.

Proof of Lemma 4.2. The argument has three parts: the discrete trajectory follows an Airy trajectory, that trajectory is almost exactly Ai, and its phase is read in the frame at m_R.

Step 1 (*the Airy flow*)**.** Let $\mathcal{U}(x,x')$ be the propagator of $(X,Y)' = -(Y,xX)$ in the decreasing variable x, which is the system satisfied by $(y,-y')$ for $y''=xy$. Then $\|\mathcal{U}(x_{m+1},x_m) - I - h_m\mathrm{B}(x_m)\| \leq C(K)h_m^2$. With Lemma A.3(a), telescoping gives $\|\prod_{m'<m} M_{m'} - \mathcal{U}(x_m,x_{m_L})\| \leq e^{C(K)}\big(C(K)\ell^{-1}+o(1)\big) \to 0$ uniformly in m and in the initial direction. The discrete vectors therefore stay uniformly close to an Airy trajectory and

away from 0, and the principal steps ϱ_m stay within $o(1)$ of the continuous lifted angle of that trajectory.

Step 2 (*the trajectory is* Ai)**.** Every limiting trajectory starts at $x = K$ with $X > 0$ and $Y/X =: \varsigma \geq 0$, so it is a multiple of $y = \mathrm{Ai} + \epsilon\,\mathrm{Bi}$ with $\epsilon = (\varsigma\,\mathrm{Ai}(K) + \mathrm{Ai}'(K))/(-\mathrm{Bi}'(K) - \varsigma\,\mathrm{Bi}(K))$. Since $\mathrm{Bi}, \mathrm{Bi}' > 0$, the mediant inequality gives $|\epsilon| \leq \max(\mathrm{Ai}(K)/\,\mathrm{Bi}(K), |\,\mathrm{Ai}'(K)|/\,\mathrm{Bi}'(K)) \leq C_0\eta_A(K)$ for every $\varsigma \in [0,\infty]$ ([15], 9.7.5–9.7.8; equation numbers of the form 9.7.5 refer to that source throughout). The Wronskian $\mathrm{Ai}\,\mathrm{Bi}' - \mathrm{Ai}'\,\mathrm{Bi} = 1/\pi$ gives $\mathrm{Ai}(K) + \epsilon\,\mathrm{Bi}(K) = \pi^{-1}/(\mathrm{Bi}'(K) + \varsigma\,\mathrm{Bi}(K)) > 0$, and since Ai / Bi decreases on $[0,\infty)$, $y > 0$ on $[0,K]$. So ϱ stays in $(-\pi/2, \pi/2)$ on the hyperbolic part of the window.

Step 3 (*reading the phase at* m_R)**.** By definition $\chi_{m_R} = \widehat{\widetilde{E}}(\mu_{m_R})^{-1}(\widehat{D}_{m_R}^{-1}(\psi_{2m_R-1}))$, so $\chi_{m_R} - m_R\pi$ is the image of ϱ_{m_R} under the lift of $\widetilde{E}(\mu_{m_R})^{-1}C_\ell$ that sends 0 to $-\pi/4$. Since $\widetilde{E}^{-1}$ multiplies e_- by e^{-s} and e_+ by e^s, this map sends (X,Y) to a positive multiple of $Xe_- + qYe_+$ with $q = e^{2s}/(4\ell)$, and $q = \sqrt{4-u}/(2\sqrt{|x_{m_R}|}) \to 1/\sqrt{K}$. Hence $\chi_{m_R} - m_R\pi \to -\pi/4 + \Omega_\epsilon(K)$, where $\Omega_\epsilon(\zeta)$ is the continuous lift of $\arg\big(y(-\zeta) - iy'(-\zeta)/\sqrt{\zeta}\big)$ on $(0,K]$, equal to $\pi/2$ at 0^+. The Wronskian and DLMF 9.7.9, 9.7.11 give $|(y, y'/\sqrt{K})(-K)| \geq cK^{-1/4}$, so $|\Omega_\epsilon(K) - \Omega_0(K)| \leq C|\epsilon|$. □

Lemma 4.3 (Airy phase). Let Ω be the continuous lift on $(0,\infty)$ of $\arg\big(\mathrm{Ai}(-\zeta) - i\,\mathrm{Ai}'(-\zeta)/\sqrt{\zeta}\big)$ with $\Omega(0^+) = \pi/2$. Then $\Omega(\zeta) = \pi/4 - {}^2\!/_3\zeta^{3/2} + O(\zeta^{-3/2})$.

Proof. By DLMF 9.7.9 and 9.7.10, $\mathrm{Ai}(-\zeta) = \pi^{-1/2}\zeta^{-1/4}(\cos\phi + O(\zeta^{-3/2}))$ and $\mathrm{Ai}'(-\zeta) = \pi^{-1/2}\zeta^{1/4}(\sin\phi + O(\zeta^{-3/2}))$ with $\phi = {}^2\!/_3\zeta^{3/2} - \pi/4$. Hence

$$\mathrm{Ai}(-\zeta) - i\,\mathrm{Ai}'(-\zeta)/\sqrt{\zeta} = \pi^{-1/2}\zeta^{-1/4}\big(e^{-i\phi} + O(\zeta^{-3/2})\big),$$

and $\Omega(\zeta) = -\phi + 2\pi n_0 + O(\zeta^{-3/2})$ for a fixed integer n_0. Writing $\mathrm{Ai}(-\zeta) = r\cos\Omega$, $-\mathrm{Ai}'(-\zeta)/\sqrt{\zeta} = r\sin\Omega$, the Airy equation gives $\Omega' = -\sqrt{\zeta} - \sin(2\Omega)/(4\zeta)$, so Ω is decreasing whenever $\cos\Omega = 0$, and the k-th zero a_k of Ai has $\Omega(|a_k|) = \pi/2 - k\pi$. By DLMF 9.9.6 and 9.9.18, ${}^2\!/_3|a_k|^{3/2} = (4k-1)\pi/4 + O(1/k)$, so $-\phi = \pi/2 - k\pi + o(1)$ there, and $n_0 = 0$. □

Lemma 4.4 (Turning sum). $T_K = \sum_{m_L<m<m_R}(\pi - \varphi_m) \to {}^2\!/_3K^{3/2}$ as $\lambda \to \infty$ at fixed K.

Proof. Blocks with $x_m \geq 0$ have $\varphi_m = \pi$. For $-K < x_m < 0$, $\pi - \varphi_m = 2\arccos(1 - u_m/2) = 2\sqrt{u_m}(1 + O(u_m)) = \ell^{-1}\sqrt{|x_m|}\,(1 + O(K/\ell^2))$. With grid steps $h_m = \ell^{-1}(1 + o(1))$ this is a Riemann sum of $\sqrt{\zeta}$ on $[0, |x_{m_R}|]$, and $|x_{m_R}| \to K$. □

Proposition 4.5 — Near field

Assume Hypothesis 4.1. If $\kappa' = 0$ let $M = \infty$; if $\kappa' \neq 0$ let $M = M(\lambda)$ satisfy $N_M/N_\lambda \to \infty$ and $N_M^{\alpha-1}/\lambda \to 0$. Then

$$\chi_M = \sum_{m<M} \varphi_m - \frac{\kappa'\pi}{2\alpha} + o(1) \qquad (\lambda \to +\infty),$$

where for $M = \infty$ the left side is $\psi(\lambda)$. The $o(1)$ is uniform for κ' in compact subsets of $(-(\alpha-1)/2, (\alpha-1)/2)$ when the other data are fixed.

Proof. This is (17) made precise. Apply Proposition 3.3 from $m_1 = m_R$ to M with the split

$$\bar\varepsilon_m := -2\kappa'/N_m, \qquad \bar\sigma_m := s(\mu_m) - s(\mu_m e^{-\bar\ell_m}),$$

which keeps the drift and the smooth part of the frame change. By Lemma A.4 the hypotheses of Proposition 3.3 hold for $\lambda \geq \lambda_0(K)$, and the error $\sum_{m_R\leq m<M} \mathscr{E}_m$ together with the boundary terms $|y_{m_R} - \chi_{m_R}|$ and $|y_M - \chi_M|$ is at most $CK^{-3/2} + o(1)$.

Step 1 (*assembly*)**.** Blocks $m \leq m_L$ have $\varphi_m = \pi$, including the finitely many small m once $\lambda > 2\max_{m<m_0} A_m$. So $\sum_{m<m_R} \varphi_m = m_R\pi - T_K$. By Lemma 4.2, Lemma 4.4 and Proposition 3.3,

$$\limsup_{\lambda\to\infty} \Big|\chi_M - \sum_{m<M} \varphi_m - \sum_{m_R\leq m<M} \bar\varepsilon_m \tan\tfrac{\varphi_m}{2}\Big| \leq \big|\Omega(K) + \tfrac{2}{3}K^{3/2} - \tfrac{\pi}{4}\big| + C_0\eta_A(K) + CK^{-3/2},$$

and the right side tends to 0 as $K \to \infty$ by Lemma 4.3.

Step 2 (*the drift*)**.** $\tan(\varphi_m/2) = (\mu_m/2)(1-\mu_m^2/4)^{-1/2}$ and $\mu_m/2 = (N_\lambda/N_m)^\alpha(1+o(1))$ on the oscillatory side. The drift sum is therefore a Riemann sum with step 2 in N of

$$-\kappa' \int_{N_\lambda}^{\infty} \frac{(N_\lambda/N)^\alpha}{\sqrt{1-(N_\lambda/N)^{2\alpha}}} \frac{dN}{N} = -\frac{\kappa'}{\alpha}\int_0^1 \frac{dw}{\sqrt{1-w^2}} = -\frac{\kappa'\pi}{2\alpha}$$

($w = (N_\lambda/N)^\alpha$, $dw/w = -\alpha\, dN/N$). The integrand has an integrable singularity at N_λ; the part below N_{m_R} is $O((\ell K/N_\lambda)^{1/2})$ and the part above N_M is $O((N_\lambda/N_M)^\alpha)$.

Step 3 (*uniformity*)**.** Every constant above, and in Lemma A.4, depends on κ' only through $|\kappa'|$ and through the bound $|\tilde\varepsilon_m| \leq (C + C|\kappa'|)N_m^{-1-\delta}$, so it is bounded for κ' in compact sets. □

5 THE FIRST-ORDER LAW FOR BALANCED LINKS

For balanced links the near field reaches all the way to infinity, and Proposition 4.5 with $M = \infty$ is already the discrete Bohr–Sommerfeld rule. This section states it (Theorem 5.1), evaluates the sum S in closed form (Proposition 5.2), and draws the consequences for eigenvalues, counting functions, and the growth of the Nevanlinna matrix. Rates follow in Section 8.

Theorem 5.1 — Discrete Bohr–Sommerfeld rule

Assume Hypothesis 2.2 and $|b_n| \le C_b(1+n)^\beta$ with $\beta < \alpha - 1$. Then

$$\psi(\lambda) - S(\lambda) \to 0 \quad (\lambda \to +\infty), \qquad \psi(\lambda) + S(|\lambda|) \to 0 \quad (\lambda \to -\infty).$$

Proof. For $\lambda \to +\infty$ this is Proposition 4.5 with $\kappa' = 0$ and $M = \infty$, since $\chi_M \to \psi$ (Section 3) and $\varphi_m \to 0$ fast enough that $\sum_{m<M} \varphi_m \to S$. For $\lambda \to -\infty$ apply this to $J_0 - \mathrm{diag}(b)$, which satisfies the same hypotheses, and use $\psi_J(\lambda) = -\psi_{J_0-\mathrm{diag}(b)}(-\lambda)$ (Theorem 2.2(iii)). □

5.1 THE SUM IN CLOSED FORM

Proposition 5.2 — Evaluation of S

(a) If $r_n \to 0$ then $S(\lambda) = N_\lambda I_\alpha(1 + o(1))$.

(b) If $r_n = O(1/n)$ then $S(\lambda) = N_\lambda I_\alpha + O(1)$.

(c) If $r_n = \kappa/n + O(n^{-1-\delta'})$ for some $\delta' > 0$, then $S(\lambda) = \Phi(\lambda) + o(1)$.

(d) If $a_n = c(n+s)^\alpha$ exactly, then $S(\lambda) - \Phi(\lambda) = O(N_\lambda^{-1/2})$ with $\kappa = \alpha s$, and this cannot be improved (Proposition 8.5).

Proof. Since S is nonincreasing in each link A_m, it suffices to evaluate it for shifted powers and to sandwich. For $A_m = c(2m+1+s)^\alpha$ put $g(x) := 2\arcsin\big(\min(1, \lambda/(2c(2x+1+s)^\alpha))\big)$, so $S = \sum_{m\ge 0} g(m)$ and, by $y = 2x+1+s$,

$$\int_{-1/2}^{\infty} g\,dx = \int_{s}^{\infty} \arcsin\Big(\min\big(1, (N_\lambda/y)^\alpha\big)\Big)dy = N_\lambda I_\alpha - \frac{s\pi}{2}.$$

On the intervals $[m - 1/2, m + 1/2]$ away from the kink of g the midpoint error is at most $1/24 \sup|g''|$, and $|g''| \le C/(N^2 u^{3/2})$ there, so these errors sum to $C\sum_{d\ge 1} N_\lambda^{-2}(N_\lambda/d)^{3/2} = O(N_\lambda^{-1/2})$; the far side contributes $O(N_\lambda^{-1})$ and the at most two intervals at the kink contribute $O(N_\lambda^{-1/2})$. This proves (d). For (c), $r_n - \kappa/n = O(n^{-1-\delta'})$ gives $c(n+\kappa/\alpha-\eta)^\alpha \le a_n \le c(n+\kappa/\alpha+\eta)^\alpha$ for $n \ge n(\eta)$; for large λ the finitely many smaller blocks have $\varphi_m = \pi$

under every comparison, so $\limsup |S-\Phi| \le \eta\pi/2$ for every $\eta>0$. Parts (a) and (b) are the same sandwich with $(1\pm\eta)c\,n^{\alpha}$ and with $c(n\pm s_0)^{\alpha}$. □

The phase (7) is thus $\Phi(\lambda)=\int_0^{\infty}\arcsin(\min(1,\lambda/(2c(y+\kappa/\alpha)^{\alpha})))dy$, the classical action of the shifted power with the absent link a_0 placed at $y=0$. Proposition 5.2(d) says that the discrete sum S carries an oscillating term of size $N_\lambda^{-1/2}$ that ψ does not have; Proposition 8.5 identifies it.

5.2 EIGENVALUES AND COUNTING FOR EVERY EXTENSION

Corollary 5.3. Assume Hypothesis 2.2 with $|b_n|\le C_b(1+n)^{\beta}$, $\beta<\alpha-1$. Let $\lambda_j=\lambda_j(t)$ be the eigenvalues of H_t labelled as in Theorem 2.2(v).

(a) $S(\lambda_j)=j\pi+t+o(1)$ as $j\to+\infty$ and $S(|\lambda_j|)=|j|\pi-t+o(1)$ as $j\to-\infty$.

(b) If $r_n=\kappa/n+O(n^{-1-\delta'})$, then $\lambda_j^{1/\alpha}=(2c)^{1/\alpha}\frac{\pi}{I_\alpha}\big(j+t/\pi+\frac{\kappa}{2\alpha}\big)+o(1)$.

(c) If $b\equiv 0$, then $\mathcal{N}_t^{+}(\lambda):=\#(\operatorname{spec}H_t\cap(0,\lambda])$ satisfies $\mathcal{N}_t^{+}(\lambda)=(S(\lambda)-t)/\pi+1/2-[t=0]+r(\lambda)$ with $|r(\lambda)|\le 1/2+o(1)$, where $[t=0]$ is 1 if $t=0$ and 0 otherwise. Under the hypothesis of (b) this reads

$$\mathcal{N}_t^{+}(\lambda)=C_\alpha(\lambda/c)^{1/\alpha}-\frac{\kappa}{2\alpha}-\frac{t}{\pi}+\frac12-[t=0]+r(\lambda),$$
$$C_\alpha=\frac{\sin(\pi/2\alpha)}{2\pi}\,\mathrm{B}\big(\tfrac{1}{2\alpha},1-\tfrac{1}{\alpha}\big)=\frac{2^{-1/\alpha}I_\alpha}{\pi}.$$

On the negative half-line $\mathcal{N}_t^{-}=\mathcal{N}_{\pi-t}^{+}$ for $t>0$ and $\mathcal{N}_0^{-}=\mathcal{N}_0^{+}$.

Proof. (a) is Theorem 2.2(v) with Theorem 5.1; (b) inserts Proposition 5.2(c). (c) For $b\equiv0$, $\psi(0)=0$ and the count is $\lfloor(\psi(\lambda)-t)/\pi\rfloor+1-[t=0]$; the floor differs from its argument by at most one, which gives the remainder $1/2$ around the midpoint. The identity for C_α follows from $\mathrm{B}(x,y)=\Gamma(x)\Gamma(y)/\Gamma(x+y)$ with Legendre's duplication formula and the reflection formula $\Gamma(z)\Gamma(1-z)=\pi/\sin\pi z$. The negative half-line follows from the reflection identity. □

Corollary 5.3(b) shows the structure of the law: the slope $(2c)^{1/\alpha}\pi/I_\alpha$ is determined by c and α, the offset $\kappa/(2\alpha)$ by the $1/n$ term of the links, and the extension enters additively through t/π in the equal-amplitude normalisation of Proposition 2.1.

Example 5.1 (The Berg–Valent chain). The quartic birth–death rates $\lambda_n=(4n+1)(4n+2)^2(4n+3)$, $\mu_n=(4n-1)(4n)^2(4n+1)$ give the symmetric links $a_k=2k\sqrt{4k^2-1}$ (by the symmetrisation of Section 6.5), so $\alpha=2$, $c=4$, $\kappa=0$. The Krein and Friedrichs spectra are known in closed form [9, 39]: in the Stieltjes variable $x_n=(2n\pi/K_0)^4$ and $((2n+1)\pi/K_0)^4$ with $K_0=K(1/\sqrt2)=\Gamma(1/4)^2/(4\sqrt{\pi})$. (Valent's text [39] writes

$K_0 = \theta(1) = \int_0^1 (1-u^4)^{-1/2}\,du$; only $K_0 = \sqrt{2}\,\theta(1)$ is consistent with the type and with Chen–Ismail's closed form [5].) In the symmetric variable $\lambda = \sqrt{x}$ and with $\mathrm{B}(1/4, 1/2) = 2\sqrt{2}K_0$, Corollary 5.3(b) predicts $\lambda_j^{1/2} = (2\pi/K_0)(j + t/\pi)$, which is exact for $t = 0$ and $t = \pi/2$. This chain, together with the cubic family of [24, Remark 3 after Prop. 14] (whose leading term is quoted there from [23], outside our balanced class), is the only previous instance we know of an eigenvalue law with an explicit constant in the limit-circle case.

5.3 ORDER, TYPE AND INDICATOR OF THE NEVANLINNA MATRIX

Corollary 5.4. Under Hypothesis 2.2 with a diagonal as in Corollary 5.3, the entries of the Nevanlinna matrix have order $1/\alpha$, are of completely regular growth, and have the indicator $\mathfrak{h}(\theta) = \frac{\pi\mathfrak{n}}{\sin(\pi/\alpha)}\big(\cos\frac{\theta-\pi}{\alpha} + \cos\theta/\alpha\big)$ on $[0,\pi]$ (and $\mathfrak{h}(-\theta) = \mathfrak{h}(\theta)$), with $\mathfrak{n} = C_\alpha c^{-1/\alpha}$. In particular the type is $T = {}^1\!/_2 c^{-1/\alpha}\,\mathrm{B}(\frac{1}{2\alpha}, 1 - 1/\alpha)$.

Proof. The entries are entire of genus zero with the same order, type and indicator [4]. The zeros of the entry B are the spectrum of one self-adjoint extension, so by Corollary 5.3(a) and Proposition 5.2(a) they are real with $n_\pm(r) \sim \mathfrak{n}\, r^{1/\alpha}$ on each half-line: an angular density of non-integral order $1/\alpha$ with point masses at $\arg z = 0, \pi$. Levin's theorem [28, Ch. II §1, Thm. 1] (for zeros on rays: Ch. I, Thms. 25 and 29) gives completely regular growth with the stated indicator, whose maximum $\pi\mathfrak{n}/\sin(\pi/(2\alpha))$ equals T. □

For $a_n = n^p$ this is Bochkov's theorem [11], here proved through the distribution of the zeros rather than through Taylor coefficients, and extended to the class of Hypothesis 2.2 with a small diagonal. For polynomial birth–death rates the two results overlap without nesting: Bochkov's theorem with the Berg–Szwarc reduction [6, Thm. 1.11, Rem. 1.10] covers all admissible rates, balanced or not. The indicator part of Valent's conjecture [38, Conj. 1] also follows from those results by a Tauberian argument: the positivity of the Taylor coefficients of the Stieltjes entry gives $\log M(r) \sim \tau r^{1/p}$, a theorem of Titchmarsh–Valiron type [10, §7.2] turns this into the density of the positive zeros, and Levin's theorem gives the indicator. Corollary 5.4 and its dimerised analogue (Corollary 6.5) give a second route through the eigenvalue law; their new content is not the indicator but the $O(1)$ terms.

5.4 WHAT THE SLOWLY VARYING HYPOTHESIS DOES AND DOES NOT GIVE

Theorem 5.1 is stated under a condition on first differences because that is what the near-field sums need. The three remarks below record what this hypothesis does not give: uniformity, an offset, and a rate for chains that are not close to a smooth symbol. None of them is used later.

Remark 5.1 (No uniformity over Hypothesis 2.2). The proof of Theorem 5.1 gives the error $C(K^{-3/2}+(\ell K)^{-1})+\eta(\lambda)$ with $\eta(\lambda) = C(N_\lambda^{-\delta}\log N_\lambda + N_\lambda^{1+\beta-\alpha} + N_\lambda^{-1}\log N_\lambda)$, and the constant depends on the chain through $\sum|\tilde{\varepsilon}_m|$. Nothing is uniform over Hypothesis 2.2, and for small δ the convergence can be invisible at every accessible λ: for $r_n = 0.4(-1)^n n^{-1.1}$ ($\delta = 0.1$) the frame normalisation ρ_m approaches 1 only like $m^{-0.1}$, so the error terms that involve $\tilde{\varepsilon}_m$ decay only like $N_\lambda^{-0.1}$. Such chains do have a rate; Theorem 8.1 gives $O(N_\lambda^{-0.1})$. Hypothesis 2.2 is sufficient but not necessary: the proof uses it only through the four sums estimated in Lemma A.4.

Remark 5.2 (No offset without the $1/n$ term). Without the κ/n hypothesis no offset need exist: there are chains satisfying Hypothesis 2.2 for which $S(\lambda) - N_\lambda I_\alpha$ is unbounded, for example $r_n = 0.3\, n^{-0.15}\sin(2\log n)$. Theorem 5.1 still holds for them, which is why it is stated with S.

Remark 5.3 (Chains with differences only). Under Hypothesis 2.2 with $\delta \in (3/4, 1]$, mollify the piecewise linear interpolant of r_n at the variable scale x^ϑ, $\vartheta \in (1-\delta/3, 1)$. One derivative falls on the interpolant and each further one costs $x^{-\vartheta}$, so the mollified symbol satisfies Hypothesis 8.1, and the remainder is $O(n^{\vartheta-1-\delta})$. Theorem 8.1 then gives $\psi - \bar{\Phi} = O(N_\lambda^{-\min(1,(4\delta-3)/3)+\epsilon} + N_\lambda^{1+\beta-\alpha})$ for every $\epsilon > 0$, where $\bar{\Phi}$ is the smooth phase (28) of the mollified symbol. For $\delta \le 3/4$ the method gives no rate for a general chain in Hypothesis 2.2, and no single rate holds over Hypothesis 2.2, which fixes neither δ nor the constant.

6 DIMERISED LINKS: A BESSEL FAR FIELD

Throughout this section $b \equiv 0$ except in Theorem 6.4, the links satisfy Hypothesis 2.1 with $2|\kappa'| < \alpha - 1$, and

$$\nu := \frac{1}{2} - \frac{\kappa'}{\alpha - 1} \in (0, 1).$$

Two solutions that decay at different rates. In a balanced chain the two solutions at $\lambda = 0$ have the same size at infinity. In a dimerised chain they do not: by Proposition 2.1, $P_{2m}^2 \asymp N_m^{-\alpha-2\kappa'}$ and $R_{2m+1}^2 \asymp N_m^{-\alpha+2\kappa'}$. Both are square-summable, which is the limit-circle property, but for $\kappa' > 0$ the solution P has the smaller block envelope, and for $\kappa' < 0$ the solution R does. We call it *the solution of smaller block-envelope size*. The boundary condition it defines will turn out to be distinguished, and every other self-adjoint boundary condition contains some of the larger solution. The mechanism is a Bessel problem: the far field of the chain is a first-order Bessel system of order ν, and the end $N = \infty$ sits at its regular singular point $\tau = 0$ (Section 6.2). After one component is eliminated the system becomes the scalar operator

$$-\frac{d^2}{d\tau^2} + \frac{\nu^2 - 1/4}{\tau^2} \qquad \text{near } \tau = 0, \quad 0 < \nu < 1,$$

whose solutions near the end behave like $\tau^{1/2+\nu}$ and $\tau^{1/2-\nu}$. Sizes are not preserved by that elimination: the principal branch $\sqrt{\tau}\,J_\nu(\tau)$ of the scalar operator corresponds to the second coordinate axis, that is to R, for both signs of κ'. The scalar picture explains the order ν and the phases; which extension is distinguished is decided by the amplitude ratio of the two coordinates at the end, computed in (21).

Two laws. At high energy this distinction decides the eigenvalues. Up to the place where the far field begins, the solution of energy λ oscillates with equal amplitude in the frame of Section 3; beyond it the solution freezes into a fixed combination of the two solutions at $\lambda = 0$. In a dimerised chain that place is the site $N_{dec} \asymp N_\lambda^{\alpha/(\alpha-1)}$ found below, and there the ratio of the sizes of $\widetilde{R}$ and $\widetilde{P}$ is $N_{dec}^{2\kappa'}$. So a solution of balanced size at N_{dec} reaches infinity with coordinates whose ratio is of order

$$\frac{\beta_\infty}{\alpha_\infty} \asymp N_{dec}^{-2\kappa'} \asymp \lambda^{-2\kappa'/(\alpha-1)},$$

times an oscillating factor of size one. For $\kappa' > 0$ this tends to 0. The boundary condition $\tan t = \beta_\infty/\alpha_\infty$ of H_t (Theorem 2.2(iv)) can then be met only near a zero of α_∞, whatever $t \neq 0$ is, and all these extensions share the law of $H_{\pi/2}$. The exception is $t = 0$, the extension defined by $\widetilde{P}$, the solution of smaller block-envelope size: it has its own law, given by the zeros of β_∞. For $\kappa' < 0$ the roles of $\widetilde{P}$ and $\widetilde{R}$, and of H_0 and $H_{\pi/2}$, are exchanged. In the balanced case $\kappa' = 0$, $\nu = 1/2$, the two solutions have the same size at infinity, the coordinates stay of comparable size, neither solution is distinguished, and every t has its own law (Corollary 5.3).

Where the offsets come from. The Bessel functions $J_{\pm\nu}(\tau)$ oscillate like $\cos(\tau \mp \frac{\nu\pi}{2} - \pi/4)$ for large τ. Matching the near-field phase Φ to these two phases moves the offsets of H_0 and $H_{\pi/2}$ from the balanced values 0 and $\pi/2$ to

$$\theta_0 = \Big(\frac{1}{2} - \nu\Big)\frac{\pi}{2} = \frac{\kappa'\pi}{2(\alpha-1)} \qquad \text{and} \qquad \frac{\pi}{2} - \theta_0.$$

Where the far field begins. On the oscillatory side a block rotates by $\varphi_m \approx \mu_m \approx 2\Lambda N_m^{-\alpha}$, while the frame drifts by $|\varepsilon_m| \approx 2|\kappa'|/N_m$. The adiabatic normal form needs the rotation to dominate the drift, and their ratio is

$$\frac{\varphi_m}{|\varepsilon_m|} \approx \frac{\Lambda N_m^{1-\alpha}}{|\kappa'|} = \frac{\alpha-1}{|\kappa'|}\,\tau(N_m), \qquad \tau(N) := \frac{\Lambda N^{1-\alpha}}{\alpha-1}.$$

So the Bessel variable τ measures rotation against drift. For $\tau \gg 1$ the chain is in the near field of Proposition 4.5, and for $\tau \lesssim 1$ it is in the far field. The crossover is where τ is of order one, that is on the scale

$$N_{dec} := \Big(\frac{\Lambda}{\alpha-1}\Big)^{1/(\alpha-1)} \asymp N_\lambda^{\alpha/(\alpha-1)}$$

(Figure 1). The two descriptions overlap on a range of N, and we match them there.

The rest of the section makes this picture precise. We derive the Bessel system from the block product (Section 6.2), match it to the near field (Proposition 6.1, Theorem 6.2), and read off the eigenvalues of every extension (Corollary 6.3) and of birth–death processes (Corollary 6.5).

6.2 FROM THE BLOCK PRODUCT TO BESSEL'S EQUATION

The block as a step of an equation. By Lemma 3.1 the block is $T_m = D_m T_1(\mu_m) D_m^{-1}$ with $D_m = \operatorname{diag}(\rho_m^{-1/2}, \rho_m^{1/2})$, and $T_1(\mu) = I + \mu \mathrm{J} - \mu^2 e_1 e_1^{\mathsf{T}}$, where J generates the rotations. In the far field μ_m is small. The dimerisation enters only through the frame normalisation, and by Proposition 2.1(b) $\rho_m = N_m^{-2\kappa'}(1 + O(N_m^{-\delta}))$, so $D_m = D(N_m)(1 + O(N_m^{-\delta}))$ with

$$D(N) := \operatorname{diag}(N^{\kappa'}, N^{-\kappa'}).$$

Since $D(N)$ is unbounded, the comparison has to be made in the weighted coordinates $D(N_m)^{-1}(\cdot)D(N_m)$: there $X := D(N_m)^{-1}D_m = I + O(N_m^{-\delta})$ is diagonal and $T_1 - I = O(\mu_m)$, so $XT_1X^{-1} = T_1 + O(\mu_m N_m^{-\delta})$ and

$$D(N_m)^{-1}\big[T_m - I - \mu_m\, D(N_m)\mathrm{J}D(N_m)^{-1}\big]D(N_m) = O(\mu_m^2 + \mu_m N_m^{-\delta}),$$

where

$$D(N)\mathrm{J}D(N)^{-1} = \begin{pmatrix} 0 & -N^{2\kappa'} \\ N^{-2\kappa'} & 0 \end{pmatrix}.$$

In an ordinary matrix norm the remainder is only $O(\mu_m^2 + \mu_m N_m^{2|\kappa'|-\delta})$: the off-diagonal entry $-\mu_m/\rho_m$ of T_m differs from $-\mu_m N_m^{2\kappa'}$ by $\mu_m N_m^{2\kappa'} O(N_m^{-\delta})$, which for $\kappa' > 0$ need not be

small. The proof of Proposition 6.1 in Appendix A compares the true and the model step in exactly these weighted coordinates. A block spans two sites, and $\mu_m = \lambda/A_m = 2\Lambda N_m^{-\alpha}(1 + O(N_m^{-1}))$ is the integral of $\Lambda N^{-\alpha}$ over $[N_m - 1, N_m + 1]$ up to the same error. So T_m is, to first order in μ_m, the step of length two of

$$\frac{dv}{dN} = \Lambda N^{-\alpha} \begin{pmatrix} 0 & -N^{2\kappa'} \\ N^{-2\kappa'} & 0 \end{pmatrix} v. \tag{20}$$

For $\kappa' = 0$ this is a pure rotation, which is the content of the normal form. For $\kappa' \neq 0$ the rotation is conjugated by the frame drift $D(N)$.

Why the order is ν. The natural time of the rotation is the total angle still to come, $\tau(N) = \int_N^\infty \Lambda N'^{-\alpha} dN' = \Lambda N^{1-\alpha}/(\alpha - 1)$, which decreases from ∞ to 0 and is the variable introduced above. In this time (20) reads $dv_1/d\tau = N^{2\kappa'} v_2$ and $dv_2/d\tau = -N^{-2\kappa'} v_1$, and the frame drift becomes a power of τ:

$$N = N_{dec}\, \tau^{-1/(\alpha-1)}, \qquad N^{2\kappa'} = N_{dec}^{2\kappa'}\, \tau^{-2\kappa'/(\alpha-1)} = N_{dec}^{2\kappa'}\, \tau^{2\nu-1}.$$

After rescaling by the constant $D(N_{dec})^{-1}$, that is $(w_1, w_2) := (N_{dec}^{-\kappa'} v_1, N_{dec}^{\kappa'} v_2)$, the system is

$$\frac{dw_1}{d\tau} = \tau^{2\nu-1} w_2, \qquad \frac{dw_2}{d\tau} = -\tau^{1-2\nu} w_1, \qquad \text{so} \qquad w_1'' + \frac{1-2\nu}{\tau}\, w_1' + w_1 = 0.$$

This is Bessel's equation in Lommel's form, solved by $w_1 = \tau^\nu J_{\pm\nu}(\tau)$; with $u := \tau^{1/2-\nu} w_1$ it becomes $-u'' + (\nu^2 - 1/4)\tau^{-2} u = u$, the operator of the guiding picture with energy 1. The order is the exponent of the drift measured in the time of the rotation: the frame weight $N^{2\kappa'}$ becomes $\tau^{2\nu-1}$, and $2\nu - 1 = -2\kappa'/(\alpha - 1)$. Balanced links give $\nu = 1/2$, where $J_{\pm 1/2}$ are the sine and cosine and there is no singular potential.

The solutions. The second components follow from $w_2 = \tau^{1-2\nu} w_1'$ and $\frac{d}{d\tau}(\tau^\nu J_\nu) = \tau^\nu J_{\nu-1}$, $\frac{d}{d\tau}(\tau^\nu J_{-\nu}) = -\tau^\nu J_{1-\nu}$. Undoing the rescaling, with $N^{\kappa'} \tau^{1/2} = N_{dec}^{\kappa'} \tau^\nu$, the two solutions of (20) are

$$y_-(N) = \big(N^{\kappa'} \tau^{1/2} J_{-\nu}(\tau),\, -N^{-\kappa'} \tau^{1/2} J_{1-\nu}(\tau)\big), \qquad y_+(N) = \big(N^{\kappa'} \tau^{1/2} J_\nu(\tau),\, N^{-\kappa'} \tau^{1/2} J_{\nu-1}(\tau)\big).$$

Let $Y(N) := [y_-, y_+]$ and $\hat{Y}(N) := D(N)^{-1} Y(N)$, the solutions in the balanced frame. We use four facts about them.

(B1) $\det Y = \det \hat{Y} = \omega_0 := 2/\pi \sin \nu\pi \neq 0$. Indeed $\det Y = \tau\big(J_\nu J_{1-\nu} + J_{-\nu} J_{\nu-1}\big)$, and this equals $2\sin(\nu\pi)/\pi$ by the Wronskian $\mathscr{W}\{J_\nu, J_{-\nu}\}(\tau) = -2\sin(\nu\pi)/(\pi\tau)$ (DLMF 10.5.1) and the recurrences $J_\mu' = J_{\mu-1} - (\mu/\tau) J_\mu = -J_{\mu+1} + (\mu/\tau) J_\mu$ (DLMF 10.6.2).

(B2) As $\tau \to \infty$, uniformly for ν in compact subsets of $(0, 1)$ (DLMF 10.17.3), $\hat{Y} = \sqrt{2/\pi}\, Y_0 + O(1/\tau)$ with $Y_0 = \left(\begin{smallmatrix} \cos a & \cos b \\ -\sin a & -\sin b \end{smallmatrix}\right)$, $a = \tau - \pi/4 + \frac{\nu\pi}{2}$, $b = \tau - \pi/4 - \frac{\nu\pi}{2}$, and $\det Y_0 = -\cos a \sin b + \sin a \cos b = \sin(a - b) = +\sin \nu\pi$. This is the near field seen from the far field: equal amplitudes, phases $\tau \pm \frac{\nu\pi}{2} - \pi/4$.

(B3) As $N \to \infty$, $Y(N) \to \mathrm{diag}(d_1, d_2)$ with $d_1 = 2^{\nu} N_{dec}^{\kappa'}/\Gamma(1-\nu)$ and $d_2 = 2^{1-\nu} N_{dec}^{-\kappa'}/\Gamma(\nu)$, by $J_\mu(\tau) \sim (\tau/2)^\mu/\Gamma(\mu+1)$ (DLMF 10.2.2) and $N^{\pm\kappa'}\tau^{1/2} = N_{dec}^{\pm\kappa'}\tau^{1/2\pm(\nu-1/2)}$. The other entries tend to 0 like $N^{1-\alpha\pm 2\kappa'}$, which is where $2|\kappa'| < \alpha - 1$ is used. This is the limit-circle end: the two Bessel solutions become the two solutions at $\lambda = 0$.

(B4) $|\hat{y}_\pm(N)| \le C \max(1, \tau^{-|\nu-1/2|})$.

The ratio of the two limits in (B3) is the amplitude ratio of the guiding picture: the factor by which the far field multiplies the second coordinate at infinity relative to the first. It carries the whole dichotomy between the extensions,

$$Q(\lambda) := \frac{d_2}{d_1} = 2^{1-2\nu}\frac{\Gamma(1-\nu)}{\Gamma(\nu)} N_{dec}^{-2\kappa'} = \Big(\frac{2(\alpha-1)}{\Lambda}\Big)^{1-2\nu}\frac{\Gamma(1-\nu)}{\Gamma(\nu)} \asymp \lambda^{-2\kappa'/(\alpha-1)}. \tag{21}$$

6.3 MATCHING AND THE BOUNDARY VALUES AT INFINITY

The model (20) is exact up to errors of relative size μ_m and $N_m^{-\delta}$, and its solutions are bounded in the balanced frame except for the factor in (B4) where $\tau \le 1$. A Gronwall argument in the basis Y therefore follows the chain from a block M in the overlap to infinity; its proof is in Appendix A.

Proposition 6.1 — Far field

Let $M = M(\lambda)$ satisfy $N_M^{2\alpha-1}/N_\lambda^{2\alpha} \to \infty$ and $N_M^{\alpha-1+\delta}/N_\lambda^{\alpha} \to \infty$, and put $c_m := Y(2m)^{-1} v_{2m-1}$ for $m \ge M$. Then $c_\infty := \lim c_m$ exists, $v_\infty = Y(\infty)c_\infty$, and $|c_\infty - c_M| \le \epsilon_{far}|c_M|$ with

$$\epsilon_{far} \le C\big(\Lambda^2 N_M^{1-2\alpha} + \Lambda N_M^{1-\alpha-\delta} + N_{dec}^{-\delta}\big) \to 0.$$

The two conditions on M and the near-field condition $N_M^{\alpha-1}/\lambda \to 0$ of Proposition 4.5 are compatible: the far field needs $N_M \gg N_\lambda^{\max(2\alpha/(2\alpha-1),\, \alpha/(\alpha-1+\delta))}$, the near field $N_M \ll N_\lambda^{\alpha/(\alpha-1)}$, and both exponents on the left are smaller than $\alpha/(\alpha-1)$ because $\delta > 0$. We fix $N_M := N_\lambda^{\gamma}$ with γ strictly between them.

Theorem 6.2 — Dimerised boundary values

Assume Hypothesis 2.1, $b \equiv 0$ and $0 < \nu < 1$. Let $(\alpha_\infty, \beta_\infty) = v_\infty$ be the coordinates at infinity of the regular solution (Theorem 2.2(i)). There are a real function $\mathscr{A}(\lambda) \neq 0$ and functions $r_1, r_2 \to 0$ such that, as $\lambda \to +\infty$,

$$\alpha_\infty = \mathscr{A}(\lambda)\Big[\sin\big(\tfrac{\pi}{4} + \tfrac{\nu\pi}{2} - \Phi(\lambda)\big) + r_1(\lambda)\Big], \qquad \beta_\infty = \mathscr{A}(\lambda)Q(\lambda)\Big[\sin\big(\Phi(\lambda) - \tfrac{\pi}{4} + \tfrac{\nu\pi}{2}\big) + r_2(\lambda)\Big].$$

The functions r_1, r_2 tend to 0 uniformly for κ' in compact subsets of $(-(\alpha-1)/2, (\alpha-1)/2)$ when the remaining data are fixed. For $\lambda \to -\infty$, $v_\infty(\lambda) = \Sigma_3 v_\infty(|\lambda|)$.

Proof. The near field delivers the phase Φ at block M; (B2) splits it between the two Bessel solutions; (B3) carries each to infinity with its amplitude d_1 or d_2.

Step 1 (*near field*)**.** With M as above, Proposition 4.5 gives $\chi_M = \sum_{m<M} \varphi_m - \kappa'\pi/(2\alpha) + o(1)$. The odd links are $A_m = c\,N_m^{\alpha}(1 + (\kappa - \kappa')/N_m + O(N_m^{-1-\delta}))$, so Proposition 5.2(c) applied to them gives $S = N_\lambda I_\alpha - (\kappa - \kappa')\pi/(2\alpha) + o(1)$. The tail is $\sum_{m\geq M} \varphi_m = \sum_{m\geq M} \mu_m + O(\Lambda^3 N_M^{1-3\alpha}) = \tau(2M) + o(1)$, by the midpoint rule on $[2m, 2m+2]$ and because $\tau(2M)/N_M = \Lambda N_M^{-\alpha}/(\alpha - 1) \to 0$. Hence $\chi_M + \tau(2M) = \Phi(\lambda) + o(1)$: the drift converts the odd-link constant $\kappa - \kappa'$ into κ.

Step 2 (*matching*)**.** Write $z := \widetilde{E}(\mu_M)^{-1} D_M^{-1} v_{2M-1} = r(\cos\chi_M, \sin\chi_M)$, $r > 0$. Then $c_M = Y(2M)^{-1} v_{2M-1} = \hat{Y}(2M)^{-1}\,[D(2M)^{-1} D_M]\,\widetilde{E}(\mu_M)\,z = r\,\hat{Y}(2M)^{-1}\big[(\cos\chi_M, \sin\chi_M) + O(N_M^{-\delta} + \mu_M)\big]$. By (B2), $\hat{Y}(2M)^{-1} = \sqrt{\pi/2}\,Y_0^{-1} + O(1/\tau(2M))$, and solving $Y_0 c = (\cos\chi, \sin\chi)$ by Cramer's rule gives

$$c_- = -\frac{\sin(\chi + b)}{\sin \nu\pi}, \qquad c_+ = \frac{\sin(\chi + a)}{\sin \nu\pi}.$$

With $\chi = \chi_M$ and $\tau = \tau(2M)$, $\chi + b = \Phi - \pi/4 - \frac{\nu\pi}{2} + o(1)$ and $\chi + a = \Phi - \pi/4 + \frac{\nu\pi}{2} + o(1)$. All errors are $o(1)$ relative to r, and Proposition 6.1 gives $v_\infty = \mathrm{diag}(d_1, d_2) c_\infty$ with $c_\infty = c_M(1 + o(1))$. This is the statement with $\mathscr{A} = r\sqrt{\pi/2}\,d_1/\sin\nu\pi$. The frame angle enters only modulo π here: replacing χ_M by $\chi_M + \pi$ changes (c_-, c_+) into $-(c_-, c_+)$, so it multiplies α_∞ and β_∞ by -1, which is absorbed into the sign of $\mathscr{A}$. The integer labels are fixed separately, by the homotopy in the proof of Corollary 6.3.

Step 3 (*uniformity*)**.** The near-field error is uniform by Proposition 4.5. In the far field the constants depend on κ' through $1/\sin\nu\pi$, through the constants in (B2)–(B4), which are continuous for ν in compact subsets of $(0, 1)$, and through the tail bound for $\log(\rho_m N_m^{2\kappa'})$, which is at most $\sum_{i\geq m} |\tilde{\varepsilon}_i| \leq (C + C|\kappa'|)N_m^{-\delta}$. The window exponent γ can be fixed for all κ' in a compact set. The negative half-line is the reflection identity of Theorem 2.2(iii). □

6.4 EIGENVALUES OF EVERY EXTENSION

Corollary 6.3. Assume Hypothesis 2.1, $b \equiv 0$ and $0 < \nu < 1$, and put $\theta_0 := \kappa'\pi/(2(\alpha - 1)) = (1/2 - \nu)\pi/2$. Let $\lambda_j(t)$ be the eigenvalues of H_t, labelled by $\psi(\lambda_j(t)) = t + j\pi$.

(i) $\Phi(\lambda_j(0)) = j\pi + \theta_0 + o(1)$ and $\Phi(\lambda_j(\pi/2)) = j\pi + \pi/2 - \theta_0 + o(1)$ as $j \to +\infty$.

(ii) For $t \notin \{0, \pi/2\}$ the eigenvalues solve $\tan t = Q(\lambda)\,\frac{\sin(\vartheta - \pi/4 + \nu\pi/2) + r_2}{\sin(\pi/4 + \nu\pi/2 - \vartheta) + r_1}$ with $\vartheta = \Phi(\lambda)$. Since $Q \to 0$ if $\kappa' > 0$ and $Q \to \infty$ if $\kappa' < 0$, $\Phi(\lambda_j(t)) = j\pi + \pi/2 - \theta_0 + o(1)$ if $\kappa' > 0$, and $\Phi(\lambda_j(t)) = j\pi + \theta_0 + \pi[t > \pi/2] + o(1)$ if $\kappa' < 0$, where $[t > \pi/2]$ is 1 if $t > \pi/2$ and 0 otherwise.

(iii) In terms of λ: $\lambda_j^{1/\alpha} = (2c)^{1/\alpha}\frac{\pi}{I_\alpha}\big(j + \frac{\kappa}{2\alpha} + \frac{\theta_t}{\pi}\big) + o(1)$, where θ_t is the offset of H_t in (i) and (ii). For the counting function $\mathcal{N}_t^+(\lambda) := \#(\operatorname{spec} H_t \cap (0,\lambda])$ this gives

$$\mathcal{N}_t^+(\lambda) = \frac{\Phi(\lambda) - \theta_t}{\pi} + \frac{1}{2} - [t=0] + r_t(\lambda), \qquad |r_t(\lambda)| \le \tfrac{1}{2} + o(1);$$

the indicator $[t=0]$ accounts for the eigenvalue 0 of H_0, which is there whatever θ_0 is.

Proof. **Step 1** (*zeros*)**.** The eigenvalues of H_0 are the zeros of β_∞, those of $H_{\pi/2}$ the zeros of α_∞ (Theorem 2.2(iv)). By Theorem 6.2, for λ large β_∞ has exactly one sign change near each solution of $\Phi = k\pi + \theta_0$ and none where $|\sin(\Phi - \theta_0)|$ is bounded below; near such a zero $\alpha_\infty = \mathscr{A}\big[(-1)^k \sin \nu\pi + o(1)\big] \neq 0$, because $\pi/4 + \frac{\nu\pi}{2} - \theta_0 = \nu\pi$, and since ψ increases and passes $\pi/2 + \pi\mathbb{Z}$ exactly at zeros of α_∞, it cannot pass two multiples of π there. So for $\lambda \ge \lambda_0$ the zeros of β_∞ correspond bijectively to integers $k \ge k_0$ with $\Phi = k\pi + \theta_0 + o(1)$, and $j = k + j_1$ for a constant integer j_1; likewise for α_∞.

Step 2 (*the label*)**.** Replace κ' by $s\kappa'$, $s \in [0,1]$, keeping the other data. The parameter $s\kappa'$ stays in a compact subset of $(-(\alpha-1)/2, (\alpha-1)/2)$, which contains the balanced value 0 at $s = 0$ (where $\nu = 1/2$), so the uniformity in Theorem 6.2 applies on the whole path. At a level λ^* with $\Phi(\lambda^*) = K\pi + \frac{3\pi}{4}$ and K large, it keeps $\beta_\infty^{(s)}(\lambda^*) \neq 0$ for all s. The function $\beta_\infty^{(s)}(\lambda)$ is jointly continuous in (s,λ): the shear weights depend continuously on s, and the tails $\sum_{k>K'} |\gamma_k^{(s)}|$ are small uniformly in s and in λ on compact sets, because the bounds of Proposition 2.1 are uniform for κ' in compact sets. Its zeros are simple, since they are eigenvalues of $H_0^{(s)}$, and the zero at $\lambda = 0$ is fixed for every s because $b \equiv 0$; so no zero enters $(0,\lambda^*]$ through either end. Hence the number of eigenvalues of $H_0^{(s)}$ in $(0,\lambda^*]$ does not depend on s. At $s = 0$ it is K by Theorem 5.1 and Proposition 5.2(c), so $j_1 = 0$; the same argument with α_∞ fixes the labels of $H_{\pi/2}$.

Step 3 (*other extensions*)**.** For $t \in (0,\pi/2)$, $\lambda_j(t)$ lies between $\lambda_j(0)$ and $\lambda_j(\pi/2)$ and solves $\tan t = \beta_\infty/\alpha_\infty$, which is the displayed equation. If $Q \to 0$ the right side can equal $\tan t$ only where the denominator is small, which is the law of $H_{\pi/2}$; if $Q \to \infty$, only where the numerator is small, the law of H_0. The case $t \in (\pi/2,\pi)$ is the same with $\lambda_j(\pi/2) < \lambda_j(t) < \lambda_{j+1}(0)$. □

Remark 6.1 (Finite Q and uniformity). The remainders r_1, r_2 of Theorem 6.2 are uniform for κ' in compact sets, but the terminal offsets of Corollary 6.3(ii) are not: they need $Q \to 0$ or $Q \to \infty$, and $Q \asymp \lambda^{-2\kappa'/(\alpha-1)}$ moves slowly when κ' is small. Keeping Q, and writing $x := \Phi(\lambda_j(t)) - j\pi$ and $a := \theta_0$, the boundary condition of H_t reads $Q\cos t\, \sin(x-a) - \sin t\, \cos(x+a) = o(1)$, that is

$$\tan x = \frac{Q\cos t\, \sin a + \sin t\, \cos a}{Q\cos t\, \cos a + \sin t\, \sin a} + o(1),$$

on the branch fixed by interlacing. At balance ($Q = 1$, $a = 0$) this is $x = t$. For a family with $\kappa' \to 0$ and $\kappa' \log\lambda \to g$ it gives $Q \to e^{-2g/(\alpha-1)}$ and $a \to 0$, so the limiting offset

is the branch of $\operatorname{atan2}(\sin t, Q\cos t)$: the balanced limit and the high-energy limit do not commute. A fixed $t \neq 0, \pi/2$ eventually leaves the transition region where $|\tan t|$ is comparable to Q, since $Q \to 0$ or $Q \to \infty$, but a t that depends on the energy need not. Uniformity up to $\nu = 0$ or $\nu = 1$ is not claimed.

Figure 2 shows both laws on computed eigenvalues. On a balanced chain each extension keeps its own offset t. On a dimerised chain with $\kappa' > 0$ the offsets of H_0 and $H_{\pi/2}$ move to θ_0 and $\pi/2 - \theta_0$, and the generic extension $t = 1$ joins the law of $H_{\pi/2}$.

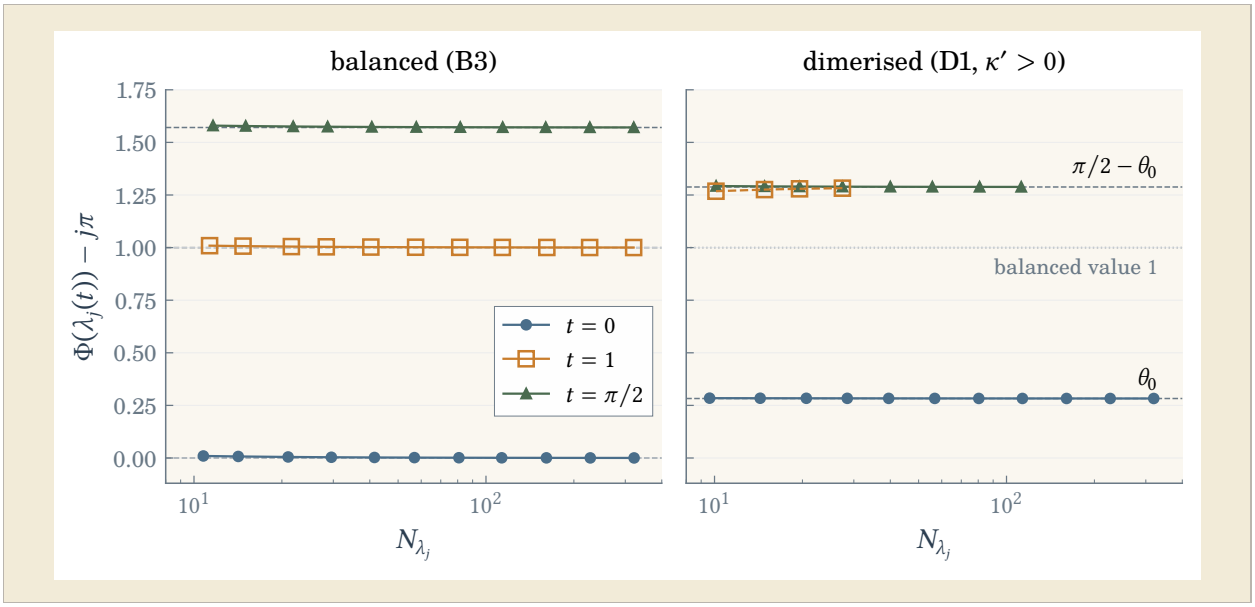


FIG. 2. *The phase $\Phi(\lambda_j(t)) - j\pi$ of computed eigenvalues against N_{λ_j}, for three extensions of the balanced chain B3 (left) and of the dimerised chain D1 with $\kappa' = 0.45$, $\nu = 0.32$ (right), from the data of Section 12. Thin dashed horizontal lines are the predicted offsets. On the balanced chain the three extensions keep the offsets 0, 1 and $\pi/2$. On the dimerised chain H_0 and $H_{\pi/2}$ move to θ_0 and $\pi/2 - \theta_0$, and the generic extension $t = 1$ (hollow squares, dashed) leaves the balanced value 1 (dotted) and lies on the law of $H_{\pi/2}$ (filled triangles), as Corollary 6.3(ii) predicts.*

Theorem 6.4 — Small diagonal

Assume Hypothesis 2.1, $0 < \nu < 1$ and $|b_n| \le C_b(1+n)^\beta$ with $\beta < \alpha - 1 - 2|\kappa'|$. Then J is in the limit-circle case and Theorem 6.2 holds verbatim for $\lambda \to +\infty$; for $\lambda \to -\infty$ it holds for $J_0 - \operatorname{diag}(b)$ at $|\lambda|$. Consequently the offsets of Corollary 6.3 hold modulo π, with labels shifted by an integer that depends on the chain.

The diagonal costs $N_\lambda^{1+\beta-\alpha}$ in the near field and is summable against the weights of (B4) in the far field; the proof is in Appendix A. Theorem 6.4 is tested numerically in Section 12 on one chain.

Let $\lambda_n = \prod_{j=1}^{p}(pn+e_j)$ $(n \ge 0)$ and $\mu_n = \prod_{j=1}^{p}(pn+d_j)$ $(n \ge 1)$, $\mu_0 = 0$, with positive rates, $p \ge 3$, $\mathfrak{e} := \sum e_j$, $\mathfrak{d} := \sum d_j$. The Stieltjes moment problem is indeterminate if and only if $p < \mathfrak{e} - \mathfrak{d} < p(p-1)$ [38, 6]. The associated symmetric Hamburger problem [13], [6, Prop. 3.2] has the zero-diagonal links

$$a_{2n+1} = \sqrt{\lambda_n}, \qquad a_{2n+2} = \sqrt{\mu_{n+1}},$$

and expanding the square roots shows that Hypothesis 2.1 holds with $\delta = 1$, $\alpha = p/2$, $c = (p/2)^{p/2}$, and

$$\kappa = \frac{\mathfrak{d}+\mathfrak{e}}{2p} - \frac{p}{4}, \qquad \kappa' = \frac{\mathfrak{d}-\mathfrak{e}}{2p} + \frac{p}{4}, \qquad \nu = \frac{(\mathfrak{e}-\mathfrak{d})/p - 1}{p-2}.$$

Indeed, with $N = 2n+1$ the odd link is $\prod_j (pn+e_j)^{1/2} = (p/2)^{p/2} N^{p/2} \prod_j \big(1 + (2e_j/p - 1)/N\big)^{1/2}$, so the odd links see $\kappa - \kappa' = \mathfrak{e}/p - p/2$; with $N = 2n+2$ the even link is $(p/2)^{p/2} N^{p/2} \prod_j (1 + 2d_j/(pN))^{1/2}$, so the even links see $\kappa + \kappa' = \mathfrak{d}/p$. The limit-circle condition $2|\kappa'| < \alpha - 1$ is exactly Valent's indeterminacy range, and Proposition 2.1(b) evaluates explicitly to $\varpi = p^{p/2} 2^{2\kappa'} \prod_j \Gamma(1 + d_j/p)/\Gamma(e_j/p)$, since $P_{2m}^2 = \prod_{i \le m} \lambda_{i-1}/\mu_i = \prod_j \Gamma(m+e_j/p)\Gamma(1+d_j/p)/\big(\Gamma(e_j/p)\Gamma(m+1+d_j/p)\big)$ and $\Gamma(m+a)/\Gamma(m+b) \sim m^{a-b}$. Balanced rates are those with $\mathfrak{e} - \mathfrak{d} = p^2/2$; this is the only case covered by Corollary 5.3.

Symmetric extensions and N-extremal solutions. The Stieltjes problem in the variable $x = \lambda^2$ has its own one-parameter family of N-extremal solutions, and only two of them are images of extensions H_t: for $b \equiv 0$, $\Sigma_3 H_t \Sigma_3 = -H_{\pi - t}$ with $\Sigma_3 = \mathrm{diag}((-1)^k)$, so only H_0 and $H_{\pi/2}$ have spectral measures symmetric under $\lambda \mapsto -\lambda$, and their images are the Krein solution μ_K and the Friedrichs solution μ_F. The other N-extremal solutions of the Stieltjes problem are not images of the H_t with $t \ne 0, \pi/2$. In Valent's modified parametrisation [39, eq. (54)] they are the zeros of $\mu(B - D/\alpha_V) - D$ with a real constant μ, which in the coordinates of Theorem 2.2 reads $\tan\psi(\lambda) = \mu'/\lambda$: the boundary angle tends to 0 like $1/\lambda$ instead of being fixed. That is why part (c) below gives the Krein law for every admissible ν, while Corollary 6.3(ii) gives the law of $H_{\pi/2}$, the Friedrichs law, for the generic H_t when $\kappa' > 0$. There is no contradiction: the equation of part (c) reads $Q(\lambda)q(\vartheta) = \mu'/\lambda$, and $\lambda Q(\lambda) \asymp \lambda^{2\nu} \to \infty$ for both signs of κ', so its right side is small compared with Q and the solutions sit near the zeros of the numerator, the Krein side.

Corollary 6.5. In this setting, with x the Stieltjes variable and $\tau_p := \int_0^1 (1-u^p)^{-2/p} du = 1/p\, \mathrm{B}(1/p, 1-2/p)$:

(a) the Krein spectrum (the support of the Krein solution μ_K, the zeros of the Nevanlinna entry D) satisfies $\tau_p \sin(\pi/p)\, x_k^{1/p} = \pi\big(k + \frac{\mathfrak{d}+\mathfrak{e}}{2p^2} - \nu/2\big) + o(1)$, $k = 0, 1, 2, \dots$;

(b) the Friedrichs spectrum (the support of the Friedrichs solution μ_F) satisfies the same with $+\nu/2$ in place of $-\nu/2$;

(c) every other N-extremal measure has its large mass points asymptotic to the Krein law, for every admissible ν.

Proof. As explained before the corollary, H_0 and $H_{\pi/2}$ are the extensions with symmetric spectral measures, and their images under $x = \lambda^2$ are the two positively supported N-extremal solutions at the ends of Valent's interval [13, 30, 39]. H_0 has the eigenvalue 0 (eigenvector P), so its image is the Krein solution μ_K, whose support contains 0. $H_{\pi/2}$ is the limit of the even truncations, whose eigenvalues squared are the zeros of the Stieltjes polynomials, and by the Markov-type theorem of [39, Prop. 5] its image is the Friedrichs solution μ_F (see also [30]). With $\lambda = \sqrt{x}$, $N_\lambda = (2/p)(x/4)^{1/p}$ and $(2/p)4^{-1/p}I_{p/2} = \tau_p \sin(\pi/p)$ (both sides are Beta values, and the identity follows from the duplication and reflection formulas for Γ), so $\Phi = \tau_p \sin(\pi/p)x^{1/p} - \kappa\pi/p$. With $\theta_0 = (1/2 - \nu)\pi/2$ and $\kappa\pi/p = \pi\big(\frac{\mathfrak{d}+\mathfrak{e}}{2p^2} - 1/4\big)$, $j\pi + \theta_0 + \kappa\pi/p = \pi\big(j + \frac{\mathfrak{d}+\mathfrak{e}}{2p^2} - \nu/2\big)$. This and Corollary 6.3(i) give (a) and (b). For (c), the entries $D(\lambda^2)$ and $\lambda\beta_\infty(\lambda)$ are even entire functions of order < 1 in λ; both vanish to second order at $\lambda = 0$ (the eigenvalue 0 of H_0 is simple and is a simple zero of D in x) and otherwise have the same simple zeros, hence they are proportional; likewise $B - D/\alpha_V$ (Valent's modified entry, α_V his parameter) and α_∞, which do not vanish at 0. In Valent's modified parametrisation [39, eq. (54)] the other N-extremal spectra are the zeros of $\mu(B - D/\alpha_V) - D$, that is $\tan\psi = \mu'/\lambda$. By Theorem 6.2 this reads $Q(\lambda)q(\vartheta) = \mu'/\lambda$ with $q(\vartheta)$ the ratio of the two sines, and since $\lambda Q(\lambda) \asymp \lambda^{2\nu} \to \infty$ for every $\nu \in (0,1)$, the solutions approach $\Phi \equiv \theta_0$. □

Corollary 6.5 reproduces every closed form we know. For the Berg–Valent quartic rates ($\nu = 1/2$, $(\mathfrak{d}+\mathfrak{e})/(2p^2) = 1/4$) it is Example 5.1. For Valent's cubic rates $\lambda_n = (3n+1)(3n+2)^2$, $\mu_n = (3n)^2(3n+1)$ [38, Prop. 1] it gives the offsets $1/6$ (Krein) and $1/2$ (Friedrichs), which are the zeros of $\sigma_2(\theta_3(1)z^{1/3})$ and $\sigma_0(\theta_3(1)z^{1/3})$ in his closed form. For the cubic families P1 and P2 of Gilewicz, Leopold, Ruffing and Valent [24] ($\nu = 2/3$ and $1/3$) it gives the Krein offsets $c+5/6$ (P1, with the first index shifted by one) and $c+1/6$ (P2), and the Friedrichs offset $c + 1/2$ for both. That the leading coefficient does not depend on c, which [24, Remark 3] asked, also follows from their Proposition 13 by a Tauberian argument; the offsets are what the general theorem adds. They can be checked directly from the integral representations in [24, Props. 8 and 10]: by Watson's lemma at $u = 0$, $\int_0^1 u^{3c-1}(1-u^3)^{-2/3}\sigma_l(\zeta\hat\theta(u))du = 2/3\,\Gamma(3c)\zeta^{-3c}e^{\zeta\theta_*/2}\big[\cos(\frac{\sqrt3}{2}\theta_*\zeta - \phi_l - c\pi) + O(\zeta^{-3})\big]$ with $\phi_0 = 0$, $\phi_1 = \pi/3$, $\phi_2 = 2\pi/3$, where θ_* is the constant written θ_0 in [24] (unrelated to our θ_0).

7 A DISCRETE OLVER BOUND

The first-order law needed only a qualitative Airy limit on a window of fixed size K (Lemma 4.2). Rates, the second-order term and the application to singular limits in Section 10 need more: a comparison of the Jacobi recurrence with Airy functions on a window of macroscopic size around the turning point, with an error bound that is uniform across the turning point. This section proves such a bound, Theorem 7.1, by a discrete variation of parameters modelled on Olver's error bounds for the Liouville–Green–Airy approximation [29, Ch. 11]. It is an exact inequality for an arbitrary three-term recurrence and an arbitrary pair of comparison sequences of Airy type; all the analysis of a particular chain goes into the size of one sum V, which Lemma 7.4 estimates for the Langer comparison functions of a power-like link function.

Discrete analogues of Olver's theorem exist [22, 21], and the uniform bounds of Fischer, Burgarth and Lonigro [17] rest on them. The error analysis there weights the error by an auxiliary function built from the Hankel function $H^{(1)}_{1/3}$: $\tilde{w}_1$ in [17, Def. A.2], who take it from [22, eq. (4.4)], and $\tilde{u}_j$, $\tilde{w}^{(j)}$ in [21, (2.22), (3.51)]. On the real line this function is $\mathrm{Ai} - i\,\mathrm{Bi}$ for negative argument and $-2i\,\mathrm{Ai}$ for positive argument [17, (A.14)–(A.16)], so it is not a solution of the Airy equation on any interval that contains the turning point. The error analysis nevertheless treats it as one: [17, (B.43)–(B.44)] bound its residual as that of an Airy solution, citing [22, p. 119], and the proof in [21, App. A, near (A.43)] uses its Wronskian with Ai_1. No Airy solution can replace it, since an Airy solution that is recessive for positive argument is a multiple of Ai, which vanishes for negative argument. We have read [21] and [17] at these places but not [22] itself. Theorem 7.1 avoids the auxiliary function: its kernel is built from the two real Airy functions, and its only weights are Olver's envelopes. Monotone envelopes are what make this possible, as explained after the statement.

7.1 ENVELOPES AND THE THEOREM

Let $c_* \approx -0.366046$ be the largest root of $\mathrm{Ai} = \mathrm{Bi}$. Olver's envelopes [29, Ch. 11, §2] are

$$\begin{aligned} \mathsf{E}(x) &:= \sqrt{\mathrm{Bi}(x)/\,\mathrm{Ai}(x)}, & \mathsf{M}(x) &:= \sqrt{2\,\mathrm{Ai}(x)\,\mathrm{Bi}(x)} && (x \ge c_*), \\ \mathsf{E}(x) &:= 1, & \mathsf{M}(x) &:= \sqrt{\mathrm{Ai}(x)^2 + \mathrm{Bi}(x)^2} && (x \le c_*). \end{aligned}$$

We use four elementary facts about them.

(A1) E is continuous and nondecreasing, since $(\mathrm{Bi}\,/\,\mathrm{Ai})' = 1/(\pi\,\mathrm{Ai}^2) > 0$.

(A2) $|\,\mathrm{Ai}\,| \le \mathsf{E}^{-1}\mathsf{M}$ and $|\,\mathrm{Bi}\,| \le \mathsf{E}\mathsf{M}$; for $x \ge c_*$ the right sides are $\sqrt{2}\,\mathrm{Ai}$ and $\sqrt{2}\,\mathrm{Bi}$.

(A3) $\mathsf{m}_0 := \sup_x \mathsf{M}(x)^2(1+|x|)^{1/2} < \infty$, because $\mathsf{M}(x)^2|x|^{1/2} \to 1/\pi$ as $x \to \pm\infty$ (DLMF 9.7.5, 9.7.7, 9.8.20).

(A4) With $\mathsf{k}(x) := \max\big(|\operatorname{Ai}'(x)|/(\mathsf{E}^{-1}\mathsf{M})(x),\, |\operatorname{Bi}'(x)|/(\mathsf{E}\mathsf{M})(x)\big)$, $\mathsf{k}_0 := \sup_x \mathsf{k}(x)(1+|x|)^{-1/2} < \infty$, by the same asymptotics (DLMF 9.7.6, 9.7.8, 9.8.21).

Numerically $\mathsf{m}_0 = 0.48483$ (at $x = {}^1\!/_2$) and $\mathsf{k}(x)(1+|x|)^{-1/2} \uparrow 1$ as $x \to -\infty$ (Appendix B); only finiteness is used.

Consider a three-term operator on the sites $n_0 - 1 \le n \le n_1 + 1$,

$$(\mathscr{L} f)(n) := \mathsf{a}_{n+1} f(n+1) + \mathsf{a}_n f(n-1) - \mathsf{d}_n f(n), \qquad \mathsf{a}_n > 0, \quad \mathsf{d}_n \in \mathbb{C}, \tag{22}$$

together with real numbers x_n, strictly decreasing in n, and $g_n > 0$. Put $\psi_A(n) := g_n \operatorname{Ai}(x_n)$, $\psi_B(n) := g_n \operatorname{Bi}(x_n)$ and $W_n := \mathsf{a}_{n+1}\big(\psi_A(n)\psi_B(n+1) - \psi_A(n+1)\psi_B(n)\big)$, and assume $W_n \neq 0$. Since the residual $\mathscr{L}(g\chi(x))(n)$ is linear in the Airy solution χ and $W_n \neq 0$, there are unique numbers $\mathfrak{p}_n, \mathfrak{q}_n \in \mathbb{C}$ with

$$\mathscr{L}\big(g\,\chi(x)\big)(n) = \chi(x_n)\,\mathfrak{p}_n + \chi'(x_n)\,\mathfrak{q}_n \qquad \text{for every solution } \chi \text{ of } \chi'' = x\chi. \tag{23}$$

With $\mathsf{E}_n := \mathsf{E}(x_n)$, $\mathsf{M}_n := \mathsf{M}(x_n)$ and $\mathsf{k}_n := \mathsf{k}(x_n)$ put

$$\nu_n := \frac{(|\mathfrak{p}_n| + \mathsf{k}_n|\mathfrak{q}_n|)\, g_n \mathsf{M}_n^2}{|W_n|}, \qquad V := \sum_{n=n_0}^{n_1} \nu_n, \qquad \theta_D := \max_{n_0 \le n \le n_1} \frac{g_n|\mathfrak{q}_n|}{\pi |W_n|}.$$

The kernel of the comparison below contains the factors $\mathsf{E}_n^2/\mathsf{E}_j^2$ between the sites n and j of a summation. Olver's choice of directions (the recessive function is started at the end where it is smallest and summed towards decreasing x, the dominant one from the other end) together with the monotonicity of E makes every such factor at most one. That is the only property of the weights the proof uses, and it is the reason no auxiliary Airy-type function is needed.

Theorem 7.1 — Discrete Olver bound

In this setting, let f_A be the solution of $\mathscr{L} f = 0$ with $f_A = \psi_A$ at $n_0 - 1$ and n_0, and, if $\theta_D < 1$, let f_B be the solution with $f_B = \psi_B$ at n_1 and $n_1 + 1$. Then for $n_0 - 1 \le n \le n_1 + 1$

$$|f_A(n) - \psi_A(n)| \le \big(e^{2V} - 1\big)\, g_n \mathsf{E}_n^{-1} \mathsf{M}_n, \qquad |f_B(n) - \psi_B(n)| \le \big(e^{2V/(1-\theta_D)} - 1\big)\, g_n \mathsf{E}_n \mathsf{M}_n.$$

Proof. **Step 1** (*variation of parameters*)**.** Let $F(n) := (f(n), f(n+1))^{\mathsf{T}}$ and let $\Psi(n)$ be the matrix with columns $(\psi_A(n), \psi_A(n+1))^{\mathsf{T}}$ and $(\psi_B(n), \psi_B(n+1))^{\mathsf{T}}$, so $\det \Psi(n) = W_n/\mathsf{a}_{n+1}$. Put $\beta_i(n) := (\mathscr{L}\psi_i)(n)$. If T_n is the transfer matrix of $\mathscr{L} f = 0$ from $F(n-1)$ to $F(n)$, then $\Psi(n) = T_n \Psi(n-1) + e_2(\beta_A(n), \beta_B(n))/\mathsf{a}_{n+1}$. Writing a solution as $F(n) = \Psi(n)c(n)$ and inverting $\Psi(n)$ gives the exact recursion

$$c(n) = c(n-1) + \frac{\sigma(n)}{W_n} \begin{pmatrix} \psi_B(n) \\ -\psi_A(n) \end{pmatrix}, \qquad \sigma(n) := \beta_A(n)c_1(n-1) + \beta_B(n)c_2(n-1), \tag{24}$$

and $f(n) = \psi_A(n)c_1(n-1) + \psi_B(n)c_2(n-1) = \psi_A(n)c_1(n) + \psi_B(n)c_2(n)$, the increment in (24) being orthogonal to $(\psi_A(n), \psi_B(n))$.

Step 2 (*the residual of a moving Airy function*)**.** $\sigma(n)$ is the residual (23) of $\chi = c_1(n-1)\,\mathrm{Ai} + c_2(n-1)\,\mathrm{Bi}$. By (A2) and the definition of k, $|\chi(x)| \le \mathsf{E}^{-1}\mathsf{M}(|c_1| + \mathsf{E}^2|c_2|)$ and $|\chi'(x)| \le \mathsf{k}\,\mathsf{E}^{-1}\mathsf{M}(|c_1| + \mathsf{E}^2|c_2|)$, so

$$|\sigma(n)| \le (|\mathfrak{p}_n| + \mathsf{k}_n|\mathfrak{q}_n|)\,\mathsf{E}_n^{-1}\mathsf{M}_n\,\Lambda(n), \qquad \Lambda(n) := |c_1(n-1)| + \mathsf{E}_n^2|c_2(n-1)|.$$

Step 3 (*forward growth*)**.** By (24) and (A2), $|c_1(n) - c_1(n-1)| \le \nu_n\Lambda(n)$ and $\mathsf{E}_n^2|c_2(n) - c_2(n-1)| \le \nu_n\Lambda(n)$. Since $x_{n+1} < x_n$, (A1) gives $\mathsf{E}_{n+1} \le \mathsf{E}_n$, hence $\Lambda(n+1) \le (1+2\nu_n)\Lambda(n)$. For f_A, $c(n_0-1) = (1,0)$, so $\Lambda(n_0) = 1$ and $\Lambda(n) \le \prod_{j<n}(1+2\nu_j)$.

Step 4 (*the error of* f_A)**.** For $n \ge n_0$, $f_A(n) - \psi_A(n) = \psi_A(n)(c_1(n-1)-1) + \psi_B(n)c_2(n-1)$, and both brackets are sums of increments over $n_0 \le j < n$. The first term is at most $g_n\mathsf{E}_n^{-1}\mathsf{M}_n\sum_j \nu_j\Lambda(j)$. In the second, $|\psi_B(n)|\,|c_2(j) - c_2(j-1)| \le g_n\mathsf{E}_n\mathsf{M}_n\mathsf{E}_j^{-2}\nu_j\Lambda(j) \le g_n\mathsf{E}_n^{-1}\mathsf{M}_n\nu_j\Lambda(j)$, because $x_j > x_n$ gives $\mathsf{E}_j \ge \mathsf{E}_n$. Summing, $|f_A(n) - \psi_A(n)| \le g_n\mathsf{E}_n^{-1}\mathsf{M}_n\sum_j 2\nu_j\Lambda(j) \le g_n\mathsf{E}_n^{-1}\mathsf{M}_n\big(\prod_j(1+2\nu_j) - 1\big)$.

Step 5 (*backward, for* f_B)**.** Write (24) as $c(n) = (I + u_n w_n^{\mathsf{T}})c(n-1)$ with $u_n = (\psi_B(n), -\psi_A(n))^{\mathsf{T}}/W_n$ and $w_n = (\beta_A(n), \beta_B(n))^{\mathsf{T}}$. From (23) and the Wronskian $\mathrm{Ai}\,\mathrm{Bi}' - \mathrm{Ai}'\,\mathrm{Bi} = 1/\pi$, $\beta_A\psi_B - \beta_B\psi_A = -g_n\mathfrak{q}_n/\pi$, so $|w_n^{\mathsf{T}}u_n| \le \theta_D < 1$ and $c(n-1) = c(n) - u_n\,w_n^{\mathsf{T}}c(n)/(1 + w_n^{\mathsf{T}}u_n)$. Now $w_n^{\mathsf{T}}c(n)$ is the residual of $c_1(n)\,\mathrm{Ai} + c_2(n)\,\mathrm{Bi}$, bounded as in the second step by $(|\mathfrak{p}_n| + \mathsf{k}_n|\mathfrak{q}_n|)\mathsf{E}_n\mathsf{M}_n\Lambda_2(n)$ with $\Lambda_2(n) := \mathsf{E}_n^{-2}|c_1(n)| + |c_2(n)|$. Hence the two increments satisfy $\mathsf{E}_n^{-2}|\Delta c_1|, |\Delta c_2| \le \nu_n\Lambda_2(n)/(1-\theta_D)$, and since $\mathsf{E}_{n-1} \ge \mathsf{E}_n$, $\Lambda_2(n-1) \le (1 + 2\nu_n/(1-\theta_D))\Lambda_2(n)$, with $\Lambda_2(n_1) = 1$. The error $f_B(n) - \psi_B(n) = \psi_A(n)c_1(n) + \psi_B(n)(c_2(n) - 1)$ is a sum over $n < j \le n_1$; now $\mathsf{E}_j \le \mathsf{E}_n$, so each increment of c_1 contributes at most $g_n\mathsf{E}_n\mathsf{M}_n\nu_j\Lambda_2(j)/(1-\theta_D)$, and the fourth step's summation gives the bound. □

The theorem is an exact inequality: no expansion in a small parameter enters it. The coefficients d_n enter only through the residuals, which is why they may be complex. Two consequences are used repeatedly. Put $d := e^{2V/(1-\theta_D)} - 1$. Then:

(D1) where $x_n \ge c_*$, $f_A(n) = \psi_A(n)(1 + O_{\le\sqrt{2}d})$ and $f_B(n) = \psi_B(n)(1 + O_{\le\sqrt{2}d})$; where $x_n \le c_*$, $|f_A(n) - \psi_A(n)|, |f_B(n) - \psi_B(n)| \le d\,g_n\mathsf{M}_n$;

(D2) the coefficient vectors satisfy $|c_1^{(A)}(n) - 1| \le d$, $|c_2^{(A)}(n)| \le \mathsf{E}_n^{-2}d$, $|c_2^{(B)}(n) - 1| \le d$, $|c_1^{(B)}(n)| \le \mathsf{E}_n^2 d$, so the Casoratian of f_A and f_B is $W_n(1+\omega)$ with $|\omega| \le 2d + 2d^2$.

Here $O_{\le c}$ denotes a quantity of modulus at most c; (D2) is read off from the third and fifth steps of the proof.

7.2 LANGER COMPARISON FUNCTIONS

We now construct x_n and g_n for a recurrence whose coefficients are samples of a smooth function, and estimate V. Let $h \in (0, h_0]$ be a small parameter, T a compact interval, and $a(t,h) > 0$ a family of functions with

$$a(\cdot,h) \to a_0 \text{ in } C^4(T)\ (h \to 0), \qquad a_0' > 0 \text{ on } T, \qquad 2a_0(t_p^0) = 1 \text{ for some interior } t_p^0. \tag{25}$$

The comparison operator is $(\mathscr{L}_a f)(n) := a((n+1)h)f(n+1) + a(nh)f(n-1) - f(n)$. Put $\hat{a}(t) := a(t + h/2, h)$ and $Z := 1/(2\hat{a})$, let t_p be the root of $Z = 1$, and define the Langer variable ζ and the amplitude g by

$$\tfrac{2}{3}\zeta^{3/2} := \int_t^{t_p} \operatorname{arccosh} Z \quad (t \le t_p), \qquad \tfrac{2}{3}(-\zeta)^{3/2} := \int_{t_p}^{t} \arccos Z \quad (t \ge t_p),$$

$$g := \Big(\frac{\zeta}{\hat{a}^2(Z^2-1)}\Big)^{1/4},$$

and $x_n := h^{-2/3}\zeta(nh)$ [21, (2.2)–(2.3)].

Lemma 7.2 (Langer data). Assume (25), and let $T' \subset T$ be a compact interval containing t_p^0 in its interior. For $h \le h_0$: t_p is unique in T' and $t_p \to t_p^0$; ζ and g are C^3 on T' with norms bounded uniformly in h; $\zeta' < 0$, $g_* \le g \le G_*$ and $|\zeta(t)| \ge r_0|t - t_p|$ on T' for positive constants. The *eikonal identity* $\cosh(\sqrt{\zeta}\,\zeta') = Z$ and the *amplitude identity* $\hat{a}\,g^2 \sinh(\sqrt{\zeta}\,\zeta')/\sqrt{\zeta} = -1$ hold on T', read with cos and sin for $t > t_p$. If a is real-analytic in (t,h) near $T' \times \{0\}$, so are ζ and g.

The proof follows the $h = 0$ data through a chain of continuous operations (Appendix A).

Lemma 7.3 (Local residuals). Under (25), the coefficients (23) of $\mathscr{L}_a$ with these x_n and $g_n := g(nh)$ satisfy, for $nh \in T'$,

$$|\mathfrak{p}_n| \le C_R h^2, \qquad |\mathfrak{q}_n| \le C_R h^{1/3} h^2, \qquad W_n = -\frac{h^{1/3}}{\pi}\big(1 + O(h)\big).$$

If a is analytic, $|\mathfrak{q}_n| \le C_R h^{1/3} h^3$ and $W_n = -(h^{1/3}/\pi)(1 + O(h^2))$.

Proof. **Step 1** (*the stencil*)**.** Fix $t = nh$, put $s_\pm := (\zeta(t \pm h) - \zeta(t))/h$ and $Y(s) := \chi(x_n + h^{1/3}s)$. Then $Y'' = (\zeta(t) + hs)Y$, so $\chi(x_{n\pm1}) = \chi(x_n)X_1(s_\pm) + h^{1/3}\chi'(x_n)X_2(s_\pm)$, where X_1, X_2 solve $Y'' = (\zeta(t) + hs)Y$ with $(X_1, X_1')(0) = (1,0)$ and $(X_2, X_2')(0) = (0,1)$. Since $a((n \pm 1)h)$ and $a(nh)$ are $\hat{a}(t \pm h/2)$, this gives

$$\mathfrak{p}_n = \hat{a}(t + \tfrac{h}{2})g(t+h)X_1(s_+) + \hat{a}(t - \tfrac{h}{2})g(t-h)X_1(s_-) - g(t),$$
$$\mathfrak{q}_n = h^{1/3}\big[\hat{a}(t + \tfrac{h}{2})g(t+h)X_2(s_+) + \hat{a}(t - \tfrac{h}{2})g(t-h)X_2(s_-)\big].$$

Step 2 (*freezing and parity*)**.** Keep $\zeta, \hat{a}, g$ fixed and replace the step h by a free variable η in all four places (the stencil points, $s_\pm$, the half-shifts, and the equation $Y'' = (\zeta + \eta s)Y$);

call the results $\mathscr{P}(t,\eta)$ and $h^{1/3}\mathscr{Q}(t,\eta)$. Since $\zeta \in C^3$ and $g, \hat{a} \in C^2$, both are C^2 in η with uniform bounds. The equation is invariant under $(s,\eta) \mapsto (-s,-\eta)$, so $X_1(-s;-\eta) = X_1(s;\eta)$ and $X_2(-s;-\eta) = -X_2(s;\eta)$; and $\eta \mapsto -\eta$ exchanges the two stencil points, with $s_\pm(-\eta) = -s_\mp(\eta)$. Hence $\mathscr{P}$ is even and $\mathscr{Q}$ is odd in η.

Step 3 (*two identities*)**.** At $\eta = 0$, $s_\pm = \pm\zeta'$ and $X_1(s) = \cosh(\sqrt{\zeta}s)$, so $\mathscr{P}(t,0) = g(2\hat{a}\cosh(\sqrt{\zeta}\zeta') - 1) = 0$ by the eikonal identity. The coefficient $\partial_\eta \mathscr{Q}(t,0)$ vanishes by the amplitude identity and its derivative; this is the discrete transport equation, derived in Section A.6. By Taylor's theorem, $\mathscr{P} = O(\eta^2)$ and $\mathscr{Q} = O(\eta^2)$; take $\eta = h$.

Step 4 (*the Casoratian*)**.** By the stencil form and $\mathrm{Ai}\,\mathrm{Bi}' - \mathrm{Ai}'\,\mathrm{Bi} = 1/\pi$,

$$W_n = \hat{a}(t + \tfrac{h}{2})\, g(t)\, g(t+h)\, X_2(s_+)\, h^{1/3}/\pi.$$

At $\eta = 0$ the factor $\hat{a}g^2 X_2(\zeta')$ equals $\hat{a}g^2 \sinh(\sqrt{\zeta}\zeta')/\sqrt{\zeta} = -1$. In the analytic case Cauchy estimates replace Taylor's theorem: $\mathscr{Q}$, odd with vanishing linear term, is $O(\eta^3)$, and the Casoratian factor expanded about the midpoint $t + h/2$ is even in η. □

A perturbed operator

$$(\mathscr{L} f)(n) = a((n+1)h)(1+e_{n+1})f(n+1) + a(nh)(1+e_n)f(n-1) - (1-d_n)f(n), \tag{26}$$

with $e_n > -1$ and $d_n \in \mathbb{C}$, has the Casoratian $W_n(1+e_{n+1})$ for the same pair, and coefficients $\mathfrak{p}_n^{ch}, \mathfrak{q}_n^{ch}$ with

$$|\mathfrak{p}_n^{ch} - \mathfrak{p}_n| + h^{-1/3}|\mathfrak{q}_n^{ch} - \mathfrak{q}_n| \le C\eta_n, \qquad \eta_n := |e_n| + |e_{n+1}| + |d_n|, \tag{27}$$

because $(\mathscr{L} - \mathscr{L}_a)\psi(n) = a((n+1)h)e_{n+1}\psi(n+1) + a(nh)e_n\psi(n-1) + d_n\psi(n)$ and $\psi(n \pm 1)$ expands as in the first step above with bounded coefficients.

Lemma 7.4 (Size of V)**.** Assume (25) and let $[n_0, n_1]$ be a set of sites with $nh \in T'$. For the operator (26) with $\bar{\eta} := \max_{n_0 \le n \le n_1} \eta_n \le h_0$,

$$V \le C\big(h + \bar{\eta}/h\big), \qquad \theta_D \le C\big(h^2 + \bar{\eta}\big).$$

Proof. By Lemma 7.3, (27) and (A4), $|\mathfrak{p}_n^{ch}| + \mathsf{k}_n|\mathfrak{q}_n^{ch}| \le C(h^2 + \eta_n)(1 + h^{1/3}(1+|x_n|)^{1/2})$, and $h^{1/3}(1+|x_n|)^{1/2} \le h^{1/3} + |\zeta(nh)|^{1/2}$ is bounded. With $|W_n^{ch}| \ge h^{1/3}/(4\pi)$, $g \le G_*$ and (A3), $\nu_n \le C(h^2 + \eta_n)h^{-1/3}(1+|x_n|)^{-1/2}$. Since $|x_n| \ge r_0 h^{-2/3}|nh - t_p|$ and the sites are spaced by h, $\sum_n (1+|x_n|)^{-1/2} \le C(1 + h^{-1}\int_0^{|T|}(h^{-2/3}r_0 s)^{-1/2}ds) \le Ch^{-2/3}$. Hence $V \le C(h^2 + \bar{\eta})h^{-1}$. The bound on θ_D is Lemma 7.3 and (27) divided by $|W_n|$. □

For the power-like chains of Section 8, $\bar{\eta} \le C(h^{1+\delta''} + h^{\alpha-\beta})$, and Lemma 7.4 gives $V \le C(h + h^{\delta''} + h^{\alpha-1-\beta})$. For the singular limits of Section 10, $\bar{\eta} \le C(h^2 + h^\alpha)$ and $V \le C(h + h^{\alpha-1})$: there $\alpha > 1$ is exactly the condition that the perturbation is negligible across the Airy window.

8 RATES FOR BALANCED LINKS

The first-order law of Section 5 has no rate, and none is uniform over Hypothesis 2.2 (Remark 5.1). This section identifies the hypothesis that controls the rate, which is closeness of the links to a *smooth* power-like function rather than a condition on differences, and proves the rate with the discrete Olver bound. The phase that converges fast is not the block sum S, which carries the oscillating term of Proposition 8.5, but the integral of the smooth link function.

Hypothesis 8.1 (smooth symbol). $a_n = A(n)(1+e_n)$ for $n \ge k_0$, where $A \in C^4[x_0,\infty)$, $A(x) = c\,x^\alpha(1+r(x))$ with $\alpha > 1$, $c > 0$ and $x^j r^{(j)}(x) \to 0$ for $0 \le j \le 4$, and $|e_n| \le C_e n^{-1-\delta''}$ for some $\delta'' > 0$. The diagonal satisfies $|b_n| \le C_b(1+n)^\beta$ with $\beta < \alpha - 1$. Finitely many entries are arbitrary.

The hypothesis implies $xA'/A \to \alpha$ and $|(\log A)^{(j)}(x)| \le Cx^{-j}$ for $1 \le j \le 4$, so the links are eventually increasing and $|\varepsilon_m| \le C(N_m^{-2} + N_m^{-1-\delta''})$. Proposition 2.1(b)–(d) and Theorem 2.2 therefore hold, with the same proofs and $\delta := \min(1,\delta'')$. Hypothesis 8.1 and Hypothesis 2.2 are not nested: $r(x) = 0.3/\log(x+2)$ satisfies Hypothesis 8.1 but has $|r_{n+1} - r_n| \asymp 1/(n\log^2 n)$, while chains in Hypothesis 2.2 with small δ need not be close to any smooth symbol (Remark 5.3). Let X_λ be the largest root of $2A(X) = \lambda$, so $X_\lambda = N_\lambda(1+o(1))$, and define the *smooth phase*

$$\Phi_A(\lambda) := \int_0^\infty \arcsin\Big(\min\big(1, \tfrac{\lambda}{2A(x)}\big)\Big)dx = \frac{X_\lambda \pi}{2} + \int_{X_\lambda}^\infty \arcsin\frac{\lambda}{2A(x)}\,dx, \tag{28}$$

which does not depend on A below x_0 once λ is large. For $A(x) = c(x+\kappa/\alpha)^\alpha$ it is exactly Φ (Proposition 5.2).

Theorem 8.1 — Rate

Under Hypothesis 8.1, as $\lambda \to +\infty$,

$$|\psi(\lambda) - \Phi_A(\lambda)| \le C\big(N_\lambda^{-1} + N_\lambda^{-\delta''} + N_\lambda^{1+\beta-\alpha}\big),$$

and $|\psi(-\lambda) + \Phi_A(\lambda)|$ satisfies the same bound. The constant depends only on α, c, x_0, the moduli of convergence in Hypothesis 8.1, C_e, δ'', C_b, β and the finitely many initial entries, and is uniform for α in compact subsets of $(1,\infty)$ when these data are.

Idea of the proof. The regular solution is compared with Airy functions on a window of macroscopic size around the turning point, by Theorem 7.1; the phase is read off in the frame of Section 3 at a block where the chain is already oscillatory; and the adiabatic normal form carries it to infinity. These steps determine ψ modulo π. The integer is fixed by deforming the chain to $c\,n^\alpha$, where Theorem 5.1 already knows it; this homotopy is the one point that is not an estimate.

8.1 THE WINDOW AND THE ESTIMATE THAT MATTERS

Define $h = h(\lambda)$ by $A(1/h + 1/2) = \lambda$, so $h \asymp N_\lambda^{-1}$, and put $a(t,h) := A(t/h)/\lambda$. Dividing the recurrence by λ, the regular solution $f(n) := p_n$ solves the operator (26) with e_n the rough part and $d_n := b_n/\lambda$. By Hypothesis 8.1, $a(\cdot,h) \to t^\alpha$ in C^4 on compact subsets of $(0,\infty)$, so (25) holds with $a_0(t) = t^\alpha$ and $t_p^0 = 2^{-1/\alpha}$. The turning point of the Langer construction is $t_p = h(X_\lambda - 1/2)$. We fix

$$t_{in} := t_p^0/2, \qquad t_1 := 1, \qquad t_{fi} := 2, \qquad \delta_w := (t_1 - t_p^0)/3,$$

and the sites $n_0 := \lceil t_{in}/h \rceil - 1$, $n_1 := \lfloor t_{fi}/h \rfloor + 1$. On the window, $n \asymp 1/h$ and $\lambda \asymp h^{-\alpha}$, so $\bar\eta \le C(h^{1+\delta''} + h^{\alpha-\beta})$, and by Lemma 7.4, which is the estimate that sets the rate,

$$d := e^{2V/(1-\theta_D)} - 1 \le C\Gamma, \qquad \Gamma := N_\lambda^{-1} + N_\lambda^{-\delta''} + N_\lambda^{1+\beta-\alpha}. \tag{29}$$

Let f_A, f_B be the solutions of Theorem 7.1 for this operator and window.

The comparison yields three facts, proved in Section A.3. First, for λ large the regular solution is, on the window, a real multiple of $f_A + \eta_K f_B$ with $|\eta_K| \le Ce^{-c/h}$ (Lemma A.5): it enters the window as the recessive Airy solution up to an exponentially small admixture. Second, on the oscillatory part of the window it has the WKB form $p_n = c_K \pi^{-1/2} g_n |x_n|^{-1/4}(\cos(\Xi_n - \pi/4) + O(\Gamma))$ with the exact phase $\Xi_n := \Xi(n + 1/2)$, $\Xi(X) := \int_{X_\lambda}^{X} \arccos(\lambda/2A(x))\,dx$ (Lemma A.6), and read in the block frame this gives $\chi_m \equiv -\Xi(2m) + O(\Gamma) \pmod\pi$ at every block of the window with $u_m \ge 1$ (Lemma A.7). Third, beyond such a block the adiabatic normal form with no drift adds $\int_{2m}^{\infty} \arcsin(\lambda/2A)$ up to $O(\Gamma)$ (Lemma A.8).

8.2 PROOF OF THE THEOREM

Proof of Theorem 8.1. **Step 1** (*modulo* π)**.** Let m^* be the least block with $N_{m^*} \ge X_\lambda$ and $u_{m^*} \ge 1$. Then $A_{m^*} = \lambda(1 + O(h))$, so its sites satisfy $nh = t_1 + o(1) = 1 + o(1)$ and lie in the range of Lemma A.6; and $u_m \ge 1$ for all $m \ge m^*$ because the links increase. Lemmas A.7 and A.8 give

$$\psi \equiv -\Xi(2m^*) + \int_{2m^*}^{\infty} \arcsin\frac{\lambda}{2A} + O(\Gamma) = \Phi_A - m^*\pi + O(\Gamma) \pmod\pi,$$

using $\arccos = \pi/2 - \arcsin$ on $[X_\lambda, 2m^*]$. So $\psi - \Phi_A = j(\lambda)\pi + O(\Gamma)$ with $j(\lambda) \in \mathbb{Z}$, and since ψ and Φ_A are continuous, j is constant on $[\lambda_0,\infty)$.

Step 2 (*the integer*)**.** For $s \in [0,1]$ let $J^{(s)}$ have links $(1-s)a_n + sc\,n^\alpha$ and diagonal $(1-s)b_n$. For $n \ge k_0$ the links are $A_s(n)(1 + e_n^{(s)})$ with $A_s(x) = c\,x^\alpha(1 + (1-s)r(x))$ and $|e_n^{(s)}| \le C|e_n|$, so every $J^{(s)}$ satisfies Hypothesis 8.1 with the same data, and the first step holds uniformly in s: $|\psi_s - \Phi_{A_s} - j_s\pi| \le C\Gamma$. Both $\psi_s(\lambda)$ and $\Phi_{A_s}(\lambda)$ are continuous in s

(for ψ_s: $\varpi^{(s)}$ is continuous because the series defining it converges uniformly, each shear increment is continuous in its weight, and the tails $\sum_{k>K}|\gamma_k^{(s)}|$ are small uniformly in s). Hence j_s is independent of s. At $s=1$ the chain is $c\,n^\alpha$ with $b\equiv 0$, for which Theorem 5.1 and Proposition 5.2(c) give $\psi - N_\lambda I_\alpha \to 0$, while $\Phi_{A_1} = N_\lambda I_\alpha$. So $j=0$. This step uses no estimate beyond the uniformity of the first step in s; it is what turns a statement modulo π into a statement about labelled eigenvalues.

Step 3 (*negative λ and uniformity*)**.** $\psi_J(-\lambda) = -\psi_{J_0-\mathrm{diag}(b)}(\lambda)$ by Theorem 2.2(iii), and $J_0 - \mathrm{diag}(b)$ satisfies the same hypotheses. Every constant above comes from the $h=0$ data t^α (which depend continuously on α and are nondegenerate for $\alpha>0$), from the far-field sums of Lemma A.8 (continuous in α), and from the listed data; so it is uniform on compact α-intervals in $(1,\infty)$. □

Nothing is claimed as $\alpha \to 1^+$: the main term $\Phi_A \asymp X_\lambda/(\alpha-1)$ diverges, and for $b\neq 0$ so does the constant through $\sum_{N\ge X} N^{\beta-\alpha}$.

8.3 CONSEQUENCES AND SHARPNESS

Lemma 8.2 (Comparing phases). If A_1, A_2 satisfy Hypothesis 8.1 and $|A_1/A_2 - 1| \le Cx^{-1-\nu}$ with $\nu>0$, then $\Phi_{A_1} - \Phi_{A_2} = O(N_\lambda^{-\nu})$.

Proof. Φ_A is nonincreasing in A, so it suffices to compare A_2 with $A_2(1\pm w)$, $w = Cx^{-1-\nu}$. Along $A_\vartheta := A_2(1+\vartheta w)$ the moving turning point contributes nothing, because the integrand equals $\pi/2$ there, and $|\partial_\vartheta \Phi_{A_\vartheta}| \le \int_{X_\vartheta}^\infty \nu\, w\,(1-\nu^2)^{-1/2}dx$ with $\nu = \lambda/2A_\vartheta$. On $[X,2X]$, $1-\nu^2 \ge c(x-X)/X$, so this part is $O(X^{-\nu})$; on $[2X,\infty)$, $\nu \le C(X/x)^\alpha$ and the part is $O(X^{-\nu})$. □

Corollary 8.3. Let $a_n = c\,n^\alpha(1+\kappa/n + O(n^{-1-\delta'}))$, with no condition on differences, and $|b_n| \le C_b(1+n)^\beta$, $\beta < \alpha - 1$. Put $\gamma := \min(1,\delta',\alpha-1-\beta)$. Then $\psi(\lambda) - \Phi(\lambda) = O(N_\lambda^{-\gamma})$, and the eigenvalues of every H_t satisfy

$$\lambda_j^{1/\alpha} = (2c)^{1/\alpha}\frac{\pi}{I_\alpha}\Big(j + \frac{t}{\pi} + \frac{\kappa}{2\alpha}\Big) + O(j^{-\gamma}).$$

If $b\equiv 0$, then the remainder $r(\lambda)$ in the count of Corollary 5.3(c) satisfies $|r(\lambda)| \le {}^1\!/_2 + O(\lambda^{-\min(1,\delta')/\alpha})$.

Proof. Take $A(x) := c\,x^\alpha(1+\kappa/x)$ for $x \ge x_0 > |\kappa|+1$ and $e_n := (1+r_n)/(1+\kappa/n) - 1 = O(n^{-1-\delta'})$. Theorem 8.1 applies with $\delta'' = \delta'$, and $A/(c(x+\kappa/\alpha)^\alpha) = 1 + O(x^{-2})$, so Lemma 8.2 and Proposition 5.2 give $\Phi_A - \Phi = O(N_\lambda^{-1})$. The eigenvalue law follows from Theorem 2.2(v) since $N_{\lambda_j} \asymp j$, and the counting statement as in Corollary 5.3(c). □

This covers every link sequence with an expansion in powers of $1/n$, and in particular the Berg–Valent chain and the squeezing chains of Section 10. Of the three terms

in Theorem 8.1, the middle one is attained, by the following proposition. The first is attained on every shifted power, where Theorem 9.1 below identifies its constant. The third is attained numerically, with the constant predicted by first-order perturbation in b (Sections 12 and 13).

Proposition 8.4 — Sharpness of the rough term

Assume Hypothesis 8.1 with $\beta < \alpha - 1 - \delta''$, $0 < \delta'' < 1$, and $e_n = e(n)$ for a C^4 function with $x^j e^{(j)}(x) = O(x^{-1-\delta''})$ $(j \le 4)$ and $x^{1+\delta''} e(x) \to \kappa_e \neq 0$. Then

$$\psi(\lambda) - \Phi_A(\lambda) = -\frac{\kappa_e}{2\alpha}\,\mathrm{B}\Big(\frac{\alpha+\delta''}{2\alpha}, \frac{1}{2}\Big) N_\lambda^{-\delta''} + o(N_\lambda^{-\delta''}).$$

Proof. $A' := A(1+e)$ satisfies Hypothesis 8.1, and the chain is A' with no rough part, so Theorem 8.1 gives $\psi - \Phi_{A'} = O(N^{-1} + N^{1+\beta-\alpha}) = o(N^{-\delta''})$. The derivative in the proof of Lemma 8.2 is an identity: $\Phi_{A'} - \Phi_A = -\int_0^1 \int_{X_\vartheta}^\infty \nu_\vartheta e\,(1+\vartheta e)^{-1}(1-\nu_\vartheta^2)^{-1/2}\,dx\,d\vartheta$ with $\nu_\vartheta = \lambda/(2A(1+\vartheta e))$. Substituting $x = X_\vartheta u$ and using dominated convergence, the inner integral is $\kappa_e X^{-\delta''}\big(\int_1^\infty u^{-\alpha-1-\delta''}(1-u^{-2\alpha})^{-1/2}du + o(1)\big)$, and $w = u^{-2\alpha}$ evaluates the integral as $\mathrm{B}(\frac{\alpha+\delta''}{2\alpha}, 1/2)/(2\alpha)$. □

The block sum S of Theorem 5.1 carries an oscillating term of exact order $N_\lambda^{-1/2}$ from the kink of the summand at the turning point, which the next proposition identifies.

Proposition 8.5 — The midpoint term

Let A satisfy Hypothesis 8.1, let S_A be the sum S with links $A(2m+1)$ and Φ_A the phase (28). Put $X := X_\lambda$, $x_t := (X-1)/2$, $\theta_\lambda := \lceil x_t \rceil - x_t \in (0,1]$ (with $\theta_\lambda = 1$ if $x_t \in \mathbb{Z}$) and $\alpha_X := XA'(X)/A(X)$. Then

$$S_A(\lambda) - \Phi_A(\lambda) = -4\sqrt{\alpha_X/X}\,\zeta(-\tfrac{1}{2}, \theta_\lambda) + O(1/X),$$

with $\zeta(s,\theta)$ the Hurwitz zeta function. Since $\zeta(-1/2,\cdot)$ ranges over $[-0.2078862\ldots, 0.0933663\ldots]$ on $(0,1]$ and θ_λ sweeps $(0,1]$ infinitely often, $\limsup_\lambda \sqrt{N_\lambda}\,|S_A - \Phi_A| = 4\sqrt{\alpha}\,|\zeta(-1/2)| > 0$.

Proof. With $g(x) := 2\arcsin(\min(1, \lambda/2A(2x+1)))$ we have $\Phi_A = \int_{-1/2}^\infty g$, so $S_A - \Phi_A = \sum_m \big(g(m) - \int_{m-1/2}^{m+1/2} g\big)$, and only $m + 1/2 > x_t$ contributes. Near the kink, with $u := x - x_t \ge 0$, Hypothesis 8.1 gives $\log A(X+2u) - \log A(X) = 2\alpha_X u/X + R(u)$ with $|R^{(j)}(u)| \le Cu^{2-j}/X^2$, and $2\arcsin\nu = \pi - 2\sqrt{2(1-\nu)}(1 + O(1-\nu))$ gives $g = \pi - c_X u^{1/2}(1+q(u))$ with $c_X := 4\sqrt{\alpha_X/X}$ and $|q^{(j)}(u)| \le Cu^{1-j}/X$ $(j \le 2)$ on $0 \le u \le c'X$. For the square-root model, the sites $m_0 + k$ have $u = k + \theta_\lambda$ and the integrals cover $u \in [0, K + \theta_\lambda + 1/2]$, so the model contributes $-c_X\big[\sum_{k=0}^K (k+\theta)^{1/2} - \tfrac{2}{3}(K+\theta+1/2)^{3/2}\big] = -c_X\big[\zeta(-1/2,\theta) + O(K^{-1/2})\big]$, by the midpoint Euler–Maclaurin form of the Hurwitz function (DLMF §25.11). Taking $K =$

$\lfloor c'X \rfloor$, the correction $c_X u^{1/2} q$ and the region $u > c'X$ contribute $O(1/X)$ by the midpoint rule. The range of $\zeta(-1/2, \cdot)$ on $(0, 1]$ follows from $\partial_\theta \zeta(-1/2, \theta) = 1/2 \zeta(1/2, \theta)$ (DLMF 25.11.17), which has one zero in $(0, 1)$; the endpoint value is $\zeta(-1/2) = -0.2078862\ldots$ and the maximum is $0.0933663\ldots$, evaluated at that zero (Appendix B). □

So the discrete sum converges to ψ only at the rate $N_\lambda^{-1/2}$, while the smooth phase does much better (Theorem 8.1). The rule of Theorem 5.1 is the right statement under Hypothesis 2.2, where no smooth phase need exist; for rates one should use Φ_A.

Remark 8.1 (The block sum). By Proposition 8.5 and Theorem 8.1, if $\delta'' > 1/2$ and $\beta < \alpha - 3/2$, then $\limsup_{\lambda\to\infty} \sqrt{N_\lambda}\,|\psi(\lambda) - S(\lambda)| = 4\sqrt{\alpha}\,|\zeta(-1/2)|$, using $|S - S_A| = O(N^{-\delta''} + N^{-(1+\delta'')/2})$: the exponent $1/2$ in $\psi - S$ cannot be improved. For chains that satisfy only a difference condition see Remark 5.3.

9 THE SECOND-ORDER TERM

For a shifted power the error in Theorem 8.1 is of exact order N_λ^{-1}, and its coefficient is universal up to the curvature of the links. This section computes it. The first-order phase is evaluated with the true link function, and the remaining $1/N_\lambda$ term is the first Born term of the uniform Airy approximation on the whole half-line, whose coefficient reduces to a Hadamard finite part of an explicit local density.

Throughout this section $b \equiv 0$ and

$$a_k = \tilde{a}(k)(1 + e_k)\ (k \geq k_0), \qquad \tilde{a}(y) := c\,(y + s)^\alpha \big(1 + \kappa_2 (y + s)^{-2}\big), \qquad |e_k| \leq C_e k^{-3}, \tag{30}$$

with $\alpha > 1$, $c > 0$, $s, \kappa_2 \in \mathbb{R}$. Then Corollary 8.3 applies with $\kappa = \alpha s$.

Theorem 9.1 — Second-order Bohr–Sommerfeld rule

Under (30), as $\lambda \to +\infty$,

$$\psi(\lambda) = \Phi(\lambda) - \Big(\kappa_2 + \frac{2\alpha - 1}{24}\Big)\frac{J_\alpha}{N_\lambda} + O\big(N_\lambda^{-4/3}\big), \qquad \psi(-\lambda) = -\psi(\lambda).$$

Consequently the eigenvalues of every H_t satisfy, with $\nu_j := (j\pi + t + s\pi/2)/I_\alpha$,

$$(\lambda_j/2c)^{1/\alpha} = \nu_j + \Big(\kappa_2 + \frac{2\alpha - 1}{24}\Big)\frac{J_\alpha}{I_\alpha \nu_j} + O\big(\nu_j^{-4/3}\big).$$

The constant has two parts. The part $\kappa_2 J_\alpha$ is classical: it is the change of the first-order action when the links are bent at relative order n^{-2} (Lemma 9.3). The part $(2\alpha - 1)J_\alpha/24$ is the quantum correction (Proposition 9.2). For the Berg–Valent chain, $a_k = 2k\sqrt{4k^2 - 1} = 4k^2(1 - \frac{1}{8k^2} + O(k^{-4}))$, so $\alpha = 2$, $s = 0$ and $\kappa_2 = -1/8 = -\frac{2\alpha - 1}{24}$: the

$1/N_\lambda$ term vanishes, consistently with Valent's closed form (Example 5.1), although the exponentially small remainder seen there is a property of the exact solution and not of the rule.

9.1 SET-UP ON THE HALF-LINE

Put $h := 1/N_\lambda$ and $a(t,h) := \tilde{a}(t/h)/\lambda = {}^1\!/_2(t+hs)^\alpha(1+\kappa_2 h^2(t+hs)^{-2})$, so that p solves (26) with $d \equiv 0$. Then $Z = 1/(2\hat{a})$ has the form $Z(t,h) = t^{-\alpha}\mathcal{Z}(h/t)$ with $\mathcal{Z}$ analytic near 0 and $\mathcal{Z}(0) = 1$; at $h = 0$, $Z = t^{-\alpha}$ and $t_p = 1$. On the oscillatory side write $r := -\zeta > 0$ and $\vartheta := \arccos Z \in [0, \pi/2)$ (in this section ϑ always has this meaning), so $g^2 = r^{1/2}/(\hat{a}\sin\vartheta)$, and write $\langle t\rangle := \max(1,t)$ and $\mathcal{G}(t) := \hat{a}(t)g(t)$.

Remark 9.1 (Weyl–Moyal reading). For links $A(X \pm {}^1\!/_2)$ the Jacobi operator has Weyl symbol $H = 2A(X)\cos p$ (the shift $f \mapsto A(x+{}^1\!/_2)f(x+1)$ has symbol $A(X)e^{ip}$). The second-order Bohr–Sommerfeld term for one-dimensional Weyl symbols is ${}^1\!/_{24}\frac{d}{dE}\oint(H_{XX}H_{pp} - H_{Xp}^2)\,dt$ [14]. Applied formally to the half-orbit from the turning point to the limit-circle end, and regularised there, the curvature part $H_{XX}H_{pp} = -4AA''\cos^2 p$ contributes $+(\alpha-1)^2 J_\alpha/(24N)$ and the gradient part $-H_{Xp}^2 = -4A'^2\sin^2 p$ contributes $-\alpha^2 J_\alpha/(24N)$, for $A = cX^\alpha$ (both are regularised Beta integrals of the type evaluated in Proposition 9.2). The sum is $-(2\alpha-1)J_\alpha/(24N)$, so $2\alpha - 1 = \alpha^2 - (\alpha-1)^2$ and ${}^1\!/_{24}$ is the Moyal coefficient. Theorem 9.1 proves this formula for power symbols; it is not used in the proof, but it says what the proof computes: the $1/N$ term is the Moyal correction of the classical action.

9.2 PLAN OF THE PROOF

The first-order phase is $\tilde{\Phi}$, the classical action of the true link function $\tilde{a}$; it differs from Φ by the classical curvature term $\kappa_2 J_\alpha/N_\lambda$ (Lemma 9.3). What remains is a quantum correction, and the proof computes it as the first Born term of the uniform Airy approximation on the whole half-line. Let $n_0 := \lceil {}^1\!/_2/h\rceil$, let f_A be the solution with $f_A = \psi_A$ at $n_0 - 1, n_0$, and write $(f_A(n), f_A(n+1))^{\mathsf{T}} = \Psi(n)c(n)$ as in (24), with $c(n_0 - 1) = (1,0)$. Five steps, proved in Section A.5, lead to the constant.

(i) The Langer data of Lemma 7.2 extend analytically from a compact interval to the half-line $[{}^1\!/_4, \infty)$, with explicit growth (Lemma A.9).

(ii) The residual coefficients decay along the half-line: $\mathfrak{p}_n = h^2\mathfrak{p}_2(t) + O(h^3 t^{-2}\mathcal{G})$ and $\mathfrak{q}_n = O(h^{10/3}t^{-10/3}\mathcal{G})$, where $\mathfrak{p}_2$ is the second Taylor coefficient of the frozen residual $\mathcal{P}$ of Lemma 7.3 and $\mathcal{G} = \hat{a}g$ (Lemma A.10).

(iii) The forward induction of Theorem 7.1 runs to infinity, and the coefficient of ψ_B in f_A converges to $\mathcal{F}_2 + O(h^2)$, with the first Born term $\mathcal{F}_2 := -\sum_{n\ge n_0}\psi_A(n)(\mathcal{L}\psi_A)(n)/W_n$ (Lemma A.11).

(iv) A solution with limiting coefficients (C_1, C_2) has its coordinates at infinity in the direction $\tilde{\Phi} - \arg(C_1 + iC_2)$, where $\tilde{\Phi}(\lambda) := y_\lambda \pi/2 + \int_{y_\lambda}^{\infty} \arcsin(\lambda/2\tilde{a}(y))\, dy$ and y_λ is the largest root of $2\tilde{a} = \lambda$ (Lemma A.12).

(v) The Born term is an Airy-weighted Riemann sum: $\mathscr{F}_2 = hK_2 + O(h^{4/3})$ with $K_2 := {}^1\!/_2 \int_1^{\infty} G_2(t)\, dt$ and $G_2 := g\mathfrak{p}_2 r^{-1/2}$ at $h = 0$ (Lemma A.13). Because $\mathrm{Ai}(-X)^2$ averages to $(2\pi X^{1/2})^{-1}$, the sum becomes the integral of a local density against $r^{-1/2}$.

The turning point contributes only $O(h^{4/3})$ to $\mathscr{F}_2$, not $O(h)$, for a reason that is visible in (ii): the frozen residual is even in the step, so it is $O(h^2)$ per site, and the turning window holds only $O(h^{-1/3})$ of the h^{-1} sites that carry the density. So the whole coefficient of $1/N_\lambda$ comes from the oscillatory half-line, where it is the finite part computed next.

9.3 THE CONSTANT

Proposition 9.2 — The Born constant

On the oscillatory side at $h = 0$, with $Z = t^{-\alpha}$, $\vartheta = \arccos Z$ and $S := \sin\vartheta$,

$$G_2 = \mathscr{D}(t) - \frac{5}{16}\,\vartheta\, r^{-3}, \qquad \mathscr{D} := -\frac{Z'^2(S^4 + 17S^2 - 15) - Z'' Z(S^4 + 6S^2)}{12\, S^5 Z},$$

and $K_2 = {}^1\!/_2\, \mathrm{f.p.}\int_1^{\infty} \mathscr{D}\, dt = \frac{2\alpha - 1}{24} J_\alpha$.

Proof. **Step 1** (*the density*)**.** $\mathfrak{p}_2 = F''(0)$ for $F(\eta) = \hat{a}(t + \eta/2) g(t + \eta) X_1(s_+(\eta); \eta)$, by parity. Expanding $X_1 = X_{10} + \eta X_{11} + \eta^2 X_{12}$ with $X_{10} = \cosh(ks)$, $k = \sqrt{\zeta}$, and X_{11}, X_{12} the solutions of $Y'' - k^2 Y = s X_{1,j-1}$ with zero data, and eliminating g, $\hat{a}$ and the derivatives of ζ through the eikonal and amplitude identities and their derivatives, one obtains on the oscillatory side ($\sqrt{\zeta} = i\sqrt{r}$) exactly the two groups of terms displayed. The intermediate identities of this elimination are listed in Section A.7.

Step 2 (*the regulator*)**.** With $\mathscr{A}(t) := \int_1^t \vartheta$ we have $r^{3/2} = {}^3\!/_2 \mathscr{A}$, so $-{}^5\!/_{16} \vartheta r^{-3} = {}^5\!/_{36} (1/\mathscr{A})'$, and for $t_0 > 1$, $\int_{t_0}^{\infty} G_2 = \int_{t_0}^{\infty} \mathscr{D} - {}^5\!/_{36} \mathscr{A}(t_0)^{-1}$.

Step 3 (*finite part*)**.** Near $t = 1$, $S = (t-1)^{1/2}\cdot$(analytic, nonzero), so $\mathscr{D} = (t-1)^{-5/2}\cdot$(analytic) and $1/\mathscr{A} = (t-1)^{-3/2}\cdot$(analytic). Hence $\int_{t_0}^{\infty} \mathscr{D} = \mathrm{f.p.}\int_1^{\infty} \mathscr{D} + \sum_{j=0,1} c_j (t_0 - 1)^{j-3/2} + O((t_0 - 1)^{1/2})$, with no logarithm and no constant from $1/\mathscr{A}$. Since $G_2 = g\mathfrak{p}_2 r^{-1/2}$ is integrable at 1, the singular terms cancel and $K_2 = {}^1\!/_2\, \mathrm{f.p.}\int_1^{\infty} \mathscr{D}$.

Step 4 (*evaluation*)**.** Substitute $w = t^{-\alpha}$, so $Z' = -\alpha w^{1+1/\alpha}$, $Z'' = \alpha(\alpha+1) w^{1+2/\alpha}$ and $dt/dw = -\alpha^{-1} w^{-1-1/\alpha} < 0$. Then, exactly,

$$\mathscr{D}\,\frac{dt}{dw} = \tfrac{1}{12} w^{1/\alpha}\big[-\sigma^{-1} + (11\alpha - 6)\sigma^{-3} - 15\alpha\sigma^{-5}\big], \qquad \sigma := (1 - w^2)^{1/2},$$

and $t \in (1,\infty)$ corresponds to w decreasing from 1 to 0. The map $t \mapsto w$ is analytic with nonzero derivative at $t = 1$ and the exponents are half-integers, so the finite part is the value at $\epsilon = 0$ of the analytic continuation of $\int \mathscr{D}(t-1)^{\epsilon} dt$ and is unchanged by the substitution. With $\mathscr{M}(\gamma) := \int_0^1 w^{1/\alpha}(1-w^2)^{-\gamma} dw = {}^1\!/_2\,\mathrm{B}({}^1\!/_2 + \frac{1}{2\alpha}, 1-\gamma)$ continued in γ, the recursion $\mathrm{B}(p,q-1) = \mathrm{B}(p,q)(p+q-1)/(q-1)$ gives $\mathscr{M}({}^1\!/_2) = \alpha J_\alpha$, $\mathscr{M}({}^3\!/_2) = -J_\alpha$ and $\mathscr{M}({}^5\!/_2) = (1-2\alpha)J_\alpha/(3\alpha)$. Hence

$$\mathrm{f.p.}\int_1^\infty \mathscr{D}\, dt = -\tfrac{1}{12}\big[-\mathscr{M}(\tfrac{1}{2}) + (11\alpha - 6)\mathscr{M}(\tfrac{3}{2}) - 15\alpha\mathscr{M}(\tfrac{5}{2})\big] = -\tfrac{1}{12}(1-2\alpha)J_\alpha = \tfrac{2\alpha-1}{12}J_\alpha$$

(a direct quadrature of G_2 gives the same value; Appendix B). □

Lemma 9.3 (The classical part). $\tilde{\Phi}(\lambda) = \Phi(\lambda) - \kappa_2 J_\alpha / N_\lambda + O(N_\lambda^{-3})$.

Proof. At $\kappa_2 = 0$, $\tilde{\Phi} = \Phi$ exactly (Proposition 5.2). Differentiating $\tilde{\Phi} = y_\lambda \pi/2 + \int_{y_\lambda}^\infty \arcsin W$, $W := \lambda/2\tilde{a}$, in κ_2 at fixed λ, the boundary terms cancel because the integrand is $\pi/2$ at y_λ, and $\partial_{\kappa_2} W = -W(y+s)^{-2}(1+\kappa_2(y+s)^{-2})^{-1}$. At $\kappa_2 = 0$ this gives $-\int_N^\infty (N/v)^\alpha v^{-2}(1-(N/v)^{2\alpha})^{-1/2} dv = -J_\alpha/N$. In the variable $u = v/v_\lambda$, where v_λ is the turning point, the turning point stays at $u = 1$, each κ_2-derivative brings a factor v_λ^{-2} and keeps the integrand $O((u-1)^{-1/2})$ at $u = 1$, so the second derivative is $O(N^{-3})$ uniformly for bounded κ_2. □

Proof of Theorem 9.1. Apply Theorem 7.1 on the window $[n_0, n_1]$, $n_1 h \approx 3$, to obtain f_B; exactly as in Lemma A.5, $p = c_K(f_A + \eta_K f_B)$ with $|\eta_K| \le Ce^{-c/h}$, and the coefficients of f_B in the basis (ψ_A, ψ_B) converge and stay bounded beyond n_1 by the forward induction of Lemma A.11 (at n_1 the site is oscillatory, so $\Lambda^{(B)}(n_1) \le 1 + 2d$). So the coefficient vector of p/c_K converges to $C = c(\infty) + O(e^{-c/h})$, and by Lemma A.12 and Lemma A.11, $\psi \equiv \tilde{\Phi} - \arg(C_1 + iC_2) = \tilde{\Phi} - \mathscr{F}_2 + O(h^2)$ modulo π. Since $\psi - \Phi \to 0$ (Corollary 8.3), $\tilde{\Phi} - \Phi \to 0$ and $\mathscr{F}_2 \to 0$, the integer is 0. Lemma A.13, Proposition 9.2 and Lemma 9.3 give the formula. The reflection is Theorem 2.2(iii), and the eigenvalue formula follows by solving $\psi(\lambda_j) = t + j\pi$ for N_{λ_j}. □

The remainder $O(N_\lambda^{-4/3})$ comes from the turning-point region in Lemma A.13 and is not sharp; numerically the deviation of $N_\lambda r_j$ from its limit decays like N_λ^{-2}, that is, the remainder is $O(N_\lambda^{-3})$ for smooth chains (Section 12). If $|e_k| \le C_e k^{-2-\delta}$ with $0 < \delta < 1$, the same proof gives the remainder $O(N_\lambda^{-4/3} + N_\lambda^{-1-\delta})$.

10 APPLICATIONS

This section gives two applications in which limit-circle Jacobi matrices with power links arise from physics: the spectra of the self-adjoint extensions of higher-order squeezing operators, where Theorem 9.1 gives the eigenvalues to second order, and the essentially singular limits of Fischer, Burgarth and Lonigro, where Theorem 7.1 removes a restriction on the growth exponent.

10.1 HIGHER-ORDER SQUEEZING

The k-photon squeezing operator $a^k + a^{\dagger k}$ on $\ell^2(\mathbb{N}_0)$ commutes with the number operator modulo k, and decomposes into k Jacobi blocks: block $m \in \{0, \dots, k-1\}$ acts on the Fock states $|m + nk\rangle$, $n \geq 0$, with zero diagonal and links

$$a_n^{(m)} = \Big(\prod_{l=1}^{k} \big(m + (n-1)k + l \big) \Big)^{1/2} \qquad (n \geq 1).$$

For $k \geq 3$ each block is in the limit-circle case and the operator has deficiency indices (k, k) [16, 17]; for $k \leq 2$ it is essentially self-adjoint. The extensions that commute with the $\mathbb{Z}_k$ symmetry are exactly the direct sums of extensions $H_{t_m}^{(m)}$ of the blocks.

Corollary 10.1. Let $k \geq 3$, $0 \leq m < k$ and $t \in [0, \pi)$. The eigenvalues of $H_t^{(m)}$ satisfy, with $s_m := (2m - k + 1)/(2k)$ and $\nu_j := (j\pi + t + s_m \pi/2)/I_{k/2}$,

$$\Big(\frac{\lambda_j}{2k^{k/2}} \Big)^{2/k} = \nu_j + \frac{(k-1)^2}{48k} \, \frac{J_{k/2}}{I_{k/2}\, \nu_j} + O\big(\nu_j^{-4/3} \big) \qquad (j \to +\infty),$$

and $\lambda_{-j}(t) = -\lambda_{j-1}(\pi - t)$ for $0 < t < \pi$, $\lambda_{-j}(0) = -\lambda_j(0)$.

Proof. Write $a_n^{(m)} = k^{k/2} \big(\prod_l (x + c_l) \big)^{1/2}$ with $x = n + s_m$ and $c_l = (l - \frac{k+1}{2})/k$, so $\sum_l c_l = 0$ and $\sum_l c_l^2 = (k^2 - 1)/(12k)$. Then $\prod_l (x + c_l) = x^k (1 - \frac{k^2-1}{24k} x^{-2} + O(x^{-3}))$, so (30) holds with $c = k^{k/2}$, $\alpha = k/2$, $s = s_m$ and $\kappa_2 = -(k^2 - 1)/(48k)$. Theorem 9.1 applies, and $\kappa_2 + (2\alpha - 1)/24 = (k-1)^2/(48k)$. The negative eigenvalues follow from the reflection identity $\psi(-\lambda) = -\psi(\lambda)$ of Theorem 2.2(iii): for $0 < t < \pi$, $\psi(-\lambda_j(t)) = -j\pi - t = (-j - 1)\pi + (\pi - t)$, so $-\lambda_j(t) = \lambda_{-j-1}(\pi - t)$, which is the stated identity with j replaced by $j - 1$ and t by $\pi - t$; for $t = 0$, $\psi(-\lambda_j(0)) = -j\pi$ gives $-\lambda_j(0) = \lambda_{-j}(0)$. □

In particular the first-order offset is $\kappa/(2\alpha) = s_m/2$: the blocks of one operator are distinguished, at first order, only by their offsets $(2m - k + 1)/(4k)$. Fischer, Knapp, Burgarth and Lonigro announce the transfer of eigenvalue asymptotics from the Jacobi literature to multiphoton models as future work [18, p. 3]; as far as we know Corollary 10.1 is the first such statement with an explicit constant.

10.2 ESSENTIALLY SINGULAR LIMITS FOR $\alpha > 1$

Fischer, Burgarth and Lonigro [17] study the family $J(\lambda) = J(0) + \lambda F$, where $J(0)$ is a Jacobi matrix with links a_n and F is diagonal with entries f_n, in their indexing

$$a_n u_{n+1} + (\lambda f_n - z)u_n + a_{n-1}u_{n-1} = 0 \qquad (n \geq 1). \tag{31}$$

Their Hypothesis 2.2 asks for positivity (i), $a_n = n^{\alpha}(1 + c_a/n + O(n^{-2}))$ with $\alpha > 4/3$ (ii), $f_n = n^{\beta}(1 + c_f/n + O(n^{-2}))$ with $\beta > \alpha$ (iii), and a total-variation condition (iv). For $\lambda > 0$ the operator $J(\lambda)$ is self-adjoint, while $J(0)$ is in the limit-circle case, and their Theorem 2.7 shows that the limits of $J(\lambda)$ as $\lambda \to 0$ in the strong resolvent sense are exactly the self-adjoint extensions J_t of $J(0)$, each attained along a sequence. They write that it remains open whether this holds under $\alpha > 1$ [17, §4.7, p. 54]. The exponent $4/3$ enters their proof in two places. The first is the λ-uniform majorant $|u_n| \leq C(|u_0| + |u_1|)n^{-\alpha/2+1/6}$ of their Theorem 2.8 (restated as their Theorem 3.34), which is square-summable exactly when $\alpha > 4/3$ and is used twice to apply dominated convergence, at (3.109) and (3.120). The second is the perturbation estimate of their Lemma 4.14, where the bound $h^{\alpha} \leq h^{4/3}$ in (4.94) uses $\alpha > 4/3$; this lemma feeds their turning-point analysis (their §4.4). Their Remark 3.35 and §4.7 say that within their strategy the restriction is necessary. Our proof replaces their §§4.4–4.6, so neither use remains. The $1/6$ is real: the eigenvector has an Airy hump of relative height $h^{-1/6}$ at the turning point, $h := \lambda^{1/(\beta-\alpha)}$. But the hump is only $h^{-1/3}$ sites wide, and with Olver's envelopes its ℓ^2 mass is of order $h^{\alpha-1}$.

Theorem 10.2 — Uniform tails

Assume Hypothesis 2.2 of [17] with (ii) weakened to $\alpha > 1$, and let $\Omega \subset \mathbb{C}$ be compact. There are $\Lambda_0 > 0$ and C such that for $0 \leq \lambda < \Lambda_0$, $z \in \Omega$ and every square-summable solution u of (31),

$$|u_n| \leq C\, n^{-\alpha/2}\big(h^{2/3} + |nh - t_p|\big)^{-1/4}(|u_0| + |u_1|) \quad (\lambda > 0), \qquad \sum_{n \geq N} |u_n|^2 \leq C N^{1-\alpha}(|u_0| + |u_1|)^2,$$

where $t_p = t_p(h) \to 2^{1/(\beta-\alpha)}$. Consequently Theorem 2.7 of [17] holds for every $\alpha > 1$, and so does the bound of their Theorem 2.8.

Proof. **Step 1** (*the regions outside the window*)**.** Put $\delta := \beta - \alpha$ and use the regions of [17, Def. 4.1, Lemma 4.2], $N_1 < N_2 < N_3 < N_4$ with $N_i h \to x_i$, $x_1^{\delta} = 5/8$, $x_2^{\delta} = 7/8$, $x_3^{\delta} = 33/8$, $x_4^{\delta} = 39/8$. Their Lemma 4.5 and Proposition 4.10 (Turán determinants) give $|u_n| \leq C n^{-\alpha/2}(|u_0| + |u_1|)$ for $n \leq N_2$, and their Proposition 4.31 gives the same for all n when $\lambda = 0$; their Proposition 4.29 (a Riccati disc) gives $|u_{n+1}/u_n| \leq 3/8$ for $n \geq N_3$ (we use this together with $a_n/a_{n+1} \leq 1 + 1/8$, as in their (4.198)). None of these uses $\alpha > 4/3$ or the Airy comparison: Lemma 4.5 uses only compactness and their Lemma 4.2 ($a_{n-1}/a_n \to 1$, $a_n \to \infty$); Proposition 4.10 uses the finite total variations of $(1/a_n)$ and (a_{n-1}/a_n), the convergence of $\sum 1/a_n$ (their Lemma 3.19, where $\alpha > 1$ suffices) and their

Hypothesis 2.2(iv); Proposition 4.29 uses the Poincaré–Perron asymptotics, which need $\lambda f_n/a_n \to \infty$, that is $\beta > \alpha$, and their Lemma 4.2; and Proposition 4.31 assumes $\alpha > 1$ explicitly.

Step 2 (*the window equation*)**.** For $N_1 \le n \le N_4$ put $u_n = (-1)^n w_n/\sqrt{\tilde b_n}$ with $\tilde a(x,h) := x^\alpha(1 + c_a h/x)$, $\tilde b(x,h) := x^\beta(1 + c_f h/x)$, $\tilde a_n := \tilde a(nh,h)$, $\tilde b_n := \tilde b(nh,h)$. Multiplying (31) by $(-1)^{n+1}h^\alpha/\sqrt{\tilde b_n}$ and using $h^\alpha\lambda = h^\beta$ gives

$$\mathscr{A}((n+1)h)(1+e_{n+1})w_{n+1} + \mathscr{A}(nh)(1+e_n)w_{n-1} - (1-d_n)w_n = 0,$$
$$\mathscr{A}(s) := \frac{\tilde a(s-h,h)}{\sqrt{\tilde b(s-h,h)\tilde b(s,h)}},$$

with $1 + e_{n+1} := h^\alpha a_n/\tilde a_n$ and $d_n := (\tilde b_n - h^\beta f_n + h^\alpha z)/\tilde b_n$. On the window $n \asymp 1/h$, so $|e_n| \le Ch^2$ and $|d_n| \le C(h^2 + h^\alpha)$ uniformly in $z \in \Omega$; d_n is complex. The coefficient $\mathscr{A}(s,h)$ is real-analytic in (s,h) near $[x_1/2, x_4 + 1] \times \{0\}$, and $\mathscr{A}(s,0) = s^{-\delta}$ is decreasing. Reflecting the site index, $\tilde w(m) := w(-m)$ solves an equation of the form (26) with coefficient $a(t,h) := \mathscr{A}(h - t, h)$, increasing in t at $h = 0$; here h is both the lattice step and a parameter of the coefficient, which (25) allows. So Lemmas 7.2 and 7.3 apply, and Lemma 7.4 gives $V \le C(h + h^{\alpha-1})$ and $\theta_D \le C(h^2 + h^\alpha)$. This is where $\alpha > 1$ enters the turning point: the spectral parameter contributes $h^\alpha z$ per site over h^{-1} sites.

Step 3 (*the solution is almost recessive*)**.** In the original orientation the Langer variable x_n increases with n, and Theorem 7.1 gives exact solutions f_A, equal to $g\,\mathrm{Ai}(x_n)$ at the right end $n_1 := N_4 - 1$, and f_B, equal to $g\,\mathrm{Bi}(x_n)$ at the left end $n_0 := N_1$, with $d := e^{2V/(1-\theta_D)} - 1 \le C(h + h^{\alpha-1}) \to 0$. By (D2) they are independent, so $w = c_A f_A + c_B f_B$. At $n_1 \ge N_3$, $x_{n_1} \asymp h^{-2/3}$; by (D1) and the monotonicity of Ai, Bi there, the ratios $r_A = f_A(n_1+1)/f_A(n_1)$ and $r_B = f_B(n_1+1)/f_B(n_1)$ satisfy $|r_A| \le 1/2$ and $|r_B| \ge 3/4$, while $r_w := w_{n_1+1}/w_{n_1}$ has $|r_w| \le 1/2$ by the first step. Hence $c_A \ne 0$ and $|c_B/c_A| \le 20\,\mathrm{Ai}(x_{n_1})/\,\mathrm{Bi}(x_{n_1}) = 20\,\mathsf{E}(x_{n_1})^{-2}$.

Step 4 (*the envelope*)**.** For $n_0 \le n \le n_1$, (D1) and the monotonicity of E give $|w(n)| \le |c_A|(1+d)g_n\mathsf{E}_n^{-1}\mathsf{M}_n(1+20\,\mathsf{E}_n^2/\mathsf{E}(x_{n_1})^2) \le C|c_A|(1+|x_n|)^{-1/4}$. On $[N_1, N_2]$, $x_n \le -ch^{-2/3}$ and the phase steps $\xi_n - \xi_{n+1} = \arccos Z(nh) + O(h)$ lie in $[1.11, 1.26]$, so of two consecutive sites one has $|\,\mathrm{Ai}(x_m)| \ge \mathsf{M}(x_m)/3$, and there $|w(m)| \ge |c_A|g_m\mathsf{M}(x_m)/4 \ge c|c_A|h^{1/6}$. Combining with the first step at $m \le N_2$ and undoing the substitution ($\tilde b_n \asymp 1$ on the window, $|\zeta(t)| \ge r_0|t - t_p|$, $m \asymp n$) gives the envelope for $N_1 \le n \le N_4$.

Step 5 (*assembly*)**.** For $n < N_1$ the first step applies, and for $n > N_4$ the ratio bound $3/8$ propagates the envelope at N_4 with a geometric factor that absorbs $(n/N_4)^{\alpha/2}$. Summing the envelope over the window gives at most $Ch^\alpha\big(h^{-2/3} + h^{-1}\int_0^{x_4} s^{-1/2}ds\big) \le Ch^{\alpha-1}$ (the first term counts the $O(h^{-1/3})$ sites with $|nh - t_p| \le h^{2/3}$), which is at most $CN^{1-\alpha}$ when the window lies beyond N, that is when $h \le x_4/N$; the other parts contribute $CN^{1-\alpha}$.

The pointwise form gives $|u_n| \le Cn^{-\alpha/2}h^{-1/6} \le C'n^{-\alpha/2+1/6}$, which is the bound of their Theorem 2.8.

Step 6 (*Theorem 2.7*)**.** If $x^{(j)} \to x$ pointwise with $x \in \ell^2$ and $\sup_j \sum_{n\ge N} |x_n^{(j)}|^2 \to 0$ as $N \to \infty$, then $x^{(j)} \to x$ in ℓ^2. Their proof of Theorem 2.7 uses Theorem 2.8 exactly twice, in (3.107)–(3.110) for the Green's function columns and in (3.119)–(3.124) for the eigenvectors, both times to obtain ℓ^2 convergence from pointwise convergence by dominated convergence. The tail bound and the remark just made give that convergence directly. Elsewhere in their §3 the growth exponent enters through their Lemma 3.19, whose conclusions (finite total variation and $\sum 1/a_n < \infty$) hold for $\alpha > 1$; we have checked the places where α is invoked, not every line of their §3. □

The threshold $\alpha > 1$ is exact: for $\alpha \le 1$, $\sum 1/a_n = \infty$, so $J(0)$ is self-adjoint by Carleman's criterion, ℓ_0 is a common core, and $J(\lambda) \to J(0)$ in the strong resolvent sense [32, Thm. VIII.25]. The gain is in the range $1 < \alpha \le 4/3$ of the general class; the squeezing operators already have $\alpha = k/2 \ge 3/2$.

11 APPLICATION: CRITICAL INDICES OF INDETERMINATE MOMENT PROBLEMS

By a theorem of M. Riesz, for an N-extremal solution μ of an indeterminate moment problem the measure $(1+x^2)^{-1}\mu$ is determinate while μ is not. Berg and Szwarc [7] define the *critical index*

$$a(\mu) := \sup\{\gamma \in [0,1] : (1+x^2)^{-\gamma}\mu \text{ is indeterminate}\}$$

and put μ in class **A** if $(1+x^2)^{-a(\mu)}\mu$ is still indeterminate and in class **B** otherwise. They show $a(\mu_F) \le 1/2 \le a(\mu_\sigma)$ for the Friedrichs solution μ_F and the other N-extremal solutions μ_σ of a Stieltjes problem, exhibit members of $\mathbf{A}(1/2)$ and $\mathbf{B}(1/2)$, and note that explicit examples are known only at the value $1/2$. They ask whether both classes **A** and **B** are nonempty [7, after Prop. 1.7], and leave open a determinacy threshold built on the Berg–Valent problem [7, Ex. 4.1]. We compute the indices exactly for Valent's polynomial rates; this gives explicit members of **B** and decides the threshold. The index is read off from the residues of a Nevanlinna function at the support points, and these residues come from the Bessel boundary values of Theorem 6.2.

11.1 A WEIGHTED CRITERION

Let x be the variable of a normalised indeterminate Hamburger problem with orthonormal polynomials $\hat{p}_k$ (in the variable x; they are not the regular solution p of the earlier sections) and Nevanlinna matrix $\left(\begin{smallmatrix} A & C \\ B & D \end{smallmatrix}\right)$, in the parametrisation of [7]. For $\sigma \in \mathbb{R} \cup \{\infty\}$ the N-extremal solution μ_σ is supported by the zero set $\mathcal{Z}_\sigma$ of $G_\sigma := B + \sigma D$ ($G_\infty := D$), and $\mu_\sigma(\{x\}) = 1/\sum_k \hat{p}_k(x)^2$ [7, Prop. 1.4]. For a Stieltjes problem the Friedrichs solution is μ_F with $F \in \mathbb{R}$, and the Krein solution is μ_∞. The identity

$$B(u)D(v) - D(u)B(v) = (u - v)\sum_{k\geq 0} \hat{p}_k(u)\hat{p}_k(v) \tag{32}$$

[8, first proof of Prop. 2.3] gives $\sum_k \hat{p}_k(u)^2 = B'D - BD'$ as $v \to u$, so on $\mathcal{Z}_\sigma$

$$\mu_\sigma(\{x\})\,\mathcal{E}_\sigma(x)^2 = W_\sigma(x), \qquad \mathcal{E}_\sigma := D,\ W_\sigma := \frac{D}{G_\sigma'}\ (\sigma \in \mathbb{R}), \qquad \mathcal{E}_\infty := B,\ W_\infty := -\frac{B}{D'}, \tag{33}$$

and $W_\sigma > 0$ there. Equivalently $W_\sigma(x) = 1/(\mu_\sigma(\{x\})G_\sigma'(x)^2)$.

The criterion below rests on one fact about N-extremal measures: the polynomials have codimension one in $L^2((1+x^2)\mu_\sigma)$, and the orthogonal complement is spanned by $\mathcal{E}_\sigma/(1+x^2)$. A weighted measure $w\mu_\sigma$ is therefore indeterminate exactly when this one function, divided by w, still has finite norm, and that norm is a single series over the support.

Proposition 11.1 — Weighted determinacy

Let μ_σ be N-extremal and let w be a function on its support with $0 < w \leq C$. Then $w\mu_\sigma$ is indeterminate if and only if

$$\mathcal{I}_\sigma(w) := \int \frac{\mathcal{E}_\sigma(x)^2}{(1+x^2)\,w(x)}\,d\mu_\sigma(x) = \sum_{x\in\mathcal{Z}_\sigma} \frac{W_\sigma(x)}{(1+x^2)\,w(x)} < \infty.$$

Consequently, with $\mathcal{S}_\sigma(\gamma) := \sum_{x\in\mathcal{Z}_\sigma} W_\sigma(x)(1+x^2)^{\gamma-1}$, $a(\mu_\sigma) = \sup\{\gamma : \mathcal{S}_\sigma(\gamma) < \infty\}$, and $\mu_\sigma \in \mathbf{B}$ if and only if $\mathcal{S}_\sigma(a(\mu_\sigma)) = \infty$.

Attribution. The criterion is the case $r = 2$ of Theorem A of Borichev and Sodin [12], specialised to N-extremal supports. Their Hamburger class $\mathfrak{H}$ consists of the transcendental real entire functions of zero exponential type with simple real zeros on which $|\lambda|^n/|B_0'(\lambda)| \to 0$ for every n; it contains the entries of the Nevanlinna matrix, so the G_σ. For $B_0 \in \mathfrak{H}$ and a weight $\omega > 0$ on its zero set Λ_{B_0} with $|\lambda|^n\omega(\lambda) \to 0$, Theorem A states that the polynomials are dense in $\ell^r(\omega, \Lambda_{B_0})$, $1 < r \leq \infty$, if and only if $\sum_{\lambda\in\Lambda_F} |\omega(\lambda)F'(\lambda)|^{-r/(r-1)} = +\infty$ for every $F \in \mathfrak{H}$ with $\Lambda_F \subset \Lambda_{B_0}$ (a $\liminf$ condition replaces this for $r = 1$). By Riesz's theorem $w\mu_\sigma$ is indeterminate exactly when the polynomials are not dense in $L^2((1+x^2)w\mu_\sigma) = \ell^2(\omega, \mathcal{Z}_\sigma)$ with $\omega^2 := (1+x^2)w\,\mu_\sigma(\{x\})$,

and for $F = G_\sigma$ the series is $\mathcal{I}_\sigma(w)$, since $1/(\mu_\sigma(\{x\})G_\sigma'(x)^2) = W_\sigma(x)$. No other divisor F matters here: a convergent series for F would give an annihilator supported on Λ_F, and by the proof below every annihilator is a multiple of $\mathcal{E}_\sigma/((1+x^2)w)$, which vanishes nowhere on $\mathcal{Z}_\sigma$.

The quantifier over divisors is the point of Theorem A. Hamburger [25] stated that a measure $\sum \mu_\lambda \delta_\lambda$ on the zero set of some $B_0 \in \mathfrak{H}$ is canonical (N-extremal) if and only if it satisfies three conditions, the third being $\sum_\lambda 1/(\mu_\lambda B_0'(\lambda)^2) = +\infty$. That condition is necessary, since otherwise $1/(\mu_\lambda B_0'(\lambda))$ annihilates the polynomials, which is the "if" direction of Proposition 11.1. It is not sufficient: Berg and Pedersen found the gap in Hamburger's proof in 1989, and Koosis gave a counterexample [27]; see the account in [1, §3.5], which also gives a counterexample on a power-separated set. Corollary 1.1 of [12], which replaces the third condition by the same divergence for every divisor $F \in \mathfrak{H}$ of B_0, is the corrected statement. Our account of Theorem A and of Corollary 1.1 follows the February 1998 preprint of [12]; we have not seen the published text. Their proof of Theorem A follows an approach of de Branges; for N-extremal supports a short Hilbert-space argument suffices.

Proof of Proposition 11.1. **Step 1** (*codimension one*)**.** If μ is N-extremal, $\mathbb{C}[x]$ has codimension one in $L^2((1+x^2)\mu)$: $f \mapsto (x-i)f$ maps $L^2((1+x^2)\mu)$ unitarily onto $L^2(\mu)$ and $\mathbb{C}[x]$ onto $(x-i)\mathbb{C}[x]$, $\mathbb{C}[x] = (x-i)\mathbb{C}[x] + \mathbb{C}$ is dense in $L^2(\mu)$ by Riesz's theorem [7, Thm. 1.3], and the codimension is not zero because μ is indeterminate. This is the case $n = 1$ of Cassier's theorem [3, Thm. 2.2].

Step 2 (*the annihilator*)**.** $f_\sigma := \mathcal{E}_\sigma/(1+x^2)$ is a nonzero element of $L^2((1+x^2)\mu_\sigma)$ orthogonal to $\mathbb{C}[x]$. For $\sigma \in \mathbb{R}$, $D(x) = x\,\mathcal{K}_0(x)$ with $\mathcal{K}_0 = \sum_k \hat{p}_k(0)\hat{p}_k \in L^2(\mu_\sigma)$ (Parseval, [8, (5)–(6)]), so $\|f_\sigma\|^2 \le \sum_k \hat{p}_k(0)^2$, and $\int D\,\bar{q}\,d\mu_\sigma = 0$ for every polynomial q [8, Prop. 3.1(i)]. For $\sigma = \infty$, (32) at a real v with $D(v) \neq 0$ gives $B = (\cdot - v)\mathcal{K}_v/D(v)$ on $\mathcal{Z}_\infty$ with $\mathcal{K}_v = \sum_k \hat{p}_k(v)\hat{p}_k \in L^2(\mu_\infty)$.

Step 3 (*the criterion*)**.** If f annihilates $\mathbb{C}[x]$ in $L^2((1+x^2)w\mu_\sigma)$, then fw annihilates it in $L^2((1+x^2)\mu_\sigma)$, because $w \le C$; so $fw = c\,f_\sigma$, and $f = c\,\mathcal{E}_\sigma/((1+x^2)w)$ has finite norm exactly when $c = 0$ or $\mathcal{I}_\sigma(w) < \infty$. Riesz's theorem applied to $(1+x^2)w\mu_\sigma$ gives the equivalence, and $w = (1+x^2)^{-\gamma}$ gives the formula for $a(\mu_\sigma)$. □

The criterion also answers the question of [7] whether both classes are nonempty, but not by explicit measures. Since it is the case $r = 2$ of Theorem A of Borichev and Sodin, the following is a corollary of their theorem rather than a result of this paper; the explicit members of **B** come from Theorem 11.3 below.

Corollary 11.2. Both classes **A** and **B** of [7] are nonempty.

Proof. Fix an N-extremal μ_σ, order its support by $|x|$ as $x_0, x_1, \dots$, write $W_k := W_\sigma(x_k)$ and $S_k := W_0 + \cdots + W_k$. By Proposition 11.1 with $w = 1$, $\sum_k W_k/(1+x_k^2) < \infty$, because μ_σ

is indeterminate; with $w=(1+x^2)^{-1}$, $\sum_k W_k=\infty$, because $(1+x^2)^{-1}\mu_\sigma$ is determinate. So $S_k\to\infty$, and since $1+x_k^2$ is nondecreasing and tends to infinity, Kronecker's lemma gives $S_k=o(1+x_k^2)$; in particular $(1+x_k^2)^\delta\ge c_\delta S_k^\delta$ for every $\delta>0$. We use two facts about the increments $S_k-S_{k-1}=W_k$. First, $\sum_k W_k/(S_kL(S_k))<\infty$ whenever $\int^\infty ds/(sL(s))<\infty$ and L is positive and nondecreasing, because each term is at most $\int_{S_{k-1}}^{S_k} ds/(sL(s))$; this covers $L=\log^2(e+s)$ and $L=s^\delta$. Second, $\sum_k W_kS_k^{\delta-1}/\log^2(e+S_k)=\infty$ for $0<\delta<1$: if $S_k>2S_{k-1}$ for infinitely many k, those terms are at least $S_k^\delta/(2\log^2(e+S_k))$, which is bounded below; otherwise, from some k on, $S_{k-1}\ge S_k/2$ and each term is at least a constant times $\int_{S_{k-1}}^{S_k} s^{\delta-1}\log^{-2}(e+s)\,ds$, whose sum diverges.

Step 1 (**A** $\neq\emptyset$)**.** Let $w_k:=\min\bigl(S_k\log^2(e+S_k)/(1+x_k^2),1\bigr)$ and $\nu:=w\mu_\sigma$. Then $\mathcal{J}_\sigma(w)\le\sum_k W_k/(S_k\log^2(e+S_k))+\sum_k W_k/(1+x_k^2)<\infty$, so ν is indeterminate by Proposition 11.1. It is N-extremal: $\nu\le\mu_\sigma$, so the polynomials are dense in $L^2(\nu)$ [7, Lemma 1.1(ii)], and Riesz's theorem applies. For $0<\delta<1$ the weight $(1+x^2)^{-\delta}w$ is bounded and

$$\mathcal{J}_\sigma\bigl((1+x^2)^{-\delta}w\bigr)\ge\sum_k\frac{W_k(1+x_k^2)^\delta}{S_k\log^2(e+S_k)}\ge c_\delta\sum_k\frac{W_kS_k^{\delta-1}}{\log^2(e+S_k)}=\infty,$$

so $(1+x^2)^{-\delta}\nu$ is determinate; for $\delta\ge1$ this follows by monotonicity [7, Lemma 1.1(i)]. By [7, Prop. 1.7(i)], **A** $\neq\emptyset$.

Step 2 (**B** $\neq\emptyset$)**.** Let $w_k:=S_k/(1+x_k^2)$, which is bounded since $S_k=o(1+x_k^2)$, and $\nu:=w\mu_\sigma$. Then $\mathcal{J}_\sigma(w)=\sum_k W_k/S_k=\infty$ by the Abel–Dini theorem, so ν is determinate. For $\delta>0$ the weight $(1+x^2)^\delta w$ need not be bounded, so we compare with the bounded weight $w_k':=\min((1+x_k^2)^\delta w_k,1)$: $\mathcal{J}_\sigma(w')\le\sum_k W_k/(S_k(1+x_k^2)^\delta)+\sum_k W_k/(1+x_k^2)\le c_\delta^{-1}\sum_k W_kS_k^{-1-\delta}+C<\infty$, so $w'\mu_\sigma$ is indeterminate, and so is the larger measure $(1+x^2)^\delta\nu$ [7, Lemma 1.1(i)]. By [7, Prop. 1.7(ii)], **B** $\neq\emptyset$. □

The content below is the exact index of the natural solutions of a large class of problems, which gives explicit members of **B** with every index in $(0,1/2)\cup(1/2,1)$.

11.2 VALENT'S RATES

Let $\lambda_n=\prod_{j=1}^p(pn+e_j)$ and $\mu_n=\prod_{j=1}^p(pn+d_j)$ be indeterminate polynomial rates as in Section 6.5, so that $p\ge3$, $p<\mathfrak{e}-\mathfrak{d}<p(p-1)$ and $\nu=((\mathfrak{e}-\mathfrak{d})/p-1)/(p-2)\in(0,1)$, and consider the associated Stieltjes problem in the variable $x=\lambda^2$. Put

$$\hat\Phi(x):=\Phi(\sqrt{x})=\tau_p\sin(\pi/p)\,x^{1/p}-\frac{\kappa\pi}{p},\qquad\theta_F(x):=\hat\Phi(x)-\frac{\pi}{4}-\frac{\nu\pi}{2},$$

$$\Upsilon(x):=\sqrt{x}\,Q(\sqrt{x})=c_Q\,x^\nu,$$

with Q from (21), $c=(p/2)^{p/2}$ and $c_Q=(2(p-2)c)^{1-2\nu}\Gamma(1-\nu)/\Gamma(\nu)>0$.

Theorem 11.3 — Critical indices for Valent's rates

For every indeterminate polynomial birth–death problem of this kind:

(a) $a(\mu_F) = \frac{1-\nu}{2}$, and $a(\mu_\sigma) = \frac{1+\nu}{2}$ for every $\sigma \in (\mathbb{R} \cup \{\infty\}) \setminus \{F\}$;

(b) every N-extremal solution lies in **B**;

(c) there is $C_{\mathrm{res}} > 0$ such that, along the large support points,

$$\frac{W_F(x)\hat{\Phi}'(x)}{\Upsilon(x)} \to C_{\mathrm{res}} \sin \nu\pi, \qquad W_\infty(y)\Upsilon(y)\hat{\Phi}'(y) \to \frac{\sin \nu\pi}{C_{\mathrm{res}}},$$

$$(\sigma - F)^2 W_\sigma(y)\Upsilon(y)\hat{\Phi}'(y) \to \frac{\sin \nu\pi}{C_{\mathrm{res}}},$$

the last for $y \geq y_0(\sigma)$, where $y_0(\sigma) \to \infty$ as $\sigma \to F$. In particular, for the Friedrichs point x_k and the Krein point y_k of the same period,

$$W_F(x_k)W_\infty(y_k)\hat{\Phi}'(x_k)\hat{\Phi}'(y_k)\Upsilon(y_k)/\Upsilon(x_k) \to \sin^2 \nu\pi,$$

a law with no free constant.

The proof turns boundary values into residues with two facts about functions of positive imaginary part.

Lemma 11.4. For $\sigma \in \mathbb{R}$, $m_\sigma := -D/G_\sigma$ is a Nevanlinna function, and so is $m_\infty := B/D$. Each is real meromorphic with simple real poles exactly on $\mathcal{Z}_\sigma$, and the residue at $x \in \mathcal{Z}_\sigma$ is $-W_\sigma(x)$.

Proof. With $u = z$, $v = \bar{z}$ in (32), $\mathrm{Im}(B\bar{D})(z) = \mathrm{Im}\, z \sum_k |\hat{p}_k(z)|^2$, so B/D, $G_\sigma/D = B/D + \sigma$ and $-D/G_\sigma$ map the upper half-plane to itself. The poles are the zeros of G_σ (of D), real and simple because $AD - BC = 1$ and $W_\sigma > 0$, with residues $-D/G_\sigma'$ and B/D'. □

Lemma 11.5. Let m be a meromorphic Nevanlinna function with simple real poles x_j and residues $-r_j < 0$.

(i) If $x_- < x_k < x_+$ and $[x_-, x_+]$ contains no other pole, then

$$r_k \geq \big(m(x_-) - m(x_+)\big)\frac{(x_k - x_-)(x_+ - x_k)}{x_+ - x_-}.$$

(ii) If $x_k < x_1 < x_2$ and $(x_k, x_2]$ contains no pole, then

$$r_k \leq \big(m(x_2) - m(x_1)\big)\frac{(x_1 - x_k)(x_2 - x_k)}{x_2 - x_1}.$$

Proof. By the Herglotz representation, $m(y) - m(v) = c_1(y - v) + \sum_j r_j \frac{y-v}{(x_j - y)(x_j - v)}$ with $c_1 \geq 0$, for real y, v off the poles. In (i) take $y = x_-$, $v = x_+$: every term except the k-th is non-positive. In (ii) take $y = x_2$, $v = x_1$: every term is non-negative. □

Proof of Theorem 11.3. **Step 1** (*the ratio*)**.** By the proof of Corollary 6.5, $D(\lambda^2) = C_1\lambda\beta_\infty(\lambda)$ and $G_F(\lambda^2) = C_2\alpha_\infty(\lambda)$ with real constants $C_1, C_2 \neq 0$: in each pair both functions are entire of order less than one in $x = \lambda^2$ with the same zeros, counted with multiplicity. In $m_F = -D/G_F = -(C_1/C_2)\lambda\beta_\infty/\alpha_\infty$ the amplitude $\mathscr{A}(\lambda)$ of Theorem 6.2 cancels, and with $\sin(\Phi - \pi/4 + \frac{\nu\pi}{2}) = \sin(\theta_F + \nu\pi)$ and $\sin(\pi/4 + \frac{\nu\pi}{2} - \Phi) = -\sin\theta_F$ we get, for every $\eta \in (0, \pi/2)$,

$$m_F(x) = -C_{\mathrm{res}}\mathrm{Y}(x)\big[\cos\nu\pi + \sin\nu\pi\cot\theta_F(x) + o(1)\big], \qquad C_{\mathrm{res}} := -C_1/C_2, \tag{34}$$

uniformly on $\{\mathrm{dist}(\theta_F(x), \pi\mathbb{Z}) \geq \eta\}$, with an additive error because the bracket may vanish there. Its sign is forced: between consecutive poles a Nevanlinna function increases, while $\cot\theta_F$ decreases; comparing the points with $\theta_F \equiv \eta$ and $\equiv 2\eta$ in one period, which are separated by no pole for large x since $|\alpha_\infty| \geq |\mathscr{A}|(\sin\eta - |r_1|) > 0$ there, gives $C_{\mathrm{res}} > 0$. (This also makes the result independent of the sign conventions in Theorem 6.2.)

Step 2 (*the support points*)**.** By (34), $|m_F| \geq c\mathrm{Y} \to \infty$ at phase distance $\geq \eta$ from both $\pi\mathbb{Z}$ and $-\nu\pi + \pi\mathbb{Z}$, while m_F is large negative at $\theta_F = k\pi + \eta$, large positive at $(k+1)\pi - \eta$ and increasing between poles. Hence for large x the poles of m_F (the points of $\mathscr{Z}_F$) are one per period with $\theta_F = k\pi + o(1)$, and the points where m_F takes a given bounded value, in particular the zeros of D ($m_F = 0$) and of $G_\sigma = G_F + (\sigma - F)D$ ($m_F = 1/(\sigma - F)$), are one per period with $\theta_F = k\pi - \nu\pi + o(1)$. Since $\hat{\Phi}(x) \asymp x^{1/p}$, the k-th large support point is $\asymp k^p$ in every case.

Step 3 (*the residues*)**.** Let $x_k \in \mathscr{Z}_F$ be large, fix η, and take the points $x_\pm$ with $\theta_F(x_\pm) = \theta_F(x_k) \pm \eta$ and x_1, x_2 with $\theta_F = \theta_F(x_k) + \eta, \theta_F(x_k) + 2\eta$; for large k no other pole lies in the intervals used, and across one period Y and $\hat{\Phi}'$ change by factors $1 + O(x^{-1/p})$. By (34), $m_F(x_-) - m_F(x_+) = C_{\mathrm{res}}\mathrm{Y}(2\sin\nu\pi\cot\eta + o(1))$ and $m_F(x_2) - m_F(x_1) = C_{\mathrm{res}}\mathrm{Y}(\sin\nu\pi/\sin 2\eta + o(1))$, while the geometric factors in Lemma 11.5 are $(\eta/2 + o(1))/\hat{\Phi}'$ and $(2\eta + o(1))/\hat{\Phi}'$. Hence

$$\eta\cot\eta\,(1 + o(1)) \leq \frac{W_F(x_k)\hat{\Phi}'(x_k)}{C_{\mathrm{res}}\sin\nu\pi\,\mathrm{Y}(x_k)} \leq \frac{2\eta}{\sin 2\eta}(1 + o(1)),$$

and letting $\eta \to 0$ gives the first limit in (c). For the Krein solution, $m_\infty = B/D = -1/m_F - F$; writing $\theta_F = k\pi - \nu\pi + \epsilon$, (34) gives $m_\infty = (C_{\mathrm{res}}\mathrm{Y})^{-1}[\cos\nu\pi - \sin\nu\pi\cot\epsilon + o(1)] - F$ for $|\epsilon| \geq \eta$, and the same four points (now around the pole $\epsilon \approx 0$) give the second limit; F cancels in the differences. For $\sigma \neq F$ put $s := \sigma - F$; then $m_\sigma = m_F/(1 - sm_F) = -1/s + 1/(s(1 - sm_F))$, and where $|s|\mathrm{Y} \gg 1$ this is $-1/s + (s^2 C_{\mathrm{res}}\mathrm{Y})^{-1}[\cos\nu\pi - \sin\nu\pi\cot\epsilon + o(1)]$, which gives the third limit; the threshold $y_0(\sigma)$ is where $|\sigma - F|\mathrm{Y}(y_0)$ becomes large, so it grows as $\sigma \to F$. The product law is the product of the first two limits.

Step 4 (*the indices*)**.** By (c), $W_F(x_k) \asymp \mathrm{Y}/\hat{\Phi}' \asymp x_k^{1+\nu-1/p}$ and $W_\sigma(y_k) \asymp y_k^{1-\nu-1/p}$ for $\sigma \neq F$. With $x_k \asymp k^p$ the terms of $\mathcal{S}_\sigma(\gamma)$ are $\asymp k^{p(2\gamma-1\pm\nu)-1}$, with $+$ for F and $-$ otherwise; finitely many small or negative support points contribute a finite amount. So $\mathcal{S}_\sigma(\gamma) < \infty$ exactly

for $\gamma < (1 \mp \nu)/2$, and at the critical index the terms are $\asymp 1/k$, so $\mathcal{S}_\sigma$ diverges and $\mu_\sigma \in \mathbf{B}$ (Proposition 11.1). □

The values $(1 \mp \nu)/2$ lie on either side of $1/2$, as [7, Thm. 3.8] requires, and every value in $(0, 1/2)$ and $(1/2, 1)$ occurs as ν ranges over the admissible rates. The proof needs only the $o(1)$ boundary values of Theorem 6.2 at points a fixed phase distance from the support: the Nevanlinna property converts values into residues.

11.3 A WORKED CHECK: THE BERG–VALENT QUARTIC PROBLEM

For the quartic rates of Example 5.1 ($p = 4$, $\nu = 1/2$), Berg and Szwarc give

$$\mu_F = \frac{4\pi}{K_0^2} \sum_{n \text{ odd}} \frac{n\pi}{\sinh n\pi}\, \delta_{n^4}, \qquad \mu_\infty = \frac{\pi}{K_0^2}\, \delta_0 + \frac{4\pi}{K_0^2} \sum_{n \geq 2 \text{ even}} \frac{n\pi}{\sinh n\pi}\, \delta_{n^4}$$

[7, (29)–(30)], with the elliptic constant $K_0 = \Gamma(1/4)^2/(4\sqrt{\pi})$ of Example 5.1 (not to be confused with C_{res}), and ask for which c the measure $\mu_c := \sum_{n \text{ odd}} \frac{n\pi}{\sinh n\pi} n^c \delta_{n^4}$ is indeterminate; they show indeterminacy for $c \geq 0$ and determinacy for $c \leq -4$ [7, Ex. 4.1]. With $w := x^{1/4}$, the functions of minimal type with the right zeros and normalisation are $G_F(x) = -\cos\frac{\pi w}{2}\cosh\frac{\pi w}{2}$ and $D(x) = \frac{4d}{\pi^2} w^2 \sin\frac{\pi w}{2}\sinh\frac{\pi w}{2}$ with $d = D'(0) = K_0^2/\pi$, the value for which $1/(DG_F')$ and $-1/(BD')$ reproduce the masses. Differentiating with $d/dx = (4w^3)^{-1} d/dw$,

$$W_F(n^4) = \frac{32d}{\pi^3}\, n^5 \tanh\frac{\pi n}{2} \quad (n \text{ odd}), \qquad W_\infty(n^4) = \frac{2\pi n}{d} \coth\frac{\pi n}{2} \quad (n \geq 2 \text{ even}).$$

So $\mathcal{S}_F(\gamma) \asymp \sum_{n \text{ odd}} n^{8\gamma - 3}$ and $\mathcal{S}_\infty(\gamma) \asymp \sum n^{8\gamma - 7}$, which gives $a(\mu_F) = 1/4$ and $a(\mu_\infty) = 3/4$, both in $\mathbf{B}$, as Theorem 11.3 predicts. In this normalisation $\hat{\Phi}'(x) = \pi/8x^{-3/4}$ and $\mathrm{Y}(y)/\mathrm{Y}(x) = (y/x)^{1/2}$, and the product in Theorem 11.3(c) equals $\tanh(\pi n/2)\coth(\pi m/2)$, which tends to $\sin^2 \pi/2 = 1$. For the threshold, μ_c is a constant multiple of $x^{c/4}\mu_F$; for $c \leq 0$ the weight $x^{c/4}$ is bounded on the support, and Proposition 11.1 gives indeterminacy if and only if $\sum_{n \text{ odd}} W_F(n^4) n^{-8} n^{-c} \asymp \sum n^{-3-c} < \infty$. Hence *$\mu_c$ is indeterminate if and only if $c > -2$*, and the threshold measure μ_{-2} is determinate. The closed forms, the masses and the product law are also checked numerically (Appendix B). Table 2 collects the indices that the criterion gives on the closed forms we know.

PROBLEM	PARAMETER	$a(\mu_F)$	$a(\mu_\sigma)$, $\sigma \neq F$
Valent's rates (Theorem 11.3)	$\nu \in (0,1)$	$(1-\nu)/2$	$(1+\nu)/2$
Berg–Valent quartic	$\nu = 1/2$	$1/4$	$3/4$
Valent's cubic [38], GLRV P2 [24]	$\nu = 1/3$	$1/3$	$2/3$
GLRV P1 [24]	$\nu = 2/3$	$1/6$	$5/6$
Al-Salam–Carlitz	$\gamma_q = \log a / \log(1/q)$	$(1-\gamma_q)/2$	$(1+\gamma_q)/2$ (Krein)
Stieltjes–Wigert	$0 < q < 1$	$1/4$	$3/4$ (Krein and $\sigma = 1$)

Table 2. *Critical indices of N-extremal solutions; all listed measures lie in* B*. The first four rows are Theorem 11.3. The last two rows are q-families of order zero, outside the class of this paper; their entries follow from Proposition 11.1 and the closed forms of [7, Ex. 4.2–4.3], and cover only the solutions named. For Al-Salam–Carlitz ($0 < q < 1 < a < 1/q$), $W_F \asymp x^{1+\gamma_q}$ on the Friedrichs support and $W_\infty \asymp x^{1-\gamma_q}$ on the Krein support; for Stieltjes–Wigert, $W_F \asymp x^{3/2}$ and $W_\infty, W_1 \asymp x^{1/2}$. At the critical index the terms of $\mathcal{S}$ have a positive limit along the geometric support, so $\mathcal{S}$ diverges. These growth rates were checked numerically (Appendix B).*

12 NUMERICAL CONFIRMATION

This section compares the headline formulas with eigenvalues computed by a method that is independent of the asymptotic machinery of the paper: it uses no frames, no normal form, and no Airy or Bessel functions. It does rely on the exact identification of the spectra in Theorem 2.2(i), as described next. The programs, their outputs and the computing details are listed in Appendix B.

Method. By Theorem 2.2(i), $p_{2m+1}(\lambda)$ is a nonzero multiple of the second coordinate $v_{2m,2}$ and $p_{2m+2}(\lambda)$ of the first coordinate $v_{2m+1,1}$, because $\widetilde{P}$ vanishes on odd and $\widetilde{R}$ on even sites. The eigenvalues of the truncation to the sites $0, \dots, n-1$ are the zeros of p_n, so the eigenvalues of the odd-size truncations converge to the zeros of β_∞ (the spectrum of H_0) and those of the even-size truncations to the zeros of α_∞ (the spectrum of $H_{\pi/2}$); a last diagonal entry $\varpi \tan t / P_{2m}^2$ on an odd truncation imposes the boundary condition of H_t at the cut. The k-th positive eigenvalue is located by Sturm counts, the number of negative pivots of the LDL^{T} factorisation of $J_n - \lambda$, in float64. Each eigenvalue is computed for three truncation sizes $n, 2n, 4n$ and extrapolated in n^{-p}, where p is the decay exponent of the neglected tail of the shear weights. For each eigenvalue we report the residual $r_j := \Phi(\lambda_j) - j\pi - \theta_t$, with θ_t the predicted offset (t for balanced links, θ_0 or $\pi/2 - \theta_0$ for dimerised links), reduced modulo π where the labels are not tracked.

Validation. The eigenvalues of a truncation of size ≈ 800 agree with a 40-digit bisection to relative $1.7 \cdot 10^{-15}$. The spread between the three truncation sizes is

recorded for every eigenvalue; it is small compared with the residuals in every row of Table 3.

The test chains. All parameters are new for this paper. B1–B3 are shifted powers with a curvature term, $a_k = c(k+s)^{\alpha}(1+\kappa_2(k+s)^{-2})$, with $(c,\alpha,s,\kappa_2) = (0.7, 2.25, 0.35, 0.6)$, $(1.3, 3.75, -0.2, -0.3)$ and $(0.5, 5, 0.6, 0)$. SQ5m0, SQ5m3 and SQ6m1 are blocks of the squeezing operators of Corollary 10.1 ($k = 5, 6$). RS is $a_k = k^3(1 + 0.5\,k^{-3/2})$, a rough part without smoothness. D1 and D2 are dimerised, $a_k = c\,k^{\alpha}(1 + (\kappa + \kappa'(-1)^k)/k)$ with $(c,\alpha,\kappa,\kappa') = (1, 3.5, 0.2, 0.45)$ and $(0.8, 2.75, -0.3, -0.25)$. BD6 is the symmetrised birth–death chain with $p = 6$, $e = (2,3,4,5,6,7)$ and $d = (0,1,2,3,3,3)$, for which $\nu = 0.375$ and the Krein offset of Corollary 6.5 is $(\mathfrak{d} + \mathfrak{e})/(2p^2) - \nu/2 = 0.3542$ levels.

Results. On every balanced chain $N_\lambda r_j$ converges to the second-order constant of Theorem 9.1, for all extensions tested. It agrees with it to seven decimal places at $N_\lambda \approx 320$ on B3 and SQ6m1 and to six on B2, to six digits at $N_\lambda \approx 80$ on the $k = 5$ squeezing blocks and to five digits at $N_\lambda \approx 29$ on B1, where the slowly decaying tail ($\alpha = 2.25$) limits the levels reached. The deviation decays like N_λ^{-2}, consistent with a remainder much smaller than the proved $O(N_\lambda^{-4/3})$. The extension enters only through t, as Theorem 2.2 and the normalisation by ϖ predict: the three columns for B3 coincide. On RS, $N_\lambda^{1/2} r_j$ agrees with the constant of Proposition 8.4 plus the N_λ^{-1} term to five digits. On the dimerised chains the residuals relative to θ_0 and $\pi/2 - \theta_0$ tend to zero (on D1 from $1.7 \cdot 10^{-3}$ at $N_\lambda = 10$ to $9.6 \cdot 10^{-5}$ at $N_\lambda = 320$, with local decay exponent rising from 0.6 to 0.9); the offsets are confirmed. Theorem 6.2 gives no rate, and we do not infer one: on D1 the decay approaches N_λ^{-1}, but the local exponent is still moving at the largest level, and nothing says that other dimerised chains behave the same way over an accessible range. For the generic extension $t = 1$ of D1 ($\kappa' > 0$) the eigenvalues follow the law of $H_{\pi/2}$, as Corollary 6.3(ii) predicts. D2 ($\kappa' < 0$) behaves in the same way; for its extension H_0, whose tail decays only like $n^{-1.25}$, only two levels ($N_\lambda \le 14$) were reached. On BD6 the Krein offset is confirmed in the same way.

A diagonal. Theorem 6.4 was tested on D1 with the diagonal $b_k = 0.5(1+k)^{1/2}$. Modulo π the residuals relative to θ_0 and $\pi/2 - \theta_0$ decrease to $1.36 \cdot 10^{-4}$ at $N_\lambda = 227$ and $7.7 \cdot 10^{-4}$ at $N_\lambda = 56$, against $1.32 \cdot 10^{-4}$ and $7.1 \cdot 10^{-4}$ at the same levels for D1 without diagonal. For the rate theorem with a diagonal, the chain $a_n = n^3$, $b_n = 2n^{3/2}$ gives $N_\lambda^{1/2}(\Phi_A(\lambda_j) \bmod \pi) = 2.217, 2.224, 2.225$ at $N_\lambda \approx 29, 81, 160$, against the first-order prediction $\frac{b_0}{4c\alpha}\,\mathrm{B}(\frac{\alpha-\beta-1}{2\alpha}, 1/2) = 2.2222$; so the $N_\lambda^{1+\beta-\alpha}$ term of Theorem 8.1 is attained with this constant. The truncation error of this chain decays only like $n^{-1/2}$ and limits the absolute accuracy of r_j to about 10^{-4}, which is not enough to test Conjecture 13.1.

chain	t	quantity	measured	predicted	N_λ
B1	0	$N r_j$	0.40696451	0.40696228	29
B1	$\pi/2$	$N r_j$	0.40696461	0.40696228	28
B2	0	$N r_j$	-0.01040757	-0.01040766	320
B2	1	$N r_j$	-0.01040757	-0.01040766	319
B2	$\pi/2$	$N r_j$	-0.01040757	-0.01040766	320
B3	0	$N r_j$	0.10404384	0.10404382	322
B3	1	$N r_j$	0.10404384	0.10404382	320
B3	$\pi/2$	$N r_j$	0.10404384	0.10404382	321
SQ5m0	0	$N r_j$	0.03341076	0.03341061	80
SQ5m0	$\pi/2$	$N r_j$	0.03341076	0.03341061	79
SQ5m3	0	$N r_j$	0.03341076	0.03341061	80
SQ5m3	$\pi/2$	$N r_j$	0.03341075	0.03341061	81
SQ6m1	0	$N r_j$	0.03742927	0.03742925	319
SQ6m1	$\pi/2$	$N r_j$	0.03742927	0.03742925	320
RS	0	$N^{1/2} r_j$	0.24065	0.24065	320
RS	$\pi/2$	$N^{1/2} r_j$	0.24064	0.24064	320
D1	0	r_j	$9.61 \cdot 10^{-5}$	0	319
D1	1	r_j	$-5.56 \cdot 10^{-3}$	0	27
D1	$\pi/2$	r_j	$7.12 \cdot 10^{-4}$	0	56
D2	0	r_j	$3.30 \cdot 10^{-3}$	0	14
D2	$\pi/2$	r_j	$1.64 \cdot 10^{-4}$	0	320
BD6	0	r_j	$-2.12 \cdot 10^{-4}$	0	320
BD6	$\pi/2$	r_j	$-1.99 \cdot 10^{-3}$	0	41

Table 3. *Residuals at the largest level computed. For balanced chains the prediction is the second-order constant $(\kappa_2 + (2\alpha - 1)/24)\, J_\alpha$ of Theorem 9.1 (for the squeezing blocks $(k-1)^2 J_{k/2}/(48k)$); for RS it is the constant of Proposition 8.4 plus the N^{-1} term; for the dimerised chains the residual relative to the offsets of Corollary 6.3 should tend to zero. The full sequences over $N_\lambda = 10 \dots 320$ are part of the data listed in Appendix B.*

13 OPEN PROBLEMS

We list the questions that the methods of this paper leave open, in the form in which we would attack them next.

Rates and the second term for dimerised links. Theorem 6.2 has no rate. The proof loses control in two places: the matching window, whose size is constrained from both sides by $N_\lambda^{\max(2\alpha/(2\alpha-1),\alpha/(\alpha-1+\delta))} \ll N_M \ll N_\lambda^{\alpha/(\alpha-1)}$, and the uniformity of the drift sum near the turning point. On the chain D1 of Section 12 the error decays roughly like N_λ^{-1} over the accessible range; we do not know whether this is the general rate. A proof would combine the Airy comparison of Section 7, which does not care about κ', with a Bessel comparison of Olver type on the far field. The second-order constant should then contain, besides the terms of Theorem 9.1, a contribution of the Bessel order ν.

Larger diagonals. The hypothesis $\beta < \alpha - 1$ (or $\beta < \alpha - 1 - 2|\kappa'|$) is exactly what makes the coordinates of Theorem 2.2 exist, and the cost of the diagonal in Theorem 8.1 is $N_\lambda^{1+\beta-\alpha}$. For a smooth diagonal $b_n = b(n)$ the natural phase is

$$\Phi_{A,b}(\lambda) := \int_0^\infty \arcsin\Big(\min\Big(1, \frac{\lambda - b(x)}{2A(x)}\Big)\Big)dx.$$

Conjecture 13.1. Under Hypothesis 8.1 with $b_n = b(n)$, b smooth with $x^j b^{(j)}(x) = O(x^\beta)$, $\beta < \alpha - 1$, $\psi(\lambda) - \Phi_{A,b}(\lambda) = O(N_\lambda^{-\min(1,\delta'')})$.

To first order in b, $\Phi_{A,b} - \Phi_A = -\frac{b_0}{4c\alpha}\,\mathrm{B}(\frac{\alpha-\beta-1}{2\alpha}, 1/2) N_\lambda^{1+\beta-\alpha}$ for $b(x) = b_0 x^\beta$, $A = cx^\alpha$, and Section 12 confirms this as the leading diagonal term of $\psi - \Phi_A$ on one chain to about 0.2%; our truncation data are not accurate enough to decide whether $N_\lambda(\psi - \Phi_{A,b})$ stays bounded. For $\beta \geq \alpha - 1$ the diagonal is no longer a perturbation of the coordinates at infinity, and even the limit-circle property depends on finer structure.

Leading-order dimerisation. If the even and odd links have different leading coefficients, $a_n = c_e n^\alpha(1 + \kappa_e/n + O(n^{-2}))$ for even n and $a_n = c_o n^\alpha(1 + \kappa_o/n + O(n^{-2}))$ for odd n with $c_e \neq c_o$, the matrix is in the limit-point case for every $\alpha > 0$ (one solution at $\lambda = 0$ grows geometrically; for birth–death rates this is [38, §2, Remark 3]), and for $\alpha > 1$ its spectrum is discrete [37, Thm. A]. The spectrum then has two turning points, at the edges $|o - e| = \lambda$ and $o + e = \lambda$ of the local band, where $o(y) := c_o y^\alpha(1 + \kappa_o/y)$ and $e(y) := c_e y^\alpha(1 + \kappa_e/y)$. Put

$$\Phi_r(\lambda) := \int_{-1/2}^\infty \arccos\Big(\mathrm{clip}\Big(\frac{|o|^2 + |e|^2 - \lambda^2}{2|o||e|}\Big)\Big)dx, \qquad o = o(2x+1),\ e = e(2x+2),$$

with clip the projection onto $[-1, 1]$.

Conjecture 13.2. The positive eigenvalues $\lambda_1 < \lambda_2 < \cdots$ satisfy $\Phi_r(\lambda_j) = (j - 1/2)\pi + O(1/j)$, whether or not 0 is an eigenvalue.

The absolute values matter: with o and e themselves the integrand jumps where o or e changes sign, and the conjecture fails for $\kappa_o < 0$. The two Maslov contributions $1/4$ come from the two simple turning points; a proof needs Airy windows for a period-two system. The leading term $\#\{j : \lambda_j \le \lambda\} = \Phi_r(\lambda)/\pi + O(N_\lambda^{2/3})$ follows from a Dirichlet decoupling into cells of length $N_\lambda^{1/3}$. Section 12 reports a test.

Sharper remainders and general symbols. In Theorem 9.1 the remainder $O(N_\lambda^{-4/3})$ comes from the turning region (Lemma A.13); numerically it is $O(N_\lambda^{-3})$ for analytic symbols. We expect an expansion of $\psi - \Phi$ in odd powers of N_λ^{-1} for links with a convergent expansion in $1/n$. For a general symbol in Hypothesis 8.1 the natural candidate for the second term is the regularised Weyl–Moyal term of Remark 9.1; we have proved it only for power symbols.

Rough perturbations. Proposition 8.4 shows that a non-oscillating rough part $e_n \sim \kappa_e n^{-1-\delta''}$ shifts the phase by an amount of exact order $N_\lambda^{-\delta''}$. An alternating rough part $(-1)^n n^{-1-\delta''}$ is a decaying dimerisation, and cancellation between neighbouring links should make its effect on the phase smaller than $N_\lambda^{-\delta''}$; its true order, and a rate for Hypothesis 2.2 with $\delta \le 3/4$, are open.

The limits $\alpha \to 1^+$ and $2|\kappa'| \to \alpha - 1$. All constants in this paper blow up as $\alpha \to 1^+$, where the limit-circle property is lost through Carleman's condition, and in the dimerised case as $\nu \to 0$ or $\nu \to 1$, where it is lost at the edge of Valent's indeterminacy range. A description that is uniform up to these boundaries would describe the transition from discrete spectrum of an extension to the spectrum of a self-adjoint operator.

14 MOTIVATION AND THE USE OF AI

The question of this paper came from quantum geometry, and I reached quantum geometry from an unexpected direction. Since late June 2026 I had been building an attention layer whose edges carry operators on the Albert algebra, the exceptional Jordan algebra, and whose pair interactions are classified in advance: the tensor square of a traceless element splits into five irreducible channels, $26 \otimes 26 = 1 \oplus 26 \oplus 52 \oplus 273 \oplus 324$ [19]. In October that work led me to the quantum tetrahedron. The five channels turned out to label a basis of the shapes of a quantum tetrahedron whose four faces carry the exceptional group F_4. For SU(2) faces the orientation operator of that tetrahedron is

the triple product $J_1 \cdot (J_2 \times J_3)$, the four-valent volume operator of loop quantum gravity. The volume operator then took me to the dynamics of loop quantum gravity, where it enters the construction of the Hamiltonian constraint. There I tried self-dual variables, in which the edge labels belong to $\mathfrak{su}(1,1)$ instead of $\mathfrak{su}(2)$. In a recoupling basis the triple product becomes the zero-diagonal Jacobi matrix with cubically growing links mentioned in the introduction. It is in the limit-circle case. So the volume operator is not determined until a boundary condition at infinity is chosen, and different choices give different spectra.

That raised the question this paper answers in general: given a Jacobi matrix in the limit-circle case, where exactly are the eigenvalues of each of its self-adjoint extensions? For the cubic chains the answer first appeared numerically, as a Bohr–Sommerfeld rule in which the extension angle enters as a phase. The rest of the paper is what was needed to prove that rule, and to see how far it extends: to every growth exponent $\alpha > 1$, to dimerised links, and to the moment problems of Section 11. The companion paper on the volume gap [20] grew from the same operator, with $\mathfrak{su}(2)$ labels. Appendix C describes how the results were reached.

Use of AI. I posed the questions, chose the problems and the order in which to pursue them, and set the standard a result had to meet. I directed the computations and decided what this paper claims. Large language models were used throughout. Anthropic's Claude was used in two ways. In sessions on claude.ai it carried out the earlier calculations on quantum tetrahedra and began the calculation on the dynamics, from which this paper's question came. Through Claude Code it wrote and ran the programs of Appendix B, ran the agent campaigns, and drafted and revised the text. It also gave the first written proofs of the results here, among them the rotation identity and the discrete Bohr–Sommerfeld rule (Theorem 5.1), the discrete Olver bound (Theorem 7.1), the second-order term (Theorem 9.1), the dimerised law (Theorem 6.2) and the critical indices (Theorem 11.3). Separate Claude agents refereed each result as it was written, and an independent Claude referee read the whole paper. OpenAI's GPT models were used through ChatGPT and Codex. ChatGPT produced the first document on the tetrahedron in the Albert algebra. A Codex session independently checked the rate and the counting law, and another compared Corollary 10.1 with directly computed spectra. GPT also reviewed the paper. Google's Gemini reviewed only the companion paper and had no part in this one. Review comments were adopted only after they had been checked against the source and our own computations. The claims rest on the following, and not on any model's assertion: the written proofs in this paper, none of which depends on a computation; exact symbolic checks of the identities listed in Appendix B; the independent eigenvalue computations of Section 12; and cited results. Responsibility for the content, and for any error that remains, is mine.

A TECHNICAL ESTIMATES AND AUXILIARY PROOFS

This appendix contains the estimates on which the main results rest but whose ideas are already explained in the main text: the geometry of the blocks near the turning point and the discretisation of the Airy equation (for Section 4), the error sums of the normal form, the comparison with the Bessel system in weighted coordinates and the cost of a diagonal (for Section 6), the window lemmas of the rate theorem (Section 8), the Langer functions on the half-line and the steps of the second-order theorem (Section 9), and two algebraic derivations.

A.1 BLOCKS NEAR THE TURNING POINT

Throughout this subsection Hypothesis 4.1 holds, m_λ is the least $m \ge m_0$ with $\mu_m < 2$ and $N'_\lambda := N_{m_\lambda}$, where m_0 is fixed so large that A is increasing on $[m_0, \infty)$; ℓ, x_m, m_L and m_R are as in Section 4.

Lemma A.1 (Block geometry). There are m_0 and C, and for each λ large the following hold.

(a) $\log A_{m+1} - \log A_m = (2\alpha/N_m)(1 + O(N_m^{-\delta}))$ for $m \ge m_0$; every link is increasing in n from some n_1 on; and $|\bar\ell_{m+1} - \bar\ell_m| \le C N_m^{-2}$.

(b) $N'_\lambda = N_\lambda(1 + o(1))$. For $m \ge m_\lambda$: $u_m \ge \min(1, \alpha/4(N_m - N'_\lambda)/N'_\lambda)$; $u_m \le C(N_m - N'_\lambda + 2)/N_\lambda$ when $N_m \le 2N'_\lambda$; $\mu_m - \mu_{m+1} \le C\mu_m/N_m$; $\mu_m \le C(N_\lambda/N_m)^\alpha$; and $u_m \ge 1$ when $N_m \ge 2N'_\lambda$.

Proof. (a) $\log A_{m+1} - \log A_m = \alpha \log(1 + 2/N_m) + \Delta\hat e_m$. For the links, $\log a_{n+1} - \log a_n = (\alpha \mp 2\kappa')/n + o(1/n) > 0$ because $\alpha > 2|\kappa'|$.

(b) $N'_\lambda = N_\lambda(1 + o(1))$ because $A_m = c\, N_m^\alpha(1 + o(1))$. Summing (a) from m_λ to $m - 1$, with the factors $1 + O(N_i^{-\delta}) \ge 1/2$ and $2/N_i \ge \int_{N_i}^{N_i+2} dN/N$, gives $\log(A_m/A_{m_\lambda}) \ge \alpha/2 \log(N_m/N'_\lambda)$, and $1 - e^{-x} \ge x/2$ on $[0,1]$ gives the lower bound on $u_m = 2 - \lambda/A_m \ge 2(1 - A_{m_\lambda}/A_m)$ for $N_m \le 2N'_\lambda$. The upper bound uses $\lambda \ge 2A_{m_\lambda - 1}$ and the upper estimate $\log(A_m/A_{m_\lambda - 1}) \le 4\alpha(N_m - N'_\lambda + 2)/N'_\lambda$. The bound on $\mu_m - \mu_{m+1}$ is (a). For $N_m \ge 2N'_\lambda$, (a) gives $\log(A_m/A_{m_\lambda}) \ge \alpha(1 - CN_\lambda^{-\delta}) \log 2$, and since $\lambda < 2A_{m_\lambda}$ this yields $u_m \ge 2 - 2 \cdot 2^{-\alpha(1 - CN_\lambda^{-\delta})} \ge 1$ once $\alpha(1 - CN_\lambda^{-\delta}) \ge 1$, which holds for λ large because $\alpha > 1$. □

Lemma A.2 (Start of the window). For $\lambda \ge \lambda_0(K)$ the hypothesis of Lemma 4.1 holds with $k_1 = 2m_L$, and $\psi_{2m_L - 1} \in (m_L\pi - \pi/2, m_L\pi)$.

Proof. At block m_L, $\lambda - 2A_{m_L} = A_{m_L} x_{m_L}/(4\ell^2) \ge c\lambda K N_\lambda^{-2/3}$. For $n_1 \le k \le 2m_L$ the links increase and $a_{2m_L} \le A_{m_L}(1 + C/N_\lambda)$, so $a_k + a_{k+1} \le 2A_{m_L}(1 + C/N_\lambda)$; the finitely many

$k < n_1$ are covered for λ large; and $|b_k| \le CN_\lambda^\beta = o(\lambda N_\lambda^{-2/3})$ because $\beta < \alpha - 2/3$. The location of ψ_{2m_L-1} is Lemma 4.1 with $k = 2m_L - 1$. □

Lemma A.3 (Airy coordinates). Fix $K \ge 1$. For $\lambda \ge \lambda_0(K)$ and $m_L \le m < m_R$:

(a) $(X,Y)_{m+1} = M_m(X,Y)_m$ with $M_m = I + h_m \mathrm{B}(x_m) + \mathscr{E}_m$, where $h_m := x_m - x_{m+1} = \ell^{-1}(1 + o(1))$ and $\|\mathscr{E}_m\| = o(1/\ell)$, uniformly in m;

(b) $\|M_m - I\| \le C(K)/\ell < 1$, and $\varrho_{m+1} = \widehat{M}_m(\varrho_m)$ with $\widehat{M}_m$ the principal lift;

(c) $\varrho_{m_L} \in (-\pi/2, \pi/2)$ is the principal angle of $(X,Y)_{m_L}$, $X_{m_L} > 0$, and $Y_{m_L}/X_{m_L} \ge -C\ell/N_\lambda$.

Proof. **Step 1** (*block matrices*)**.** By Lemma 3.1, $D_{m+1}^{-1} T_m D_m = e^{\varepsilon_m \Sigma_3} T_1(\mu_m)$. Hence $(X,Y)_{m+1} = M_m(X,Y)_m$ with $M_m = C_\ell^{-1}(-e^{\varepsilon_m \Sigma_3} T_1(\mu_m)) C_\ell$, and with $\delta_m := u_m/2 = -x_m/(8\ell^2)$ the exact part is

$$C_\ell^{-1}\big(-T_1(2 - 2\delta_m)\big)C_\ell = \begin{pmatrix} 1 - 4\delta_m + 2\delta_m^2 & (1-\delta_m)(2-\delta_m)/(2\ell) \\ -8\ell\delta_m(1-\delta_m) & 1 - 4\delta_m + 2\delta_m^2 \end{pmatrix}$$

(by direct multiplication). Since C_ℓ^{-1} has norm at most 4ℓ, the mismatch contributes at most $C\ell|\varepsilon_m| \le C\ell/N_\lambda$, and a diagonal contributes at most $C\ell(|b_{2m}| + |b_{2m+1}|)/A_m \le C\ell N_\lambda^{\beta-\alpha}$ by the computation before (12); both are $o(1/\ell)$ because $\ell^2 \asymp N_\lambda^{2/3}$ and $\beta < \alpha - 2/3$. The grid steps are $h_m = 4\ell^2 \mu_m (1 - e^{-(\log A_{m+1} - \log A_m)}) = 16\alpha\ell^2 N_\lambda^{-1}(1 + o(1)) = \ell^{-1}(1 + o(1))$ uniformly on the window, by Lemma A.1(a) and the choice of ℓ. Comparing entries, the diagonal of the exact part is $1 + O(K/\ell^2)$, its $(1,2)$ entry differs from h_m by $o(1/\ell)$, and its $(2,1)$ entry equals $h_m x_m + o(K/\ell)$. This is (a).

Step 2 (*lifts*)**.** Write $\Theta_m := \widehat{D}_m^{-1}(\psi_{2m-1})$, a lift of the angle of z_m. We claim $\Theta_{m+1} = \pi + \widehat{C}_\ell\big(\widehat{M}_m(\widehat{C}_\ell^{-1}(\Theta_m))\big)$. Both sides are lifts of $D_{m+1}^{-1} T_m D_m = -C_\ell M_m C_\ell^{-1}$. Deform $(\mu_m, \rho_m, \rho_{m+1}, b)$ linearly to $(2,1,1,0)$; along the path $\|M - I\| < 1$. At the end, $T_1(2)$ carries e_- (angle $-\pi/4$) through $L(2)$ to angle $\pi/4$ and through $U(2)$ to $3\pi/4$, with canonical increments $\pi/2$ each, while $M = \left(\begin{smallmatrix} 1 & 1/\ell \\ 0 & 1 \end{smallmatrix}\right)$ fixes the angle 0; so both sides send $-\pi/4$ to $3\pi/4$, and they agree throughout. Since lifts of linear maps commute with translation by π, this is (b).

Step 3 (*the start*)**.** At m_L, Lemma A.2 and the fact that a positive diagonal matrix maps each open quadrant into itself give $\Theta_{m_L} \in (m_L\pi - \pi/2, m_L\pi)$; since $\widehat{C}_\ell$ maps $(-\pi/2, \pi/2)$ onto $(-3\pi/4, \pi/4)$, $\varrho_{m_L} \in (-\pi/2, \pi/2)$ is the principal angle of $(X,Y)_{m_L}$. By (19) and Lemma 4.1, $X_{m_L} > 0$ and $Y_{m_L}/X_{m_L} = 4\ell(r - c'_{m_L})/(r + c'_{m_L})$ with $r = p_{2m_L}/p_{2m_L-1} \ge 1$ and $c'_{m_L} = 1 + O(1/N_\lambda)$, so $Y_{m_L}/X_{m_L} \ge -C\ell/N_\lambda$. □

Lemma A.4 (Error sums beyond the window). Fix $K \ge 1$, let M be as in Proposition 4.5, and use the split of its proof. For $\lambda \ge \lambda_0(K)$ the hypotheses of Proposition 3.3 hold for $m_R \le m < M$, and

$$\sum_{m_R \le m < M} \mathscr{E}_m + |\xi_{m_R}| + |\xi_M| \le C\big(K^{-3/2} + (\ell K)^{-1}\big) + \eta(\lambda) + C\frac{\ell^2}{KN_\lambda} + C\frac{N_M^{\alpha-1}}{\lambda}[\kappa' \ne 0],$$

with $\eta(\lambda) = C(N_\lambda^{-\delta}\log N_\lambda + N_\lambda^{1+\beta-\alpha} + N_\lambda^{-1}\log N_\lambda)$ and C independent of K, bounded for κ' in compact sets. For $M = \infty$, $|\xi_M|$ is read as $\lim_M |\xi_M| = 0$.

Proof. **Step 1** (*bounds per block*)**.** Let $m \ge m_R$. By (14), $\bar\ell_m \le 2\alpha/N_m$ and monotonicity of s',

$$\bar\sigma_m \le \mu_m s'(\mu_m)\bar\ell_m \le \frac{\alpha\mu_m}{N_m u_m}, \qquad |\tilde\sigma_m| \le \frac{\mu_m|\Delta\hat e_m|}{u_m} \le \frac{C\mu_m}{N_m^{1+\delta}u_m},$$

and $e^{2s} \le 2u^{-1/2}$ gives $|\tilde\varepsilon_m|e^{2s_m} \le CN_m^{-1-\delta}u_m^{-1/2}$, $\bar\varepsilon_m^2 e^{4s_m} \le CN_m^{-2}u_m^{-1}$, $\omega_m \le CN_m^{\beta-\alpha}u_m^{-1/2}$. By (13), $|\xi_m| \le \alpha/(\sqrt{2}N_m u_m^{3/2}) + C|\kappa'|/(N_m\mu_m u_m)$. For the differences we use $|\mu_m - \mu_{m+1}| \le C\mu_m/N_m$, $|\varphi_{m+1} - \varphi_m| \le C/(N_m u_m^{1/2})$, $|\Delta(1/\sin\varphi_m)| \le C/(N_m\mu_m u_m^{3/2})$ and $|\bar\sigma_{m+1} - \bar\sigma_m| \le C\mu_m/(N_m^2 u_m^2)$; the last follows from $\partial_\mu[s(\mu) - s(\mu e^{-h})] \le Ch/u^2$, $\partial_h[\dots] \le C\mu/u$ and $|\bar\ell_{m+1} - \bar\ell_m| \le CN_m^{-2}$, and involves no second difference of r. The factor μ_m in $\bar\sigma$ cancels the $1/\mu_m$ in $\Delta(1/\sin\varphi)$, and

$$|\xi_{m+1} - \xi_m| \le \frac{C}{N_m^2 u_m^{5/2}} + \frac{C|\kappa'|}{N_m^2\mu_m u_m^2}.$$

Step 2 (*the near region* $N_m \le 2N_\lambda'$)**.** Here $\mu_m \asymp 1$ and $u_m \asymp (N_m - N_\lambda')/N_\lambda$ by Lemma A.1(b), and $N_{m_R} - N_\lambda' \ge c\ell K$ because $|x_{m_R}| \ge K$ and $N_\lambda = 16\alpha\ell^3$. Comparing sums with integrals, $\sum N_m^{-2}u_m^{-5/2} \le CN_\lambda^{1/2}(\ell K)^{-3/2} \le CK^{-3/2}$, $\sum N_m^{-2}u_m^{-2} \le C/(\ell K)$, $\sum|\tilde\sigma_m| \le CN_\lambda^{-\delta}\log N_\lambda$, $\sum(|\tilde\varepsilon_m|e^{2s_m} + \bar\varepsilon_m^2 e^{4s_m}) \le C(N_\lambda^{-\delta} + N_\lambda^{-1}\log N_\lambda)$ and $\sum\omega_m \le CN_\lambda^{1+\beta-\alpha}$. The cross terms are bounded by the same sums: $|\xi_{m+1}|\sigma_m \le CN_m^{-2}u_m^{-5/2}$ and $|\xi_{m+1}||\bar\varepsilon_m|e^{2s_m} \le CN_m^{-2}u_m^{-2}$.

Step 3 (*the far region* $N_m > 2N_\lambda'$)**.** Here $u_m \ge 1$ and $\mu_m \le C(N_\lambda/N_m)^\alpha$. The balanced terms are $O(N_m^{-2})$, $O(\mu_m N_m^{-1-\delta})$ and $O(N_m^{\beta-\alpha})$ and sum to $C(N_\lambda^{-1} + N_\lambda^{-\delta} + N_\lambda^{1+\beta-\alpha})$. The dimerised terms are $|\xi_m^{dim}| \le C/(N_m\mu_m) \le CN_m^{\alpha-1}/\lambda$, $|\Delta\xi_m^{dim}| \le CN_m^{\alpha-2}/\lambda$ and $|\xi_m||\bar\varepsilon_m| \le CN_m^{\alpha-2}/\lambda$; summed up to M they are at most $CN_M^{\alpha-1}/\lambda \to 0$. This is the only place where M must be finite when $\kappa' \ne 0$.

Step 4 (*the side conditions and the boundary terms*)**.** $\bar\sigma_{m_R} \le C/(\ell K)$, $|\varepsilon_m|e^{2s_m} \le C\ell/(\sqrt{K}N_\lambda)$ and $\omega_m \to 0$, so the hypotheses of Proposition 3.3 hold for $\lambda \ge \lambda_0(K)$. The boundary terms satisfy $|y_{m_R} - \chi_{m_R}| \le |\xi_{m_R}| \le CK^{-3/2} + C\ell^2/(KN_\lambda)$ and $|y_M - \chi_M| \le |\xi_M| \to 0$. □

A.2 THE FAR FIELD AND A DIAGONAL

Proof of Proposition 6.1. Since $v_{2m+1} = T_m v_{2m-1}$ exactly, $c_{m+1} = (I + \mathscr{E}_m)c_m$ with $\mathscr{E}_m := Y(2m+2)^{-1}(T_m - \Pi_m)Y(2m)$ and $\Pi_m := Y(2m+2)Y(2m)^{-1}$ the model propagator over one block. For 2×2 matrices $(Y^{-1}x)_1 = \det[x, y_+]/\det Y$ and $(Y^{-1}x)_2 = \det[y_-, x]/\det Y$, and det is invariant under $D(N)$; hence $|(\mathscr{E}_m)_{ij}| \le |\hat{y}_{i'}(2m+2)|\,\|\hat{\Delta}_m\|\,|\hat{y}_j(2m)|/\omega_0$ with $\hat{\Delta}_m := D(2m+2)^{-1}(T_m - \Pi_m)D(2m)$ and i' the column other than i.

The true step in balanced coordinates is $D(2m+2)^{-1}D(2m)\cdot X_m^{-1}T_1(\mu_m)X_m$ with $X_m := D_m^{-1}D(2m) = \mathrm{diag}(x_m, 1/x_m)$, and $x_m = 1 + O(N_m^{-\delta})$ by Proposition 2.1(b); since $T_1 - I = O(\mu)$, $X^{-1}T_1X = T_1 + O(\mu_m N_m^{-\delta})$. The model step is the propagator of $d\hat{z}/dN = (\Lambda N^{-\alpha}\mathrm{J} - (\kappa'/N)\Sigma_3)\hat{z}$ over $[2m, 2m+2]$: its drift part is exactly $D(2m+2)^{-1}D(2m)$, its rotation part is $I + (\int_{2m}^{2m+2}\Lambda N^{-\alpha}dN)\mathrm{J} + O(\mu_m^2)$, and the two parts commute up to $O(\mu_m/N_m)$. Since $\mu_m = 2\Lambda N_m^{-\alpha}(1 + O(N_m^{-1}))$, we get $\|\hat{\Delta}_m\| \le C(\mu_m^2 + \mu_m N_m^{-\delta})$.

By (B4), $\|\mathscr{E}_m\| \le C(\mu_m^2 + \mu_m N_m^{-\delta})\max(1, \tau(2m)^{-|1-2\nu|})$. Where $\tau \ge 1$ the sum over $m \ge M$ is at most $C(\Lambda^2 N_M^{1-2\alpha} + \Lambda N_M^{1-\alpha-\delta})$. Where $\tau \le 1$, $(\alpha-1)|1-2\nu| = 2|\kappa'|$ gives terms $C\Lambda^{1-|1-2\nu|}N^{-\alpha-\delta+2|\kappa'|}$ with exponent below $-1-\delta$; their sum from N_{dec} is $CN_{dec}^{-\delta}$ because $\Lambda N_{dec}^{1-\alpha} = \alpha - 1$, and the μ^2 terms give CN_{dec}^{-1}. So $\prod(I + \mathscr{E}_m)$ converges and $|c_\infty - c_M| \le (e^{\epsilon_{far}} - 1)|c_M|$. Finally $v_{2m-1} = Y(2m)c_m$, $Y(2m) \to Y(\infty)$ by (B3), and $\|Y(2m)\|$ is bounded. □

Proof of Theorem 6.4. $\sum |b_k|(\widetilde{P}_k^2 + \widetilde{R}_k^2) \le C\sum k^{\beta-\alpha+2|\kappa'|} < \infty$, which gives Theorem 2.2 and the limit-circle property (Proposition 2.1(d)). The near field carries the diagonal through ω_m as in Lemma A.4, at a cost $CN_\lambda^{1+\beta-\alpha} + CN_M^{1+\beta-\alpha}$. In the far field the b-shears add at most $CN_m^{\beta-\alpha}$ to $\|\hat{\Delta}_m\|$; with the weights of the proof of Proposition 6.1 (where $\tau \le 1$ the weight is $\tau^{-|1-2\nu|} \le CN^{2|\kappa'|}\Lambda^{-|1-2\nu|}$ and $\Lambda^{|1-2\nu|} \asymp N_{dec}^{2|\kappa'|}$) their sum is at most $C(N_M^{1+\beta-\alpha} + N_{dec}^{1+\beta-\alpha})$, which tends to 0; this uses that the exponent $\beta - \alpha + 2|\kappa'|$ of the terms beyond N_{dec} is below -1. The reflection identity covers $\lambda \to -\infty$. The labelling argument of Corollary 6.3 uses $\psi(0) = 0$, which fails for $b \ne 0$; hence only the offsets modulo π are claimed. □

A.3 THE WINDOW LEMMAS OF THE RATE THEOREM

The notation is that of the window subsection of Section 8.

Lemma A.5 (Connection). For λ large, $p = c_K(f_A + \eta_K f_B)$ on the window, with $c_K \in \mathbb{R}\setminus\{0\}$ and $|\eta_K| \le C\,\mathrm{Ai}(x_{n_0})/\mathrm{Bi}(x_{n_0}) \le Ce^{-c/h}$.

Proof. For large k the links increase, and for $k \le n_0 + 2$, $a_k + a_{k+1} \le \lambda\big((t_{in} + 3h)^\alpha + o(1)\big) \le \lambda(2^{-1-\alpha} + o(1))$; the finitely many small k and the diagonal ($|b_k| = o(\lambda)$) are covered for λ large. By Lemma 4.1, p is positive and increasing up to $n_0 + 2$, so $r_p := p_{n_0+1}/p_{n_0} \in [1, C]$. At n_0, $x_{n_0} \ge ch^{-2/3} > c_*$, so by (D1) $f_A = \psi_A(1 + O(d))$ and $f_B = \psi_B(1 + O(d))$ at $n_0, n_0 + 1$.

As x decreases by $|\zeta'|h^{1/3}$ per site, DLMF 9.7.5 and 9.7.7 give $r_A := f_A(n_0+1)/f_A(n_0) = e^k(1+o(1))$ and $r_B := f_B(n_0+1)/f_B(n_0) = e^{-k}(1+o(1))$ with $k := \operatorname{arccosh} Z(t_{in}) \geq c > 0$. Writing $p = c_K f_A + c_B f_B$ and using constancy of Casoratians, $c_K = \operatorname{Cas}[p, f_B]/\operatorname{Cas}[f_A, f_B]$, and at n_0 $\operatorname{Cas}[p, f_B] = a_{n_0+1} p_{n_0} f_B(n_0)(r_B - r_p) \neq 0$ because $r_B < 1 \leq r_p$. Then $\eta_K := c_B/c_K = \frac{f_A(n_0)}{f_B(n_0)} \cdot \frac{r_p - r_A}{r_B - r_p}$, and $\operatorname{Ai}(x)/\operatorname{Bi}(x) \leq Ce^{-4/3x^{3/2}}$. □

Lemma A.6 (Oscillatory form). For sites with $nh \in [t_p^0 + \delta_w, t_{fi}]$,

$$p_n = c_K\, \pi^{-1/2} g_n |x_n|^{-1/4} \big(\cos(\Xi_n - \tfrac{\pi}{4}) + O(\Gamma)\big), \qquad \Xi_n := \int_{X_\lambda}^{n+1/2} \arccos \frac{\lambda}{2A(x)}\, dx.$$

Proof. There $x_n \leq -ch^{-2/3}$, $\mathsf{E} = 1$, and by (D1) $|f_A - \psi_A| \leq dg\mathsf{M}$, while $|\eta_K f_B| \leq Ce^{-c/h} g\mathsf{M}$. DLMF 9.7.9 and 9.8.20 give $\operatorname{Ai}(x_n) = \pi^{-1/2}|x_n|^{-1/4}(\cos(\xi_n - \pi/4) + O(h))$ and $\mathsf{M}(x_n) = \pi^{-1/2}|x_n|^{-1/4}(1 + O(h))$ with $\xi_n = {}^2\!/_3 |x_n|^{3/2}$. Finally the phase is exact: $\xi_n = h^{-1}\int_{t_p}^{nh} \arccos Z(u)\, du$, and the substitution $x = u/h + {}^1\!/_2$ turns $Z(u) = \lambda/(2A(u/h + {}^1\!/_2))$ into the integrand of Ξ_n and the limits into X_λ and $n + {}^1\!/_2$. □

Lemma A.7 (Frame angle in the window). Let m be a block with $u_m \geq 1$ whose sites $2m-1, 2m$ satisfy the hypothesis of Lemma A.6. Then $\chi_m \equiv -\Xi(2m) + O(\Gamma) \pmod{\pi}$, where $\Xi(X) := \int_{X_\lambda}^{X} \arccos(\lambda/2A(x))\, dx$.

Proof. By (19), $z_m = D_m^{-1} v_{2m-1}$ is a nonzero real multiple of $(p_{2m}, -c_m' p_{2m-1})$ with $c_m' = 1 + O(h)$. Over $[2m - {}^1\!/_2, 2m + {}^1\!/_2]$, $\lambda/2A(x) = \frac{\mu_m}{2}(1 + O(h + |e_{2m+1}|)) \leq {}^5\!/_8$, and arccos is Lipschitz on $[0, {}^5\!/_8]$, so $\Xi_{2m} - \Xi_{2m-1} = \pi/2 - \frac{\varphi_m}{2} + O(\Gamma)$; also $g_{2m-1}|x_{2m-1}|^{-1/4}$ and $g_{2m}|x_{2m}|^{-1/4}$ differ by a factor $1 + O(h)$. By Lemma A.6, with $\Theta := \Xi_{2m} - \pi/4$, z_m is a real multiple of $w + O(\Gamma)$, where $w := (\cos\Theta, -\sin(\Theta + \varphi_m/2))$ and $|w|^2 \geq 1 - \sin(\varphi_m/2) = u_m/2 \geq {}^1\!/_2$. In the basis $e_\pm$, with $\Theta' := \Theta - \pi/4 + \frac{\varphi_m}{4}$,

$$w = -\sqrt{2}\sin\big(\tfrac{\pi}{4} - \tfrac{\varphi_m}{4}\big)\sin\Theta'\, e_+ + \sqrt{2}\cos\big(\tfrac{\pi}{4} - \tfrac{\varphi_m}{4}\big)\cos\Theta'\, e_-.$$

Since $\tan^2(\pi/4 - \varphi/4) = (1 - \sin\varphi/2)/(1 + \sin\varphi/2) = e^{-4s}$ by Lemma 3.1, $\widetilde{E}(\mu_m)^{-1} w$ is a positive multiple of $\cos\Theta' e_- - \sin\Theta' e_+ = R(-\Theta')e_-$, whose angle is $-\pi/4 - \Theta' = -\Xi_{2m} + \pi/4 - \frac{\varphi_m}{4}$. As $\|\widetilde{E}^{-1}\| \leq e^s \leq C$ and $|w|^2 \geq {}^1\!/_2$, the frame angle χ_m (the angle of $\widetilde{E}(\mu_m)^{-1} z_m$) is this value modulo π up to $O(\Gamma)$. Finally $\Xi_{2m} = \Xi(2m) + {}^1\!/_2 \arccos(\mu_m/2) + O(h) = \Xi(2m) + \pi/4 - \frac{\varphi_m}{4} + O(h)$. □

Lemma A.8 (Elliptic tail). Let m_1 be a block with $N_{m_1} \geq X_\lambda$ and $u_m \geq 1$ for $m \geq m_1$. Then

$$\Big|\psi - \chi_{m_1} - \sum_{m \geq m_1} \varphi_m\Big| + \Big|\sum_{m \geq m_1} \varphi_m - \int_{2m_1}^{\infty} \arcsin \frac{\lambda}{2A(x)}\, dx\Big| \leq C\Gamma.$$

This is Proposition 3.3 with no drift and the sums taken over the elliptic region, where $u_m \geq 1$; the proof is below.

Proof of Lemma A.8. Apply Proposition 3.3 from m_1 to ∞ with $\bar\varepsilon_m := 0$ and $\bar\sigma_m := s(\mu_m) - s(\mu_m e^{-\bar\ell_m})$, $\bar\ell_m := \log A(N_{m+1}) - \log A(N_m)$. Since $u_m \ge 1$, all powers of u_m are bounded, $\sin\varphi_m \ge c\mu_m$, and $\mu_m \le C(X_\lambda/N_m)^\alpha$. From $|(\log A)^{(j)}| \le Cx^{-j}$ we get $\bar\sigma_m \le C\mu_m/N_m$, $|\bar\sigma_{m+1} - \bar\sigma_m| \le C\mu_m/N_m^2$, hence $|\xi_m| \le C/N_m$ and $|\xi_{m+1} - \xi_m| \le C/N_m^2$; and $|\tilde\sigma_m| \le C\mu_m N_m^{-1-\delta''}$, $|\varepsilon_m| \le C(N_m^{-2} + N_m^{-1-\delta''})$, $\omega_m \le CN_m^{\beta-\alpha}$. Summed over $N_m \ge X_\lambda$ these give $C\Gamma$, the boundary terms $|\xi_{m_1}|$ and $\lim|\xi_M|$ are $O(1/X_\lambda)$, and $\chi_M \to \psi$. For the second difference, φ_m is the midpoint value on $[2m, 2m+2]$ of $2\arcsin(\lambda/2a_{2m+1})$; the function $\arcsin(\lambda/2A)$ has second derivative $O(x^{-2})$ there, and replacing a_{2m+1} by $A(2m+1)$ costs $O(\mu_m|e_{2m+1}|)$; the sums are $O(X_\lambda^{-1} + X_\lambda^{-\delta''})$. □

A.4 LANGER DATA

Proof of Lemma 7.2. $Z \in C^4$ and $Z - 1$ has a simple zero at t_p, so $Z(u) - 1 = (t_p - u)Y(u)$ with $Y(u) = -\int_0^1 Z'(t_p + \vartheta(u - t_p))d\vartheta \in C^3$ and $Y > 0$. Write $\operatorname{arccosh}(1+w) = \sqrt{2w}\,K(w)$ with K analytic on $\mathbb{C} \setminus (-\infty, -2]$; on the oscillatory side $w \in (-1, 0)$ the same K serves with $\operatorname{arccosh} = i\arccos$ and $\sqrt{2w} = i\sqrt{2|w|}$. Then $F := \sqrt{2Y}\,K(Z-1) \in C^3$, and on both sides of t_p

$$\int_t^{t_p} \operatorname{arccosh} Z = (t_p - t)^{3/2} H(t), \qquad H(t) := \int_0^1 \sqrt{\vartheta}\, F(t_p + (t - t_p)\vartheta)\, d\vartheta > 0,$$

so $\zeta = (t_p - t)\,(3/2 H)^{2/3}$ and $g^4 = (3/2 H)^{2/3}/(\hat a^2 Y(Z+1))$ are C^3. Each operation used (the root t_p, division by a simple zero, composition with K, the ϑ-integral, a power of a positive function) is continuous from C^4 to C^3 near the $h = 0$ data, so the bounds and positivity statements, which hold at $h = 0$, persist. The identities are the definitions differentiated: $\sqrt{\zeta}\,\zeta' = -\operatorname{arccosh} Z$ and $\sinh^2(\operatorname{arccosh} Z) = Z^2 - 1$. In the analytic case every step preserves analyticity. □

A.5 THE SECOND-ORDER TERM ON THE HALF-LINE

This subsection proves the five steps listed in the plan of the proof of Theorem 9.1; the notation is that of Section 9.1.

Lemma A.9 (The half-line). There are $h_0, c_0 > 0$ such that the Langer data of Lemma 7.2 extend analytically to t in a neighbourhood of $[1/4, \infty)$ containing the sector $\{|\operatorname{Im} t| < c_0 \operatorname{Re} t\}$ for large $\operatorname{Re} t$, jointly analytic in complex $|h| \le h_0$. For real t: $\zeta' < 0$ on $[1/4, \infty)$, $|\zeta(t)| \ge r_0 \min(|t - t_p|, |t - t_p|^{2/3})$, and for $t \ge T_1$, $r \asymp t^{2/3}$, $|\zeta'| \asymp t^{-1/3}$, $g^2 \asymp t^{1/3-\alpha}$, $\mathcal{G} \asymp t^{\alpha/2+1/6}$ and $|\vartheta - \pi/2| \le Ct^{-\alpha}$.

Proof of Lemma A.9. On a compact neighbourhood of $[1/4, T_1]$ this is Lemma 7.2 in its analytic form. In the sector, $|Z| \le 1/2$ for large $\operatorname{Re} t$, so $\vartheta = \arccos Z$ is analytic

with $|\vartheta - \pi/2| \le C|t|^{-\alpha}$; then $\mathscr{A}(t) := \int_{t_p}^{t} \vartheta$ (along a path in the domain) satisfies $\mathscr{A} = \pi/2t + O(1)$, $r = (3/2\mathscr{A})^{2/3}$ is analytic and agrees with $-\zeta$ on the real axis, and $g^2 = r^{1/2}/(\hat{a}\sin\vartheta)$ is analytic and nonvanishing. □

Lemma A.10 (Residuals with decay). Let $\mathfrak{p}_2(t) := \tfrac{1}{2}\partial_\eta^2 \mathscr{P}(t,\eta)|_{\eta=0,h=0}$, with $\mathscr{P}$ as in the proof of Lemma 7.3. For $0 < h \le h_0$ and all sites with $t = nh \ge \tfrac{1}{2} - h$:

(a) $|\mathfrak{p}_n - h^2\mathfrak{p}_2(t)| \le Ch^3\mathscr{G}(t)\langle t\rangle^{-2}$ and $|\mathfrak{p}_2(t)| \le C\mathscr{G}(t)\langle t\rangle^{-2}$;

(b) $|\mathfrak{q}_n| \le Ch^{10/3}\mathscr{G}(t)\langle t\rangle^{-10/3}$;

(c) $W_n = -(h^{1/3}/\pi)(1+\omega_n)$ with $|\omega_n| \le Ch^2\langle t\rangle^{-2}$;

(d) $\mathfrak{p}_2$ is real-analytic on $[\tfrac{1}{4},\infty)$ and $|\mathfrak{p}_2'(t)| \le C\mathscr{G}(t)\langle t\rangle^{-3}$.

Proof of Lemma A.10. On the compact part this is Lemma 7.3 in its analytic form, together with $\mathscr{P} = \eta^2\mathscr{P}_2(t;h) + O(\eta^4)$ and $\mathscr{P}_2(t;h) = \mathfrak{p}_2(t) + O(h)$ by analyticity in h. For $t \ge T_1$ take $|\eta| \le c_1 t$ with c_1 small. Then $t \pm \eta$ stays in the sector, $|\hat{a}(t \pm \eta/2)| \le C\hat{a}(t)$, $|g(t \pm \eta)| \le Cg(t)$, $|s_\pm| \le Ct^{-1/3}$, and along the segment from 0 to $s_\pm$ the coefficient of $Y'' = (\zeta + \eta s)Y$ is $O(t^{2/3})$, so $|X_1| \le C$ and $|X_2| \le Ct^{-1/3}$. Hence $|\mathscr{P}| \le C\mathscr{G}$ and $|\mathscr{Q}| \le C\mathscr{G}t^{-1/3}$ on the disc $|\eta| \le c_1 t$, also for complex h. Cauchy's estimate on this disc, with the parity and the two identities of Lemma 7.3, gives (a), (b) and the Casoratian bound, and Cauchy in h gives $|\mathscr{P}_2(t;h) - \mathfrak{p}_2(t)| \le C|h|\mathscr{G}t^{-2}$; (d) is Cauchy in t on a disc of radius $c_1 t$. The rough part adds, by (27) and $|e_k| \le C_e h^3 t^{-3}$, at most $Ch^3t^{-3}\mathscr{G}$ to $\mathfrak{p}_n$, $Ch^{10/3}t^{-10/3}\mathscr{G}$ to $\mathfrak{q}_n$ and Ch^3t^{-3} to ω_n. □

Lemma A.11 (Forward bound on the half-line). $V \le C_V h$; $c(n) \to c(\infty)$ with $|c_1(\infty) - 1|, |c_2(\infty)| \le e^{2V} - 1$; and

$$\left|c_2(\infty) - \mathscr{F}_2\right| \le V(e^{2V} - 1) = O(h^2), \qquad \mathscr{F}_2 := -\sum_{n \ge n_0} \frac{\psi_A(n)\,(\mathscr{L}\psi_A)(n)}{W_n}.$$

Proof. On the compact part, $\sum \nu_n \le Ch$ by Lemma 7.4. For $t \ge T_1$, (A3) gives $\mathsf{M}^2 \le \mathsf{m}_0 h^{1/3} r^{-1/2}$, the identity $\hat{a}g^2 r^{-1/2} = 1/\sin\vartheta \le 2$ holds, and $\mathsf{k} \le \mathsf{k}_0(1+|x|)^{1/2} \le Ch^{-1/3}r^{1/2}$; with Lemma A.10 this gives $\nu_n \le C(h^2t^{-2} + h^3t^{-3})$, whose sum over sites spaced by h is $O(h)$. Steps 1–4 of the proof of Theorem 7.1 are a forward induction that never uses a right endpoint; since x_n is strictly decreasing on the half-line and $W_n \ne 0$, they give $\Lambda(n) \le e^{2V}$ and summable increments, hence the convergence and the bounds. For the last claim write $c(n-1) = (1,0) + \delta c(n-1)$; then $c_2(n) - c_2(n-1) = -\psi_A(n)(\mathscr{L}\psi_A)(n)/W_n - \psi_A(n)\sigma_\delta(n)/W_n$, where σ_δ is the residual of $\delta c_1\,\mathrm{Ai} + \delta c_2\,\mathrm{Bi}$, bounded by $(|\mathfrak{p}_n| + \mathsf{k}_n|\mathfrak{q}_n|)\mathsf{E}_n^{-1}\mathsf{M}_n(e^{2V} - 1)$ by the second step of that proof. With $|\psi_A| \le g\mathsf{E}^{-1}\mathsf{M}$ each term is at most $\nu_n(e^{2V} - 1)$. □

Lemma A.12 (Reading the phase at infinity). Let y_λ be the largest root of $2\tilde{a} = \lambda$ and

$$\tilde{\Phi}(\lambda) := y_\lambda\pi/2 + \int_{y_\lambda}^{\infty} \arcsin(\lambda/2\tilde{a}(y))\,dy.$$

If a real solution satisfies $u(n) = C_1(n)\psi_A(n) + C_2(n)\psi_B(n)$ with $C(n) \to (C_1, C_2) \neq 0$, then its coordinates $(\alpha_\infty, \beta_\infty)$ of Theorem 2.2(i) are a positive multiple of $(\cos(\tilde\Phi - \delta_C), \sin(\tilde\Phi - \delta_C))$, where $\delta_C := \arg(C_1 + iC_2)$.

Proof. With DLMF 9.8.3, $\mathrm{Ai}(-X) = \mathsf{M}_D(X)\cos\theta_D(X)$ and $\mathrm{Bi}(-X) = \mathsf{M}_D(X)\sin\theta_D(X)$, where $\theta_D = \pi/4 - \xi(1 + O(X^{-3}))$, $\xi = {}^2\!/_3 X^{3/2}$ (DLMF 9.8.20–9.8.22). Hence $u(n) = g_n \mathsf{M}_D(|x_n|)|C|\big(\cos(\xi_n - \pi/4 + \delta_C) + o(1)\big)$. As in Lemma A.6, $\xi_n = \int_{y_\lambda}^{n+1/2} \arccos(\lambda/2\tilde a)$, and writing $\arccos = \pi/2 - \arcsin$ gives $\xi_n - \pi/4 = \frac{n\pi}{2} - \tilde\Phi + o(1)$. So $u_{2m} = (-1)^m g \mathsf{M}_D |C|(\cos(\tilde\Phi - \delta_C) + o(1))$ and $u_{2m+1} = (-1)^m g \mathsf{M}_D |C|(\sin(\tilde\Phi - \delta_C) + o(1))$. By Theorem 2.2(i) and the vanishing pattern of $\widetilde P, \widetilde R$, $\alpha_\infty = \lim u_{2m}/\widetilde P_{2m}$ and $\beta_\infty = \lim u_{2m+1}/\widetilde R_{2m+1}$, where $\widetilde P_{2m} = (-1)^m(\rho_m/A_m)^{1/2}$, $\widetilde R_{2m+1} = (-1)^m(\rho_m A_m)^{-1/2}$ and $\rho_m \to 1$. Finally $g_n^2 \mathsf{M}_D^2 = (h^{1/3}/\pi)(1 + o(1))/(\hat a \sin\vartheta)$ with $\hat a(nh) = \tilde a(n + {}^1\!/_2)/\lambda$ and $\sin\vartheta \to 1$, so both amplitudes tend to the same limit $(h^{1/3}\lambda/\pi)^{1/2}$. □

Lemma A.13 (An Airy-weighted Riemann sum). $\mathscr{F}_2 = hK_2 + O(h^{4/3})$, where $K_2 := {}^1\!/_2 \int_1^\infty G_2(t)\,dt$ and $G_2 := g\mathfrak{p}_2 r^{-1/2}$ at $h = 0$.

Proof. **Step 1** (*reduction*). $\mathscr{F}_2 = -\sum_n g_n \mathrm{Ai}(x_n)(\mathfrak{p}_n \mathrm{Ai}(x_n) + \mathfrak{q}_n \mathrm{Ai}'(x_n))/W_n$. Dropping $\mathfrak{q}$, replacing W_n by $-h^{1/3}/\pi$ and $\mathfrak{p}_n$ by $h^2\mathfrak{p}_2(t_n)$ costs $O(h^2)$, by Lemma A.10 and the bounds in the proof of Lemma A.11. So $\mathscr{F}_2 = \pi h^{5/3}\sum_{n\ge n_0} F(t_n)\,\mathrm{Ai}(x_n)^2 + O(h^2)$ with $F := g\mathfrak{p}_2$, where $|F| \le C\langle t\rangle^{-5/3}$, $|F'| \le C\langle t\rangle^{-8/3}$ and F is analytic near t_p.

Step 2 (*near the turning point*). Fix $L \ge 1$. Sites with $|t_n - t_p| \le h^{2/3}L$ number at most $CLh^{-1/3}$ and have $|\mathrm{Ai}| \le 1$, so they contribute $O(Lh^{1/3})$ to $\pi h^{2/3}\sum F\,\mathrm{Ai}^2$; on the hyperbolic side $\mathrm{Ai}(x)^2 \le e^{-4/3x^{3/2}}$ and the contribution is $O(h^{1/3})$; the matching piece of ${}^1\!/_2\int_{t_p} Fr^{-1/2}$ is $O((h^{2/3}L)^{1/2})$.

Step 3 (*the oscillatory region*). For $t_n > t_p + h^{2/3}L$, $X_n := |x_n| \ge cL$ and $\mathrm{Ai}(-X)^2 = (2\pi X^{1/2})^{-1}(1 + \sin 2\xi + O(X^{-3/2}))$. With $X_n^{-1/2} = h^{1/3}r_n^{-1/2}$ the sum becomes ${}^1\!/_2\sum_n h\,\phi(t_n)(1 + \sin 2\xi_n + O(X_n^{-3/2}))$ with $\phi := Fr^{-1/2}$. The error term sums to $O(h^{1/3}/L)$. The mean term is a Riemann sum of a function with $|\phi| \le C(t - t_p)^{-1/2}$ and $|\phi'| \le C(t - t_p)^{-3/2}$ near t_p and $|\phi'| \le C\langle t\rangle^{-3}$ far out, so it equals $\int_{t_p + h^{2/3}L}^\infty \phi + O(h^{2/3}L^{-1/2})$. For the oscillating term let $\varphi_n := \xi_{n+1} - \xi_n \in (0, \pi/2]$, which satisfies $\varphi_n \ge c\min(1, (t_n - t_p)^{1/2})$; summation by parts with $e^{2i\xi_n} = (e^{2i\xi_{n+1}} - e^{2i\xi_n})/(e^{2i\varphi_n} - 1)$ bounds it by $O(h^{1/3}/L)$.

Step 4 (*the limit*). Fixing L, $\pi h^{2/3}\sum F\,\mathrm{Ai}^2 = {}^1\!/_2\int_{t_p}^\infty Fr^{-1/2}dt + O(h^{1/3})$. With $t = t_p(h) + v^2$ the right side is analytic in h, so it equals $K_2 + O(h)$. □

A.6 THE DISCRETE TRANSPORT IDENTITY

This is the computation behind $\partial_\eta \mathcal{Q}(t,0)=0$ in the proof of Lemma 7.3. Put $k:=\sqrt{\zeta}$ (imaginary on the oscillatory side; the computation is algebraic) and $\Theta:=k\zeta'$, so that the eikonal identity reads $\cosh\Theta=Z$ and the amplitude identity reads $\hat{a}\,g^2\sinh\Theta/k=-1$. The solution of $Y''=(\zeta+\eta s)Y$ with $Y(0)=0$, $Y'(0)=1$ is $X_2=X_{20}+\eta X_{21}+O(\eta^2)$ with

$$X_{20}(s)=\frac{\sinh ks}{k},\qquad X_{21}(s)=\frac{s^2\cosh ks}{4k^2}-\frac{s\sinh ks}{4k^3},$$

the solution of $Y''-k^2Y=sX_{20}$ with zero data; X_{20} is odd and X_{21} is even in s. With $s_\pm(\eta)=\pm\zeta'+\eta\zeta''/2+O(\eta^2)$,

$$\mathcal{Q}(t,\eta)=\hat{a}(t+\tfrac{\eta}{2})g(t+\eta)X_2(s_+(\eta);\eta)+\hat{a}(t-\tfrac{\eta}{2})g(t-\eta)X_2(s_-(\eta);\eta).$$

At $\eta=0$ the two terms cancel because X_{20} is odd. Differentiating once at $\eta=0$ and using $X_{20}'=\cosh ks$,

$$\partial_\eta\mathcal{Q}(t,0)=(\hat{a}'g+2\hat{a}g')\frac{\sinh\Theta}{k}+\hat{a}g\Big[\zeta''\cosh\Theta+\frac{\zeta'^2\cosh\Theta}{2k^2}-\frac{\zeta'\sinh\Theta}{2k^3}\Big],$$

where the bracket is $\zeta''\cosh\Theta+2X_{21}(\zeta')$. Differentiating the amplitude identity, with $k'=\zeta'/(2k)$ and $\Theta'=k'\zeta'+k\zeta''$, gives

$$(\hat{a}'g+2\hat{a}g')\frac{\sinh\Theta}{k}=-\hat{a}g\Big[\cosh\Theta\Big(\zeta''+\frac{\zeta'^2}{2k^2}\Big)-\frac{\zeta'\sinh\Theta}{2k^3}\Big].$$

The two displays add up to zero. The Casoratian factor of Lemma 7.3 at $\eta=0$ is $\hat{a}g^2X_{20}(\zeta')=\hat{a}g^2\sinh\Theta/k=-1$, the amplitude identity itself.

A.7 THE LOCAL DENSITY OF THE BORN CONSTANT

These are the intermediate identities of the elimination in the first step of the proof of Proposition 9.2. Put $w:=\operatorname{arccosh}Z$, so that the eikonal identity is $\sqrt{\zeta}\,\zeta'=-w$, and express every derivative through the jets $Z,Z',Z'',Z^{(3)}$ and ζ:

$$\zeta'=-w\zeta^{-1/2},\qquad \zeta''=-w'\zeta^{-1/2}-\tfrac{1}{2}w^2\zeta^{-2},\qquad w'=\frac{Z'}{\sinh w},\qquad w''=\frac{Z''-\cosh w\,w'^2}{\sinh w},$$

and $\zeta^{(3)}$ by one more differentiation; $\hat{a}=1/(2Z)$, $\hat{a}'=-Z'/(2Z^2)$, $\hat{a}''=-Z''/(2Z^2)+Z'^2/Z^3$; and $g=\zeta^{1/4}(\hat{a}\sinh w)^{-1/2}$, differentiated by the chain rule. The solution of $Y''=(\zeta+\eta s)Y$ with $Y(0)=1$, $Y'(0)=0$ is $X_1=X_{10}+\eta X_{11}+\eta^2X_{12}+O(\eta^3)$ with $X_{10}=\cosh ks$,

$$X_{11}=\frac{s^2\sinh ks}{4k}-\frac{s\cosh ks}{4k^2}+\frac{\sinh ks}{4k^3},$$

$$X_{12}=\frac{s\big(3k^3s^3\cosh ks-10k^2s^2\sinh ks+21ks\cosh ks-21\sinh ks\big)}{96k^5},$$

where X_{1j} solves $Y''-k^2Y=sX_{1,j-1}$ with zero data. Put $F(\eta):=\hat{a}(t+\eta/2)g(t+\eta)X_1(s_+(\eta);\eta)$ with $s_+(\eta)=\zeta'+\eta\zeta''/2+\eta^2\zeta^{(3)}/6+O(\eta^3)$. Since $\mathcal{P}(t,\eta)=F(\eta)+F(-\eta)-g$ is even in η,

$\mathfrak{p}_2 = {}^1\!/_2 \partial_\eta^2 \mathscr{P}(t,0) = F''(0)$, and the constant term $2F(0) - g = g(2\hat{a}\cosh w - 1)$ vanishes by the eikonal identity. Expanding F to second order with the jets above, then passing to the oscillatory side by $\zeta = -r$, $\sqrt{\zeta} = i\sqrt{r}$, $w = i\vartheta$, the terms with $Z^{(3)}$ cancel and

$$g\,\mathfrak{p}_2\, r^{-1/2} = \mathscr{D} - \frac{5}{16}\,\vartheta\, r^{-3}, \qquad \mathscr{D} = -\frac{Z'^2(S^4 + 17S^2 - 15) - Z'' Z(S^4 + 6S^2)}{12\, S^5 Z}, \quad S = \sin\vartheta,$$

which is the formula of Proposition 9.2. The expansion is long but mechanical; it was carried out and checked in exact arithmetic (Appendix B).

B COMPUTATIONAL CROSS-CHECKS AND REPRODUCIBILITY

No proof in this paper depends on a computation. This appendix lists the computations that were run alongside the proofs as cross-checks, and how the eigenvalues of Section 12 were computed. Each item names the program that produced it; the programs and their outputs are listed in Section B.3.

B.1 CROSS-CHECKS OF CLOSED FORMS AND IDENTITIES

The following identities, all proved in the text, were also checked in exact arithmetic (SymPy) or at 40 to 60 significant digits (mpmath) by `exact_checks.py`, unless another program is named.

- The Beta-function forms of I_α and J_α in (6) and of C_α in Corollary 5.3, and the identity between Valent's density and $C_\alpha c^{-1/\alpha}$ for balanced rates.
- The drift integral $-\kappa'\pi/(2\alpha)$ of Proposition 4.5.
- The block identities of Lemma 3.1, the generator (11) and the exact Euler-step matrix in the proof of Lemma A.3.
- The Bessel facts (B1)–(B3), the matching coefficients in the proof of Theorem 6.2 and the limits defining Q in (21); the birth–death dictionary and the Krein offset of Corollary 6.5; the closed form of ϖ for birth–death chains, against direct products on the chain BD6 (`varpi_bd.py`).
- Olver's envelope constants in (A3)–(A4) of Section 7: $\mathrm{m}_0 = 0.48483$ at $x = {}^1\!/_2$, and $\mathrm{k}(x)(1 + |x|)^{-1/2}$ increasing to 1 as $x \to -\infty$ (`airy_envelopes.py`).
- The elimination behind the local density $\mathscr{D}$ of Proposition 9.2 and the discrete transport identity of Lemma 7.3 (`p2_reduction.py`; the hand derivations are in Sections A.6 and A.7); the finite part and the w-substitution in Proposition 9.2; and K_2 by direct quadrature of G_2, which agrees with $(2\alpha - 1)J_\alpha/24$ to $2 \cdot 10^{-12}$.

- The split of the Weyl–Moyal term in Remark 9.1 into $+(\alpha-1)^2 J_\alpha/(24N)$ and $-\alpha^2 J_\alpha/(24N)$.
- The range $[-0.2078862\ldots,\ 0.0933663\ldots]$ of $\zeta(-1/2,\cdot)$ on $(0,1]$ in Proposition 8.5, with the maximum at $\theta = 0.3027218\ldots$.
- The squeezing expansions of Corollary 10.1, and interval-arithmetic values of the constants.
- The Berg–Valent closed forms of Section 11: the masses $1/(DG_F')$ and $-1/(BD')$ against [7, (29)–(30)], the closed forms of W_F and W_∞, and the product law of Theorem 11.3(c) (`bv_closed_forms.py`).
- The q-family rows of Table 2 (`qfamily_indices.py`). For an order-zero problem Hadamard's theorem gives $G_\sigma = Cz^m \prod_j (1 - z/x_j)$ over the support, so $W_\sigma = 1/(\mu_\sigma(\{x\})G_\sigma'^2)$ is computable up to a constant from the masses alone. For Al-Salam–Carlitz the masses are those of [7, Ex. 4.2]; at $(q,a) = (0.5, 1.5)$ and $(0.3, 2)$ the local exponents of W_F and W_∞ agree with $1+\gamma_q$ and $1-\gamma_q$ to seven digits, with $W/x^{1\pm\gamma_q}$ converging to a positive constant. For Stieltjes–Wigert the masses are $1/\sum_n \hat{p}_n(x)^2$ with the polynomials of [7, Ex. 4.3], at the zeros of Ramanujan's function; at $q = 0.5$ and 0.3 the exponents of W_F, W_1 and W_∞ approach $3/2$, $1/2$ and $1/2$ to six digits, again with positive limits of W/x^e. The same method reproduces W_F of the Berg–Valent problem to relative $2\cdot 10^{-6}$ at $n = 7$.
- The test of Conjecture 13.2 (`conjH.py`): $\Phi_r(\lambda_j)/\pi - j$ is -0.50040, -0.50019 and -0.50032 at $j = 200$ on three chains, approaching $-1/2$ like $1/j$, while with o, e in place of $|o|, |e|$ the first chain ($\kappa_o < 0$) gives -0.70040.

B.2 THE EIGENVALUE COMPUTATIONS

The Sturm counts of Section 12 are the only long computation. They are made by `sturm.py` and `truncation_run.py`, and for chains with a diagonal by `sturm_diag.py`, `diag_run.py` and `diag_balanced.py`. The k-th positive eigenvalue was found by bisection on the counts, and for most chains by 256-section, which evaluates 256 spectral parameters per pass over the sites; on a test case 256-section left the eigenvalues bit-for-bit unchanged and cut the number of passes about fourfold. The cases B3, SQ6m1 and D1 at $t = 0$ were computed with bisection, the others with 256-section. The counts were run on a GPU (NVIDIA RTX 4090) with one spectral parameter per thread in float64. On a bounded workload ($n = 2\cdot 10^6$ sites, 64 spectral parameters, chain B3, including transfers and synchronisation) the GPU kernel took 0.54 s against 10.3 s for the vectorised CPU loop, with identical counts (`gpu_probe.py`), and the eigenvalues of a truncation of size ≈ 800 agree with a 40-digit bisection in mpmath (`validate_sturm.py`). The maximal truncation size was reduced from 2^{26} to 2^{24} and then 2^{22} sites for the chains with slowly decaying tails, which lowers the largest level

reached for them (last column of Table 3). The chain parameters are in `chains.py`; the table and Figure 2 are produced from the stored truncation data by `make_table.py` and `fig_laws.py`, without computing new eigenvalues. The exact checks of the previous subsection are small CPU computations.

B.3 CODE AND DATA AVAILABILITY

The LaTeX sources of this paper, all programs named above (folder `scripts`) and their outputs (folder `outputs`, including the per-eigenvalue truncation data behind Table 3 and Figure 2) are published in the repository `github.com/johnzfitch/limit-circle-jacobi-asymptotics`, release tag `v1.1`. The file `SHA256SUMS.txt` in the release lists a SHA-256 checksum for every file. From the repository root, with Python 3, mpmath and SymPy, the cross-checks are reproduced by

```
python scripts/exact_checks.py
python scripts/p2_reduction.py
python scripts/airy_envelopes.py
python scripts/varpi_bd.py
python scripts/bv_closed_forms.py
python scripts/qfamily_indices.py
python scripts/conjH.py
```

each of which prints its results (the stored copies are the corresponding files in `outputs`). The eigenvalue data are reproduced by `python scripts/validate_sturm.py`, `python scripts/truncation_run.py` (all chains, or one case with `--case <chain> <t>`), `python scripts/diag_run.py` and `python scripts/diag_balanced.py`; these need CuPy and a CUDA device (`sturm.py` also contains the NumPy reference implementation of the count that was used for validation). Finally `python scripts/make_table.py` and `python scripts/fig_laws.py` rebuild Table 3 and Figure 2 from the stored data (the figure script uses the plotting style module `es_figure` of the typesetting kit, and matplotlib). The computations were directed by the author and written and run by Claude Code (Section 14); their independence is from the asymptotic proofs, not from the people and tools that wrote them.

C HOW THE RESULTS WERE REACHED

The work was kept in a running findings ledger and a task log, with corrections recorded as they happened. It began on 12 October 2026, when I asked whether the tetrahedron has any place in the theory of Jordan algebras. By 17 October it had become a study of quantum tetrahedra over the four rank-three Jordan algebras, in which the volume of a quantum tetrahedron reappears as the commutator of two pair Casimirs. From there it moved to the dynamics of loop quantum gravity: whether a Lorentzian Hamiltonian constraint can close on an off-shell space. That question was not settled, and the companion paper [20] describes it. This appendix follows the branch that became the present paper.

On 20 October, testing whether self-dual variables could tell Lorentzian dynamics apart from a formal factor of i, the four-valent volume with $\mathfrak{su}(1,1)$ labels was written as a tridiagonal matrix and found not to be essentially self-adjoint. The first reading was that the extension angle was an invisible phase. A referee corrected it: for equal labels on two legs only $t = 0$ and $t = \pi/2$ respect the orientation, and they give different smallest volumes. A literature search showed that the closed form of the matrix elements was classical, and that the infinite representations behind it had been studied. What was not found anywhere was the limit-circle property and the extension it forces, so that was proved first, for four classes of chains. The novelty assessment was candid: probably new, but a short exercise for a specialist.

The same afternoon the spectra of these extensions were seen to obey a Bohr–Sommerfeld rule in which the extension angle appears as a phase. A referee found the first account's claim about the meaning of $t = 0$ and $t = \pi/2$ wrong, since it depends on how the legs are labelled. The rigorous proof rests on the factorisation of the transfer matrix into shears, which is classical [26], and on an exact rotation identity for two-site blocks. With these it reached the counting law for cubic links, but at first without a rate. The rate $O(\lambda^{-1/3})$ was first derived from Geronimo's turning-point theorem [21]. A referee then found that the printed proof of that theorem relies on an Airy weight that cannot exist, so the rate was only conditional. It was replaced by the discrete Olver bound of Theorem 7.1, proved from scratch, and a second referee judged the rate unconditional. Meanwhile the rule was proved for every exponent $\alpha > 1$, and the second-order constant $(2\alpha - 1)/24$, first seen numerically, was proved as well. A check of this stage by Codex confirmed the cubic rate. It also found an auxiliary threshold that was false, with the counterexample $a_n = n^2 + 100n$.

A novelty audit of the general law found it new but not large: its leading constant is the one Valent's conjecture predicts, and the Codex check located isolated leading terms for special cubic chains in earlier work cited by Gilewicz, Leopold, Ruffing and Valent [24]. I judged that a general law with a predicted constant was not enough, and asked for named open problems. That led to three of them. The first was the dimerised links of Section 6 and Valent's conjecture, where a referee showed that the

indicator follows in half a page from known results, so that what is new is the $O(1)$ term. The second was the question of Fischer, Burgarth and Lonigro [17], answered in Theorem 10.2. The third was the open problem of Berg and Szwarc [7] on critical indices. Our determinacy criterion for it turned out to be, in substance, a theorem of Borichev and Sodin [12], and the attribution was checked against their 1998 preprint before it was cited. That material was first planned as a separate paper and was folded into Section 11 on 21 October, because it does not stand on its own.

The paper was written from 21 October, from the primary sources and not from our working notes. I asked for a revision that explains the central mechanisms, the phase budget behind the missing quarter-period and the Bessel far field, before their proofs, with the routine estimates moved to Appendix A. An independent Claude referee then read the whole paper and checked its constants with its own Wronskian code. It found no wrong theorem, but it did find a gap in a remark on the class A, which was repaired. A GPT referee found that the reflection formula for negative eigenvalues in Corollary 10.1 was off by one in its index. It also found that the solutions defining the boundary conditions must solve the recurrence only away from the first site. Both findings were confirmed against Theorem 2.2 before they were adopted. The final revision also found and fixed an inconsistent diagonal value on one page. When the code was first published, 52 of its checksums failed because the repository had normalised line endings; this was fixed before the release was cited here.

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